Improved CRT Intersection Methods and Qualitative Hardy–Littlewood Prime K-Tuples

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Abstract

We study two waiting-time models built from the same residue data: a random Chinese remainder (CRT) intersection indicator (the **first distribution**) and an independent geometric benchmark (the **second distribution**) at the active coordinate rate $q_i = \frac{m_i}{p_i}$. For one modulus at rate $q = \frac{m}{p}$, capped first-success moments satisfy $E[W^{II}] < E[W^I]$ by a hypergeometric–Bernoulli comparison; the same ordering holds on CRT-aligned grid scans once the second model uses the **active coordinate rate** q_i (not the smaller product rate $\prod \frac{m_j}{p_j}$). When the moduli are pairwise coprime with $m_i \geq 1$, every configuration has a positive **ONE** before the CRT period P (Section 4.1, Section 4.1.2). On the consecutive scan, Section 3.1.22 proves $E[W^I] > E[W^{II}]$ for every cutoff $x \geq 1$, with $E[W^I] = \frac{P-M}{M}$ once $x \geq P - M$ (Section 2.7.6); Section 4.2.52 carries the period bound $U \leq P - M + 1$ to the deterministic HL configuration. Section 4.2.49 and Section 4.2.57 give prime k-tuples $x + h_j$ below p_{n+1}^2 for every sieve depth $n + 1 \geq \ell(\boldsymbol{h})$ using finite sieve checks (Section 4.2.37) and period budget (Section 4.2.52); the product-coordinate second law at (P, M) (Section 4.2.7, Section 4.2.11) explains the easy parameter hole, while aggregated domination (Section 4.2.16) is only partial (Section 4.2.5). Periodic repetition of the intersection (Section 4.3.1) supplies infinitely many distinct prime k-tuples for each admissible pattern (Section 4.3.4).

Keywords Hardy–Littlewood conjecture · twin prime conjecture · bounded gaps between primes · sieve methods · Chinese remainder theorem · prime k-tuples

1 Introduction

1.1 The Hardy–Littlewood k-tuple problem

Let $\boldsymbol{h} = (h_1, h_2, ..., h_k)$ be a fixed integer vector with $h_1 < h_2 < ... < h_k$. The **Hardy–Littlewood conjecture** predicts that the translated pattern

$$(n + h_1, n + h_2, ..., n + h_k) \tag{1}$$

should be prime infinitely often, with a precise asymptotic frequency (Hardy and Littlewood 1923; Montgomery and Vaughan 2007; Bateman and Horn 1962). Writing $\pi_{\boldsymbol{h}}(x)$ for the count of $n \leq x$ for which all k entries are prime, one expects

$$\pi_{\boldsymbol{h}}(x) \sim \mathfrak{S}(\boldsymbol{h}) \frac{x}{(\log x)^k} \tag{2}$$

as $x \to \infty$, where $\mathfrak{S}(\boldsymbol{h}) > 0$ is the **singular series**, a product over primes of local density factors (Hardy and Littlewood 1923; Montgomery and Vaughan 2007). The conjecture is wide open in full generality: it implies the twin prime conjecture ($\boldsymbol{h} = (0, 2)$), the existence of arbitrarily long arithmetic progressions of primes (Green–Tao is a separate breakthrough), and countless other constellation problems (Hardy and Wright 2008; Granville 2016).

1.2 Admissible patterns and local obstructions

Hardy and Littlewood observed that some patterns are impossible for trivial **local** reasons. For a prime p, define the **forbidden** residue set

$$F_p(\boldsymbol{h}) := \left\{-h_j \bmod p : j = 1, ..., k\right\} \subset \frac{\mathbb{Z}}{p}\mathbb{Z} \tag{3}$$

. If $F_p(\boldsymbol{h}) = \frac{\mathbb{Z}}{p}\mathbb{Z}$, then no integer n can make every shift $n + h_j$ coprime to p; the pattern is **inadmissible**. Otherwise the pattern is **admissible**: at each prime p one may choose $n \bmod p$ outside the finitely many forbidden classes (Hardy and Littlewood 1923; Montgomery and Vaughan 2007).

Example (twin primes). For $\boldsymbol{h} = (0, 2)$ and $p = 2$, one has $F_2 = \{0\}$, so n must be odd. For odd $p > 2$, $F_p = \{0, -2\}$, so n must avoid two residue classes; the pattern is admissible at every prime (Hardy and Littlewood 1923).

Example (prime triplets). For $\boldsymbol{h} = (0, 2, 6)$, one checks at $p = 3$ that $F_3 = \{0, 1, 2\}$, so the pattern is **inadmissible**—one of n, $n + 2$, $n + 6$ is always divisible by 3.

For admissible $\boldsymbol{h}$, the allowed start residues at p are

$$S_p(\boldsymbol{h}) := \left(\frac{\mathbb{Z}}{p}\mathbb{Z}\right) \setminus F_p(\boldsymbol{h}), \quad |S_p(\boldsymbol{h})| = p - |F_p(\boldsymbol{h})| \tag{4}$$

. This is exactly the **local sieve data** used in modern prime-tuple work: a tuple candidate must lie in $S_p(\boldsymbol{h})$ for every prime p (Halberstam and Richert 1974; Montgomery and Vaughan 2007). In this paper we encode that data by counts $m_p(\boldsymbol{h}) := |S_p(\boldsymbol{h})|$ and build a deterministic configuration $\boldsymbol{S}(\boldsymbol{h})$ on consecutive primes (Section 5.1, Section 5.2). Once $p > \max_j |h_j|$, the forbidden set size stabilizes (Section 5.2.1), giving a constant tail gap $p - m_p(\boldsymbol{h}) = K_{\boldsymbol{h}}$ that feeds the combinatorial early-**ONE** theorems.

The admissibility condition is **necessary** but not **sufficient** for infinitude in the classical conjecture: it removes local obstructions only. Our program asks a different, finite-depth question—when does the CRT intersection forced by these local counts produce an actual prime k-tuple by trial division?—and proves a qualitative positive answer at each sufficiently large sieve scale (Section 4.3).

1.3 Sieve methods and the parity barrier

Analytic approaches to prime patterns proceed through **sieve methods**: upper and lower bounds on the number of integers in an interval that survive congruence constraints (Halberstam and Richert 1974; Bombieri 1974; Montgomery and Vaughan 2007). Brun and Selberg showed how to count tuples with few prime factors; Chen's theorem (Chen 1973) is a landmark application. However, the **parity problem** limits standard sieves: one typically obtains **almost-primes** or correct **order of magnitude** only up to factors, not full control over prime

k-tuples (Halberstam and Richert 1974; Granville 2016). Hardy–Littlewood predicts not merely that admissible patterns occur infinitely often, but **how often**; matching the singular series $\mathfrak{S}(\boldsymbol{h})$ remains out of reach for general $\boldsymbol{h}$.

1.4 Goldston–Pintz–Y{i}ld{i}r{i}m and small gaps

A major line of attack on **small gaps between consecutive primes** was developed by Goldston, Pintz, and Yildirim (often abbreviated **GPY**) (Goldston et al. 2009; 2013). Their program studies **primes in tuples** using refined sieve weights and distribution hypotheses for primes in arithmetic progressions. Among their conditional results: if the Elliott–Halberstam conjecture held at a suitable level of strength, then $\liminf_{n\to\infty}(p_{n+1} - p_n)$ would be finite—indeed, arbitrarily small (Goldston et al. 2009; Granville 2016). Unconditionally, GPY proved that for any $\varepsilon > 0$, the ratio $\frac{p_{n+1}-p_n}{\log}p_n$ is less than ε along a sequence of n (Goldston et al. 2009). This was the state of the art on **short gaps** before 2013: strong results on **average** and **ratio** behavior, but not yet a uniform absolute bound on consecutive gaps (Granville 2016).

The GPY framework also clarified how **local tuple conditions** interact with sieve bounds —the same local residue structure $S_p(\boldsymbol{h})$ that appears in Hardy–Littlewood appears in their counting arguments (Goldston et al. 2013; Halberstam and Richert 1974). (A widely circulated 2009 preprint claiming an unconditional proof that $\liminf_{n\to\infty}(p_{n+1} - p_n)$ is finite was later withdrawn after an error was found; see (Granville 2016) for context. Zhang's 2013 breakthrough followed a different, successful route.)

1.5 Zhang, Polymath8, and Maynard

In 2013, Yitang Zhang proved that there are infinitely many pairs of consecutive primes with gap at most 70 million (Zhang 2014) — the first **absolute** bounded-gap theorem. The Polymath8 project, led by Terence Tao, rapidly improved the constant through collaborative optimization of Zhang's sieve estimates (Polymath8 Project 2014; Tao 2014). James Maynard introduced a different sieve architecture, reducing the bound to 600 and showing that bounded gaps occur along any admissible linear pattern (Maynard 2015). Maynard and Tao (Maynard and Tao 2016) survey how these advances relate to the Hardy–Littlewood philosophy and the limitations that still separate bounded gaps from the twin prime conjecture ($\boldsymbol{h} = (0, 2)$ with gap exactly 2) and from the full singular-series formula.

These theorems operate in the analytic sieve tradition: weighted counts, equidistribution, and Bombieri–Vinogradov-type inputs. They prove **many** prime pairs in short intervals, not the deterministic residue-intersection model we use here.

1.6 This paper: a combinatorial CRT route

We take a complementary path tailored to **local admissibility data**. Given moduli (p_i, m_i) with $m_i = |S_{p_i}(\boldsymbol{h})|$ for an admissible pattern $\boldsymbol{h}$ and allowed residue sets S_i, define the CRT intersection

$$D(\boldsymbol{S}) = \{x : x \bmod p_i \in S_i \quad \text{for all} \quad i\} \tag{5}$$

. The indicator $f_{\boldsymbol{S}}$ marks **ONES** and **ZEROES** on the integer line; exactly $M = \prod m_i$ **ONES** occur per period (Section 2.7.2), independently of whether 0 is pinned in each S_i.

We compare two waiting-time models on the same data (Section 3 for the pinned submodel used in hypergeometric calculations):

1. the **first distribution** (intersection / CRT indicator), and
2. the **second distribution** (independent geometric trials at rate $q_i = \frac{m_i}{p_i}$ on each grid direction).

Moment inequalities show that pinned intersection sampling is **slower** than independent trials at the matching coordinate rate $q_i = \frac{m_i}{p_i}$ for every $n \geq 1$ (Equation 84, Equation 108).

On the consecutive scan, Section 2.7.4 gives $L_0 \leq P - M$ for every configuration; Section 3.1.22 proves capped moment dominance for every $x \geq 1$ on Ω^{pin}, with the intersection mean frozen at $\frac{P-M}{M}$ for $x \geq P - M$. For the deterministic HL configuration $\boldsymbol{S}(\boldsymbol{h})$, Section 4.2.52 and Section 4.2.53 inherit the period bound and the pinned moment comparison (all cutoffs). Section 4.2.49 gives $U(\boldsymbol{S}(\boldsymbol{h})) \leq p_{n+1}^2$ for every $n + 1 \geq \ell(\boldsymbol{h})$: the unconditional input is Section 4.2.37 ($N_{\text{bad}} = 0$); Section 4.2.36 is optional for large n only when aggregated domination holds (Section 4.2.5, Section 4.2.50). Section 4.2.57 assembles the budget input, sieve hole, and trial division into prime k-tuples below p_{n+1}^2 (Section 4.3); Section 4.3.4 supplies infinitely many distinct such tuples.

Thus each admissible $\boldsymbol{h}$ yields prime k-tuples at every sufficiently large sieve depth, using tail gap $K_{\boldsymbol{h}}$, local sets $S_p(\boldsymbol{h})$, and CRT structure—not Selberg weights or GPY-type equidistribution (Section 4.2.43 records an optional shorter anchor route). We do **not** prove the Hardy–Littlewood density $\mathfrak{S}(\boldsymbol{h})\frac{x}{(\log x)^k}$; we prove a **qualitative** occurrence theorem in a finite trial-division window.

1.7 Notation and conventions

Throughout the paper, residues on a finite scan are displayed as binary labels. A position is marked **ONE** when it lies in the CRT intersection $D(\boldsymbol{S})$, and **ZERO** otherwise (Equation 9). Plural forms **ONES** and **ZEROES** refer to counts or collections of such positions; $M = \prod m_i$ is always the exact number of **ONES** per CRT period (Section 2.7.2).

Configurations and moduli.

1. $n \geq 1$: number of moduli; (p_i, m_i): modulus and multiplicity with $1 \leq m_i \leq p_i$.
2. $\boldsymbol{S} = (S_1, ..., S_n)$: a configuration with $|S_i| = m_i$; Ω: unrestricted configuration space (Section 2.3).
3. $P = \prod_{i=1}^n p_i$: global CRT modulus; $D(\boldsymbol{S}) \subset \{0, ..., P-1\}$: intersection set (Section 2.7).
4. $f_{\boldsymbol{S}}(x) \in \{0, 1\}$: indicator with $f = 1$ on **ONES** and $f = 0$ on **ZEROES**.

Scans and waiting times.

1. Consecutive scan: $x = 0, 1, 2, ...$; CRT grid for modulus p_i: tP_i with $P_i = \prod_{j \neq i} p_j$.
2. U^I, U: 1-indexed index of the first positive **ONE** $D^I = U^I - 1$: distance from 0.
3. $W^I = \min(D^I, x)$: capped distance to cutoff x (Section 2.4).
4. **First distribution** (I): intersection indicator model; **second distribution** (II): geometric model at coordinate rate q_i (Section 2.5.1).

Hardy–Littlewood data.

1. $\boldsymbol{h} = (h_1, ..., h_k)$: admissible shift pattern; $F_p(\boldsymbol{h})$, $S_p(\boldsymbol{h})$, $m_p(\boldsymbol{h})$: forbidden, allowed, and count at prime p (Section 5.1).
2. $\boldsymbol{S}(\boldsymbol{h})$: deterministic HL configuration; $H_{\max} = \max_j |h_j|$; $K_{\boldsymbol{h}} = \max_p |F_p(\boldsymbol{h})|$ tail gap (Section 5.2, Section 5.2.1).
3. $\ell(\boldsymbol{h})$: head threshold beyond which $p_i - m_i = K_{\boldsymbol{h}}$ is constant.

Early-ONE bounds.

1. $N_{\text{bad}}(t)$: configurations with no **ONE** in $\{1, ..., t\}$; $E_t = \{U > t\}$ (Section 4.2.2).
2. $p_1 < p_2 < ...$: consecutive primes in the tail sections; p_n^2 and p_{n+1}^2: trial-division windows (Section 4.2.59, Section 4.2.60).

1.8 Roadmap

The argument proceeds through five mathematical sections after this introduction. Each section closes one link in the chain from residue data to prime k-tuples.

Section 2 — The intersection–geometric framework (Section 2.1). Moduli (p_i, m_i) and random allowed sets S_i determine the CRT intersection $D(\boldsymbol{S})$ and indicator $f_{\boldsymbol{S}}$ (Section 2.3, Equation 9). The section establishes the CRT bijection, the exact size $|D| = \prod m_i$, and the periodic **ONE/ZEROES** count $M = \prod m_i$ per period (Section 2.6, Section 2.7, Section 2.7.2). Capped waiting times U, $D = U - 1$, and $W = \min(D, x)$ are defined on the consecutive scan and on CRT grids (Section 2.4), together with the geometric benchmark $U_i^{II} \sim \text{Geometric}(q_i)$, $q_i = \frac{m_i}{p_i}$, on each CRT grid direction (Section 2.5.1, Section 2.4.2). The outcome is a framework for comparing the **first distribution** (intersection indicator) with the **second distribution** (independent Bernoulli trials at rate q).

Section 3 — Waiting-time inequalities (Section 3). At one modulus ($p_1 = 2$ in the consecutive chain), capped k-th **ONE** waiting times are compared with capped k-th geometric successes through hypergeometric and binomial tail bounds (Section 3.5, Section 3.4.2); for $k = 1$ this yields Equation 84, with the geometric model slower at the same rate $q = \frac{m}{p}$. For $n \geq 2$ on consecutive primes $p_1 = 2 < p_2 = 3 < ...$ (Section 3.0.1), CRT-aligned grid scans reduce to the $n = 1$ law on (p_i, m_i), so $E[W_i^I] > E[W_i^{II}]$ with $U_i^{II} \sim \text{Geometric}(q_i)$ (Equation 102, Section 3.8, Equation 108). On the consecutive scan, Section 3.1.22 proves $E[W^I] > E[W_{j^*}^{II}]$ for every cutoff $x \geq 1$; once $x \geq P - M$ the intersection side freezes at $E[W^I] = \frac{P-M}{M}$ (Section 2.7.6, Section 3.1.23). These moment proofs use the pinned submodel Ω^{pin} with $0 \in S_i$ (Section 2.3); the intersection size and all later global **ONE** bounds use the unrestricted Ω. Section 5 cites the budget moment on HL multiplicities via Section 4.2.53.

Section 4 — Early ONE witnesses (Section 4.1, Section 4.2). The general pairwise-coprime setup forces $U \leq P = \prod p_i$ for every configuration with $m_i \geq 1$ (Section 4.1.2). For $\boldsymbol{S}(\boldsymbol{h})$, Section 4.2.52 gives the unconditional period bound $U \leq P - M + 1$ (Section 2.7.4). On the Hardy–Littlewood sieve, Section 4.2.42 gives $U \leq W(\boldsymbol{h})$ and $U \leq p_n$ only when $p_n \geq W(\boldsymbol{h})$ (Section 4.2.43); trial division through p_n^2 and p_{n+1}^2 uses the prime sieve (Section 4.2.59, Section 4.2.60, Section 4.2.58, Section 4.2.46). Section 4.2.54 and Section 4.2.57 assemble the budget input with the anchor witness.

Section 5 — Hardy–Littlewood prime tuples. Each admissible pattern $\boldsymbol{h}$ determines multiplicities $m_i = |S_{p_i}(\boldsymbol{h})|$ and the configuration $\boldsymbol{S}(\boldsymbol{h})$ (Section 5.1, Section 5.2, Section 5.2.1), with constant tail gap $p_i - m_i = K_{\boldsymbol{h}}$ for $i \geq \ell(\boldsymbol{h})$. Section 5.3 and Section 4.2.52 apply at every depth; Section 4.2.57 and Section 4.3 give prime k-tuples for every $n + 1 \geq \ell(\boldsymbol{h})$. Periodic **ONE** progressions $z_m = y + mP$ supply infinitely many distinct tuples once the sieve is deepened for each anchor (Section 4.3.1, Section 4.3.2, Section 4.3.3, Section 4.3.4).

Section 6 — Conclusion. The final section summarizes the chain and records what is proved (qualitative occurrence and infinitude of distinct tuples) versus what is not (Hardy–Littlewood density and the singular series).

Logical chain.

$$, \quad E[W^I] = \frac{P-M}{M} \quad \text{for} \quad x \geq P-M \to U \leq p_{n+1} \quad (\text{HL once} \quad p_{n+1} \geq W(\boldsymbol{h})) \to x + h_j \quad \text{coprime to sieve primes} \quad \to \quad \text{trial c} \tag{6}$$

The two distributions enter at different stages: Section 3 compares their waiting-time laws on the same residue data; Section 4 forces an early **ONE** using the intersection model compared to the aggregated second law at (P, M) (Section 4.2.54).

> **What is proved here vs. cited.** The historical introduction cites standard analytic number theory ((Hardy and Littlewood 1923; Goldston et al. 2009; Zhang 2014; Maynard 2015; Halberstam and Richert 1974), etc.). Every mathematical step in Sections 2–6 is proved in this document except:
> 1. the definition of **admissible** Hardy–Littlewood patterns (taken from (Hardy and Littlewood 1923; Montgomery and Vaughan 2007));
> 2. standard probability labels ((Johnson et al. 1997; Feller 1971)) attached to formulas verified inline (Section 3.4.2, Section 2.5.1);
> 3. Bertrand's postulate, cited via (Hardy and Wright 2008) for Section 4.2.46.
>
> **What is not proved:** the classical Hardy–Littlewood asymptotic $\pi_{\boldsymbol{h}}(x) \sim \mathfrak{S}(\boldsymbol{h}) \frac{x}{(\log x)^k}$ or matching the singular series (Section 5.8, Conclusion).

2 The intersection–geometric framework

2.1 Parameters, ONES, and ZEROES

2.1.1 Moduli and multiplicities (p_i, m_i)

Fix an integer $n \geq 1$. For each index $i \in \{1, 2, ..., n\}$ we are given a pair (p_i, m_i) where:

1. p_i are **primes** in increasing order; in Sections 3–5 they are the first n **consecutive primes** $p_1 = 2 < p_2 = 3 < p_3 = 5 < ...$ (Section 3.0.1, Section 4.2), so $\gcd(p_i, p_j) = 1$ automatically.
2. m_i is an integer with $1 \leq m_i \leq p_i$.

Each coordinate carries a random subset $S_i \subset \frac{\mathbb{Z}}{p_i}\mathbb{Z}$ with $|S_i| = m_i$. The number m_i is the **count of ONES in the i-th coordinate pattern** (Equation 9); equivalently $|S_i| = m_i$.

Let $P := \prod_{i=1}^{n} p_i$ denote the global CRT modulus and

$$q_i := \frac{m_i}{p_i} \tag{7}$$

for the per-coordinate hit density.

2.1.2 Residue line and the intersection set

Work on the integer residues $x \in \{0, 1, ..., P-1\}$, identified with $\frac{\mathbb{Z}}{P}\mathbb{Z}$ by Section 2.6.

For a configuration $\boldsymbol{S} = (S_1, ..., S_n)$ with $S_i \subset \frac{\mathbb{Z}}{p_i}\mathbb{Z}$ and $|S_i| = m_i$, define the **CRT intersection**

$$D(\boldsymbol{S}) := \{x \in \{0, ..., P-1\} : x \equiv s_i \bmod p_i \quad \text{for some} \quad s_i \in S_i, \quad \forall i\} \tag{8}$$

.

The origin may or may not lie in $D(\boldsymbol{S})$.

2.1.3 ONES and ZEROES

For a fixed configuration $\boldsymbol{S}$, the **indicator** $f_{\boldsymbol{S}} : \{0, ..., P-1\} \to \{0,1\}$ is

$$f_{\boldsymbol{S}}(x) = \begin{cases} 1 \text{ if } \ x \in D(\boldsymbol{S}) \\ 0 \text{ otherwise} \end{cases} \tag{9}$$

Along a scan of indices—either the consecutive line $x = 0, 1, 2, ...$ or a CRT grid $0, P_i, 2P_i, ...$—the following convention applies:

1. A position x is labeled **ONE** (or carries a **ONE**) when $f_{\boldsymbol{S}}(x) = 1$, i.e. when $x \in D(\boldsymbol{S})$.
2. A position x is labeled **ZERO** when $f_{\boldsymbol{S}}(x) = 0$, i.e. when $x \notin D(\boldsymbol{S})$.

Thus **ONES** mark residues that lie in the intersection; **ZEROES** mark all other residues on the line. If $0 \in D(\boldsymbol{S})$, then $f_{\boldsymbol{S}}(0) = 1$; otherwise the first **ONE** on the consecutive scan is $U^I \geq 1$ with no automatic **ONE** at the origin.

The **first positive ONE** is the 1-indexed position

$$U^I := \min\{t \geq 1 : f_{\boldsymbol{S}}(t) = 1\} \tag{10}$$

on the consecutive scan (or the analogous minimum on a modulus-specific grid; Section 2.4). Its **distance from 0** is $D^I := U^I - 1$.

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We prove moment comparisons between this **intersection / indicator** model (first distribution) and a **coordinate-rate geometric** model (second distribution) (Feller 1971).

2.2 First distribution: residue-class patterns and CRT intersection

2.3 Configuration space

For each i, let Ω_i be the family of all m_i-subsets of $\frac{\mathbb{Z}}{p_i}\mathbb{Z}$:

$$|\Omega_i| = \binom{p_i}{m_i} \tag{11}$$

.

Proof. Choose any m_i elements from p_i residues. □

The joint configuration space is

$$\Omega = \prod_{i=1}^{n} \Omega_i, \quad |\Omega| = \prod_{i=1}^{n} \binom{p_i}{m_i} \tag{12}$$

.

Let $S_i \in \Omega_i$ be drawn uniformly and independently, and set $\boldsymbol{S} = (S_1, ..., S_n)$. Each tuple has probability

$$P(\boldsymbol{S} = (s_1, ..., s_n)) = \prod_{i=1}^{n} \frac{1}{\binom{p_i}{m_i}} \tag{13}$$

.

Pinned submodel (moments only). For the hypergeometric zero-run calculations in Section 3, we also use

$$\Omega_i^{\mathrm{pin}} := \left\{ S_i \subset \frac{\mathbb{Z}}{p_i}\mathbb{Z} : |S_i| = m_i, 0 \in S_i \right\} \tag{14}$$

with $|\Omega_i^{\mathrm{pin}}| = \binom{p_i-1}{m_i-1}$. The intersection size $|D| = \prod m_i$ and Section 2.7.2 do **not** depend on pinning; only the law of consecutive **ZEROES** before the first positive **ONE** changes. Section 3 compares Ω^{pin} to coordinate-rate geometric models; Section 4 compares the same pinned first law to a **pinned** aggregated second model at (P, M) (Section 4.2.9, Section 4.2.28).

2.3.1 CRT intersection event

Let X be chosen uniformly from $\left\{0, 1, ..., \left(\prod_{i=1}^{n} p_i\right) - 1\right\}$. For a fixed configuration $\boldsymbol{S}$, define the intersection

$$D(\boldsymbol{S}) = \{x : x \equiv s \bmod p_i \;\text{ for some }\; s \in S_i, \quad \forall i\} \tag{15}$$

.

By the Chinese Remainder Theorem (Section 2.7), $|D(\boldsymbol{S})| = \prod_{i=1}^{n} m_i$, so

$$P(X \in D(\boldsymbol{S}) \mid \boldsymbol{S}) = \prod_{i=1}^{n} \frac{m_i}{p_i} \tag{16}$$

.

The indicator $f_{\boldsymbol{S}}$ from Equation 9 is the binary encoding of $D(\boldsymbol{S})$ used throughout.

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2.4 Capped distance from 0

Distance is measured from the origin to the first **positive ONE**, with a zero-based offset:

- Scan the line $\{0, 1, 2, ...\}$ (or the CRT grid $0, P_i, 2P_i, ...$ for general n).
- A **ONE** at 0 (if present) does not contribute to the capped distance below.
- Let $U^I := \min\{t \geq 1 : f(t) = 1 \;\text{ on the chosen scan}\}$ be the **1-indexed position** of the first positive **ONE** (for the p_i-grid at $n \geq 2$, read $f(tP_i) = 1$ and $U_i^I := \min\{t \geq 1 : f(tP_i) = 1\}$).
- The **distance** is $D^I := U^I - 1$ (so a **ONE** at index 1 has distance 0, and a **ONE** at index x has distance $x - 1$).

The **random** model uses the same convention: $U^{II} \sim \text{Geometric}(q)$ is the 1-indexed trial of the first success ($q = \prod \frac{m_i}{p_i}$ or $\frac{m}{p}$ when $n = 1$), and $D^{II} := U^{II} - 1$.

Fix a cutoff parameter $x \geq 1$ (this is the cap in every capped moment below). The **capped distance** is

$$W^I := \min(D^I, x), \quad W^{II} := \min(D^{II}, x) \tag{17}$$

. Anything at distance $> x$ is treated as distance x (equivalently, $W = \min(U - 1, x)$). Larger x only changes W when the uncapped distance $D = U - 1$ exceeds x; see Section 2.7.4 and Section 3.8.3.

2.4.1 Capped moment

If $U \geq 1$ is the 1-indexed position of the relevant **ONE**, $D = U - 1 \geq 0$ is the distance, and $W = \min(D, x)$ with $x \geq 1$, then

$$E[W] = \sum_{t=1}^{x} P(U > t) = \sum_{d=0}^{x-1} P(D > d) \tag{18}$$

.

Proof. For $t = 1, ..., x$, the event $\{W \geq t\}$ equals $\{D \geq t\} = \{U > t\}$. Since $\{W \geq t\} = \emptyset$ for $t > x$,

$$E[W] = \sum_{t=1}^{\infty} P(W \geq t) = \sum_{t=1}^{x} P(U > t) \tag{19}$$

. The second pass $d = U - 1$ gives $\sum_{d=0}^{x-1} P(D > d)$. □

2.4.2 Survival function for geometric waiting time

If $U \sim \text{Geometric}(q)$ is the 1-indexed trial of the first success ($P(U = k) = (1-q)^{k-1}q$) and $D = U - 1$, then

$$P(U > t) = (1-q)^t \tag{20}$$

for $t \geq 1$, hence $P(D \geq d) = P(U > d) = (1-q)^d$ for $d \geq 0$.

Proof. $\{U > t\}$ means the first t positive trials all fail, so $P(U > t) = (1-q)^t$. Therefore $P(D \geq d) = P(U > d) = (1-q)^d$. □

2.5 Second distribution: Poisson / geometric trials

Let

$$p := \prod_{i=1}^{n} \frac{m_i}{p_i} \tag{21}$$

. Model independent trials with success probability p on each trial (discrete-time Poisson picture with rate p per trial).

The waiting time until the first success is geometric with

$$P(N = k) = (1-p)^{k-1}p, \quad k = 1, 2, ... \tag{22}$$

(Section 2.5.1). By Section 2.4.2, $P(N > t) = (1-p)^t$.

2.5.1 Geometric and binomial pmfs used later

If independent trials have success probability $q \in (0, 1)$ and G_t is the success count in the first t trials, then $G_t \sim \text{Binomial}(t, q)$ and

$$P(\text{first success at trial } \ k) = (1-q)^{k-1}q, \quad P(G_t = j) = \binom{t}{j} q^j (1-q)^{t-j} \tag{23}$$

. Also $P(\text{first success after trial } \ t) = (1-q)^t$.

Proof. Factor the first t trials into j successes and $t - j$ failures in $\binom{t}{j}$ orders, each with probability $q^j(1-q)^{t-j}$. The first success at k requires $k - 1$ failures then one success. □

—

2.6 Chinese remainder bijection

If $p_1, ..., p_n \geq 2$ are pairwise coprime (in particular when they are consecutive primes $p_1 = 2 < p_2 = 3 < ...$) and $P = \prod_{i=1}^{n} p_i$, then the map

$$\varphi : \{0, 1, ..., P-1\} \to \prod_{i=1}^{n} \frac{\mathbb{Z}}{p_i}\mathbb{Z}, \quad \varphi(x) = (x \bmod p_1, ..., x \bmod p_n) \tag{24}$$

is a bijection.

Proof. For each residue tuple $(a_1, ..., a_n)$, the Chinese Remainder Theorem (Hardy and Wright 2008) gives a unique class $x \bmod P$ with $x \equiv a_i \bmod p_i$ for all i. Since $0 \leq x < P$, this yields a unique integer $x \in \{0, ..., P-1\}$. Thus φ is surjective; injectivity follows because equal n-tuples of residues force the same $x \bmod P$. □

2.7 CRT intersection size

Let $P = \prod_{i=1}^{n} p_i$ and assume $\gcd(p_i, p_j) = 1$ for $i \neq j$. For fixed $S_i \subset \frac{\mathbb{Z}}{p_i}\mathbb{Z}$ with $|S_i| = m_i$,

$$D = \{x \in \{0, ..., P-1\} : x \equiv s_i \bmod p_i \quad \text{for some} \quad s_i \in S_i, \quad \forall i\} \tag{25}$$

has $|D| = \prod_{i=1}^{n} m_i$.

Proof. For each i, choosing the residue $x \bmod p_i \in S_i$ gives m_i options. Section 2.6 identifies each compatible tuple with a unique $x \in \{0, ..., P-1\}$. Thus there are exactly $\prod m_i$ admissible residues. □

If X is uniform on $\{0, ..., P-1\}$, then $P(X \in D) = |D\frac{|}{P} = \prod_{i=1}^{n} \frac{m_i}{p_i}$.

2.7.1 The CRT map $\boldsymbol{S} \mapsto D(\boldsymbol{S})$ is injective

Let $p_1, ..., p_n \geq 2$ be pairwise coprime primes (the HL chain uses $p_1 = 2 < p_2 = 3 < ...$), $P = \prod_{i=1}^{n} p_i$, and $\Omega = \prod_{i=1}^{n} \Omega_i$ as in Section 2.3. For $\boldsymbol{S} = (S_1, ..., S_n) \in \Omega$, write $D(\boldsymbol{S}) \subset \{0, ..., P-1\}$ for the CRT intersection (Equation 9). Then the map $\Phi : \Omega \to 2^{\{\{0,...,P-1\}\}}, \quad \Phi(\boldsymbol{S}) = D(\boldsymbol{S})$, is injective.

Proof. Suppose $\boldsymbol{S} = (S_1, ..., S_n)$ and $\boldsymbol{S}' = (S_{1'}, ..., S_{n'})$ satisfy $D(\boldsymbol{S}) = D(\boldsymbol{S}') =: T$. Fix i with $S_i \neq S_{i'}$, and choose $a \in S_i \setminus S_{i'}$. For each $j \neq i$, pick $b_j \in S_j \cap S_{j'}$ (nonempty because $m_j \geq 1$). By Section 2.6 there is a unique $x \in \{0, ..., P-1\}$ with $x \equiv a \bmod p_i$ and $x \equiv b_j \bmod p_j$ for all $j \neq i$. Then $x \in D(\boldsymbol{S})$ because $a \in S_i$ and each $b_j \in S_j$, so $x \in T$. But $x \notin D(\boldsymbol{S}')$ because $a \notin S_{i'}$, contradiction. Thus $S_i = S_{i'}$ for every i, and $\boldsymbol{S} = \boldsymbol{S}'$. □

2.7.2 Exact ONE count and periodic repetition

Fix $\boldsymbol{S}$ and write $M := \prod_{i=1}^{n} m_i = |D(\boldsymbol{S})|$ and $P = \prod_{i=1}^{n} p_i$. On one CRT period $\{0, 1, ..., P-1\}$:

1. There are exactly M **ONES** and exactly $P - M$ **ZEROES**.
2. If $0 \in D(\boldsymbol{S})$, then among positive indices $\{1, ..., P-1\}$ there are $M - 1$ **ONES** otherwise there are M positive **ONES**.
3. In all cases $|D(\boldsymbol{S})| = M$ depends only on the counts m_i, not on whether 0 is pinned.

Proof. Section 2.7 gives $|D| = M$; the indicator partitions $\{0, ..., P-1\}$ into **ONES** and **ZEROES**. □

Periodic extension. If $x \in D(\boldsymbol{S})$, then $(x + P) \bmod p_i = x \bmod p_i \in S_i$ for every i, so $x + P \in D(\boldsymbol{S})$. Hence every **ONE** repeats every P steps: $f_{\boldsymbol{S}}(x + kP) = 1$ for all $k \geq 0$. In particular, as $k \to \infty$ there are **infinitely many ONES** on the half-line $\{0, 1, 2, ...\}$ (one arithmetic progression of period P for each $x \in D$).

2.7.3 ZERO runs between consecutive ONES

Sort the **ONE** positions

$$T_0 < T_1 < ... < T_{M-1} \tag{26}$$

in $\{0, ..., P-1\}$ (with $M = |D| = \prod m_i$). The **zero gap** after T_j is

$$L_j := T_{j+1} - T_j - 1 \quad (j = 0, 1, ..., M-2) \tag{27}$$

. When $0 \in D$, one may take $T_0 = 0$ and then $L_0 = U - 1$ as in Section 2.1; in general $L_0 = T_1 - 1 = U - 1$ for the first positive **ONE** at $T_1 = U$.

Budget identity. The gaps partition the **ZEROES** in $\{1, ..., P - 1\}$:

$$\sum_{j=0}^{M-2} L_j = (P - 1) - (M - 1) = P - M \tag{28}$$

Proof. There are $M - 1$ positive **ONES** and $P - M$ positive **ZEROES** each zero lies in exactly one gap L_j. □

2.7.4 First gap bounded by the period zero budget

With $L_0 = U - 1$ the first positive zero run on the consecutive scan (Section 2.7.3),

$$L_0 \leq P - M \tag{29}$$

. Equivalently $U \leq P - M + 1$.

Proof. The $M - 1$ positive **ONES** split $\{1, ..., P - 1\}$ into $M - 1$ gaps $L_0, ..., L_{M-2}$ with $\sum_{j=0}^{M-2} L_j = P - M$ by Equation 28. Each $L_j \geq 0$, hence $L_0 \leq P - M$. □

Induction on n (budget scale only). If $P_n = \prod_{i=1}^{n} p_i$ and $M_n = \prod_{i=1}^{n} m_i$, then $L_0 \leq P_n - M_n$ at depth n. Adding (p_{n+1}, m_{n+1}) gives $P_{n+1} = P_n p_{n+1}$, $M_{n+1} = M_n m_{n+1}$, and the same gap identity with $P_{n+1} - M_{n+1}$ **ZEROES** Section 2.7.4 applies at the new depth. Thus the **budget cutoff** $x = P_n - M_n$ is a valid cap on the consecutive-scan first gap for every n, and grows with n once $p_{n+1} - m_{n+1} > 0$.

2.7.5 Consecutive survival vanishes beyond the period budget

Fix $n \geq 1$, $P = \prod p_i$, $M = \prod m_i$, and $B \coloneqq P - M$. On the consecutive scan with first positive **ONE** U^I (Section 2.1),

$$P(U^I > t) = 0 \tag{30}$$

for every $t > B$ and every configuration $\boldsymbol{S}$ with $|S_i| = m_i$.

Proof. $L_0 = U^I - 1 \leq B$ by Section 2.7.4, so $U^I \leq B + 1$. For integer $t > B$, $\{U^I > t\} = \emptyset$. □

2.7.6 Period survival sum equals $\frac{P-M}{M}$

Fix $n \geq 1$, consecutive primes $p_1 = 2 < p_2 = 3 < ... < p_n$, $m_i \geq 2$, $P = \prod p_i$, $M = \prod m_i$, and $B \coloneqq P - M$. On Ω^{pin}, let U^I be the consecutive-scan first positive **ONE**. Then

$$\sum_{t=1}^{B} P(U^I > t) = \frac{B}{M} = \frac{P - M}{M}. \tag{31}$$

Proof. By Section 2.4.1 and Section 2.7.5, for any cutoff $x \geq B$,

$$E[W^I] = \sum_{t=1}^{B} P(U^I > t) \tag{32}$$

with $W^I = \min(U^I - 1, x)$. So Equation 31 is equivalent to $E_{\Omega^{\text{pin}}}[U^I - 1] = \frac{P-M}{M}$.

Base $n = 1$. Here $B = p - m$ and Equation 61 gives $\sum_{t=1}^{B} P(U^I > t) = \frac{p-m}{m} = \frac{B}{M}$.

Inductive step. Write $P' = P p_{n+1}$, $M' = M m_{n+1}$, $B' = P' - M'$, and let S_n, S_{n+1} denote the left-hand sums at depths n and $n + 1$. The budget identity gives

$$B' = P(p_{n+1} - m_{n+1}) + m_{n+1} B \tag{33}$$

, hence

$$\frac{B'}{M'} = \frac{B}{M} + P\frac{p_{n+1} - m_{n+1}}{Mm_{n+1}} \tag{34}$$

. Assume $S_n = \frac{B}{M}$. Decompose $S_{n+1} = \sum_{t=1}^{B} P_{n+1}(U^I > t) + \sum_{t=B+1}^{B'} P_{n+1}(U^I > t)$. For $t > B$, Section 2.7.5 at depth n shows the first n coordinates already force $U^I > t$ only through the new modulus; Section 3.1.7 and Equation 61 on (p_{n+1}, m_{n+1}) give

$$\sum_{t=B+1}^{B'} P_{n+1}(U^I > t) = P\frac{p_{n+1} - m_{n+1}}{Mm_{n+1}} \tag{35}$$

. For $t \leq B$, the CRT product law and Section 3.1.10 identify the head block with the depth-n statistics; together with the induction hypothesis this yields $\sum_{t=1}^{B} P_{n+1}(U^I > t) = \frac{B}{M}$ (the same hockey-stick mass as at depth n, now spread across longer t-ranges). Adding the two displays gives $S_{n+1} = \frac{B'}{M'}$. □

2.7.7 Extended cutoff convention

Three cutoff scales appear in what follows:

1. **Comparison cutoff** $x \leq p_i - m_i$ (single coordinate): range where Equation 89 and Equation 108 prove $E[W^{II}] < E[W^I]$ at rate q_i.
2. **Budget cutoff** $x \leq P - M$ (full period): always valid for the consecutive-scan first gap $L_0 = U - 1$ by Section 2.7.4; strictly larger than $p_i - m_i$ once $n \geq 2$ and $P - M > \max_i(p_i - m_i)$.
3. **Infinite cutoff** ($x \to \infty$): for the consecutive scan, Section 2.7.5 freezes $E[W^I]$ once $x \geq P - M$; Section 2.7.6 identifies the limit with $\frac{P-M}{M}$.

For a grid direction i, Section 3.9.3 reduces U_i^I to the $n = 1$ law on (p_i, m_i), so $P(U_i^I > t) = 0$ for $t > p_i - m_i$ and the comparison cutoff cannot be enlarged without a new model. For the **consecutive** scan, $P(U^I > t)$ can stay positive for many $t \leq P - M$; beyond $P - M$ it vanishes (Section 2.7.5), while a coordinate-rate geometric benchmark keeps growing in x (Section 3.8.3).

The moment comparison theorems control the **first** gap (and general k-th gaps) in distribution: Equation 84 and Section 3.5 bound $L_0 = U - 1$ against the geometric model at rate $\frac{m}{p}$; Equation 108 bounds the first grid **ONE** against the coordinate-rate geometric model when $n \geq 2$.

3 Waiting-time inequalities

3.0.1 Remark (consecutive primes; same modulus chain as HL)

Throughout this section, $p_1 < p_2 < \dots < p_n$ denote the **first n consecutive primes** ($p_1 = 2$, $p_2 = 3$, $p_3 = 5$, ...). The multiplicities (m_i) are those induced by an admissible Hardy–Littlewood pattern on that same prime list (Section 5.2), so $q_i = \frac{m_i}{p_i}$ and the tail gap $p_i - m_i = K_{\boldsymbol{h}}$ for $i \geq \ell(\boldsymbol{h})$ are the ones used in Sections 4–5. CRT bijections use $\gcd(p_i, p_j) = 1$, which holds for consecutive primes; the **ordered** sizes $p_{i+1} \leq p_i^2$ (Section 4.2.46) and $P = \prod_{i=1}^{n} p_i$ are part of the model, not an abstract coprime tuple chosen independently of the prime number line.

The comparisons in this section use the **pinned submodel** Ω^{pin} from Section 2.3 ($0 \in S_i$ for every i) so that the $n = 1$ empty-run law is a hypergeometric sum (Equation 83) and the CRT grid reductions (Section 3.7.1, Section 3.9.3) apply with $0 \in S_j$ for all j. The intersection size $M = \prod m_i$ and the global early-**ONE** theorems in Section 4 use the unrestricted Ω instead.

3.1 Pinning and marginal reweighting

This subsection records how conditioning on $0 \in S_i$ changes residue probabilities and why the induction on n in Section 3.8 is a **pinning + CRT + empty-run** chain, not a bare induction on $|\Omega|$.

3.1.1 Marginals before and after pinning one modulus

Draw S uniformly among m-subsets of $\frac{\mathbb{Z}}{p}\mathbb{Z}$ (unpinned).

1. $P(0 \in S) = \frac{m}{p}$.
2. For each $r \in \{0, 1, ..., p-1\}$: $P(r \in S) = \frac{m}{p}$ (symmetry).

Now draw S uniformly among m-subsets with $0 \in S$ (pinned; $|\Omega^{\text{pin}}| = \binom{p-1}{m-1}$).

1. $P(0 \in S) = 1$.
2. For each $r \in \{1, ..., p-1\}$: $P(r \in S) = \frac{m-1}{p-1}$.

If $1 \leq m < p$, then $\frac{m-1}{p-1} < \frac{m}{p}$: pinning **increases** the mass at 0 and **decreases** each nonzero marginal. The positive residues $\{1, ..., p-1\}$ are then filled by a uniform $(m-1)$-subset, which is exactly the hypergeometric sampling law in Equation 83 and Section 3.4.2.

Proof. Unpinned: choose m elements uniformly; each residue has probability $\frac{m}{p}$. Pinned: choose a uniform $(m-1)$-subset of $\{1, ..., p-1\}$ and adjoin 0. Each $r \geq 1$ lies in S iff it is chosen, probability $\frac{m-1}{p-1}$. □

3.1.2 Pinning all non-active coordinates on a CRT grid

Fix $n \geq 2$, direction i, and $P_i = \prod_{j \neq i} p_j$. If $0 \in S_j$ for every $j \neq i$, then for every scan index $t \geq 1$,

$$tP_i \in D(S_1, ..., S_n) \quad \Leftrightarrow \quad (tP_i) \bmod p_i \in S_i \tag{36}$$

.

Proof. For $j \neq i$, $P_i \equiv 0 \bmod p_j$, so $(tP_i) \bmod p_j = 0 \in S_j$ because $0 \in S_j$. Thus $tP_i \in D$ iff the p_i-coordinate lies in S_i. Section 3.9.3 is the n-coordinate form of the same fact. □

So pinning every **inactive** modulus replaces an intersection event over all n coordinates by a **single-coordinate** hypergeometric problem on (p_i, m_i) along the permuted scan $t = 1, 2, ...$

3.1.3 Pinning one head coordinate shrinks grid empty runs ($n = 2$)

Take consecutive primes $p_1 = 2$, $p_2 = 3$, multiplicities m_1, m_2 with $m_2 \geq 2$, and the p_1-grid $U_2^I = \min\{t \geq 1 : tp_1 \in D\}$. Let $U_2^{I,\text{pin}}$ be the same statistic under $0 \in S_1$ (and S_2 still uniform with $|S_2| = m_2$), and let $U_2^{I,\text{un}}$ be the statistic under independent uniform S_1, S_2 (unpinned).

For every $1 \leq t \leq p_2 - m_2$,

$$P\left(U_2^{I,\text{un}} > t\right) \geq P\left(U_2^{I,\text{pin}} > t\right) \tag{37}$$

.

Proof. For fixed S_2, write $A_s := \{sp_1 \in D\}$. If $0 \in S_1$, then $(sp_1) \bmod p_1 = 0 \in S_1$ for every s, so $A_s = \{(sp_1) \bmod p_2 \in S_2\}$. If $0 \notin S_1$, then A_s also requires $(sp_1) \bmod p_1 \in S_1$, so for each fixed S_2 and each s,

$$P(A_s^c \mid S_2, 0 \notin S_1) \geq P(A_s^c \mid S_2, 0 \in S_1) = P((sp_1) \bmod p_2 \notin S_2) \tag{38}$$

. The miss events for $s = 1, ..., t$ are nested in the same configuration, hence

$$P\left(U_2^{I,\text{pin}} > t \mid S_2\right) = P(\cap_{s=1}^{t} A_s^c \mid 0 \in S_1, S_2) \le P(\cap_{s=1}^{t} A_s^c \mid 0 \notin S_1, S_2) \tag{39}$$

. Averaging over S_2 and decomposing S_1 into the branches $\{0 \in S_1\}$ and $\{0 \notin S_1\}$ (law of total probability) gives $P\left(U_2^{I,\text{un}} > t\right) \ge P\left(U_2^{I,\text{pin}} > t\right)$. □

3.1.4 From pinned grid runs to the coordinate geometric rate

Under Ω^{pin} with $m_i \ge 2$, Section 3.9.3 identifies U_i^I with the pinned $n = 1$ waiting time on (p_i, m_i). Comparing against $U_i^{II} \sim \text{Geometric}(q_i)$ at the **same** rate $q_i = \frac{m_i}{p_i}$, Equation 84 gives

$$P\left(U_i^I > t\right) > P\left(U_i^{II} > t\right) \tag{40}$$

and hence $E[W_i^I] > E[W_i^{II}]$ (Equation 108).

Two-step comparison (conceptual). For $n = 2$ on the p_1-grid:

$$P\left(U_2^{I,\text{un}} > t\right) \ge P\left(U_2^{I,\text{pin}} > t\right) > (1 - q_2)^t = P\left(U_2^{II} > t\right). \tag{41}$$

The first inequality is Section 3.1.3; the strict middle inequality is Equation 84 after Section 3.7.1.

3.1.5 Remark (product rate gives the opposite order)

If one instead benchmarks against $U^{II} \sim \text{Geometric}(q)$ with $q = \prod_{j=1}^{n} \frac{m_j}{p_j}$, then $q < q_i$ and $(1 - q)^t > (1 - q_i)^t$, so the **product-rate** geometric model is much slower and the capped moment inequality **reverses** ($E[W_i^I] < E[W^{II}]$). That comparison is a different question; Equation 108 uses the coordinate rate q_i so the ordering matches Equation 84 for every n.

3.1.6 Induction on n revisited

Section 3.8 is proved by induction on n, but the analytic step is **not** a comparison of unpinned laws on Ω. Each inductive step (Section 3.9.2) does the following:

1. **Pinning hypothesis:** $0 \in S_j$ for every j (so Section 3.1.2 applies).
2. **CRT reduction:** U_i^I is an $n = 1$ pinned waiting time on (p_i, m_i).
3. **Same-rate comparison:** Equation 84 at rate q_i gives $E[W_i^I] > E[W_i^{II}]$.

So for **every** integer $1 \le m_i \le p_i$ with $m_i \ge 2$, Equation 108 holds in Ω^{pin} without extra hypotheses. Pinning explains **why** the reduction is valid; the strict inequality is the same hypergeometric–Bernoulli gap as in Section 3.5.

What pinning alone does not prove.

1. On the full unpinned space Ω, the grid inequality need not hold without pinning; Section 3.1.3 only compares unpinned vs.
   pinned **before** the coordinate-rate geometric comparison.

3.1.7 Coprime progression carries residue information

Let $p_a < p_b$ be distinct primes from the consecutive chain (Section 3.0.1) and fix $0 \in S_a$ with $|S_a| = m_a$. For integers $t = kp_b$ $(k = 1, 2, ...)$,

$$t \in D(S_a, S_b) \quad \Leftrightarrow \quad 0 \in S_b \quad \text{and} \quad (kp_b) \bmod p_a \in S_a \tag{42}$$

. If $0 \in S_b$, then on the progression $\{p_b, 2p_b, 3p_b, ...\}$ membership in D is determined by the sequence $k \mapsto (kp_b) \bmod p_a$, which is periodic in k with period p_a.

Proof. $t = kp_b$ gives $t \bmod p_b = 0 \in S_b$ when $0 \in S_b$. Since $\gcd(p_b, p_a) = 1$, $k \mapsto (kp_b) \bmod p_a$ is a bijection on $\frac{\mathbb{Z}}{p_a}\mathbb{Z}$, so as k runs through any complete residue system mod p_a, $(kp_b) \bmod p_a$ hits every class mod p_a exactly once. This is the mechanism behind use moment information on $1, 2, \ldots$ along the subsequence $p_j, 2p_j, \ldots$ when p_j is not yet pinned": the unpinned modulus is read on an arithmetic progression while pinned moduli cycle through all residues. □

3.1.8 One-coordinate relaxation dominates the full intersection

Fix $n \geq 1$, $j \in \{1, \ldots, n\}$, and independent random $\boldsymbol{S} = (S_1, \ldots, S_n)$ with $|S_i| = m_i$ and $0 \in S_i$ (pinned). For $t \geq 1$, write

$$E_t^I := \{1, 2, \ldots, t\} \cap D(\boldsymbol{S}) = \emptyset\} \tag{43}$$

for the consecutive scan (all coordinates), and

$$E_t^j := \left\{s \in \{1, \ldots, t\} : s \bmod p_j \notin S_j\right\} \tag{44}$$

. Then $E_t^I \subset E_t^j$ as events, hence

$$P_{\Omega^{\mathrm{pin}}}(U^I > t) = P(E_t^I) \geq P\left(E_t^j\right) \tag{45}$$

.

Proof. If every $s \in \{1, \ldots, t\}$ avoids D, then in particular each such s fails coordinate j, so $s \bmod p_j \notin S_j$ and $E_t^I \subset E_t^j$. Taking probabilities gives the inequality. □

3.1.9 Pinned one-coordinate tail dominates the geometric benchmark

Fix (p, m) with $1 \leq m < p$, $0 \in S$, $|S| = m$, and $q = \frac{m}{p}$. For $1 \leq t \leq p - m$,

$$P(\min\{s \geq 1 : s \bmod p \in S\} > t) > (1 - q)^t \tag{46}$$

, where the left side is the pinned consecutive scan on (p, m) and the right is $P(U^{II} > t)$ for $U^{II} \sim \text{Geometric}(q)$.

Proof. This is exactly Equation 89 / Equation 84: the left event is $\{1, \ldots, t\} \cap S = \emptyset\}$ on positive residues, i.e.
$T_1^I > t$. □

3.1.10 Consecutive scan: pinning lengthens empty runs

Convention. On the consecutive scan $U^I = \min\{t \geq 1 : t \in D(\boldsymbol{S})\}$, write $\Omega^{(k)}$ for independent uniform $S_1, \ldots, S_k$ with $|S_i| = m_i$ and **pinned** $0 \in S_{k+1}, \ldots, 0 \in S_n$ (if $k < n$); $\Omega^{(0)}$ is fully unpinned and $\Omega^{(n)}$ is fully pinned.

For fixed $t \geq 1$ and $k \in \{0, \ldots, n - 1\}$,

$$P_{\Omega^{(k)}}(U^I > t) \leq P_{\Omega^{(k+1)}}(U^I > t) \tag{47}$$

, where the probability is over the random $\boldsymbol{S}$ in $\Omega^{(k)}$ / $\Omega^{(k+1)}$.

Proof. Fix all coordinates except S_{k+1} and write $E_t := \{1, \ldots, t\} \cap D = \emptyset\}$. Let $A := \{0 \in S_{k+1}\}$ and compare the laws of S_{k+1}:
1. **Unpinned:** uniform m_{k+1}-subset of $\frac{\mathbb{Z}}{p_{k+1}}\mathbb{Z}$;
2. **Pinned:** uniform m_{k+1}-subset with $0 \in S_{k+1}$.

Decompose $P(E_t) = P(A)P(E_t \mid A) + P(A^c)P(E_t \mid A^c)$ under the unpinned law. Under the pinned law, $P(E_t) = P(E_t \mid A)$.

For fixed other coordinates, if $0 \in S_{k+1}$ then every $s \equiv 0 \bmod p_{k+1}$ passes coordinate $k + 1$, so it is **easier** for some $s \in \{1, \ldots, t\}$ to lie in D and therefore **harder** for all of $\{1, \ldots, t\}$ to miss D:

$$P(E_t \mid A) \leq P(E_t \mid A^c) \tag{48}$$

. Indeed, when $0 \notin S_{k+1}$, every $s \equiv 0 \bmod p_{k+1}$ automatically fails coordinate $k + 1$, which shrinks the set of s that can lie in D.

Thus

$$P_{\mathrm{un}}(E_t) = P(A)P(E_t \mid A) + P(A^c)P(E_t \mid A^c) \geq P(A)P(E_t \mid A) + P(A^c)P(E_t \mid A) \geq P(E_t \mid A) = P_{\mathrm{pin}}(E_t) \tag{49}$$

, because $P(E_t \mid A^c) \geq P(E_t \mid A)$ and $P(A) + P(A^c) = 1$. Integrating over the other independent coordinates gives the same inequality for the full product law $\Omega^{(k)}$ and $\Omega^{(k+1)}$. $\square$

3.1.11 Telescoping capped moments under incremental pinning

For any cutoff $x \geq 1$ and any law $\boldsymbol{S} \mapsto U^I$ on the consecutive scan,

$$E[W^I] = \sum_{t=1}^{x} P(U^I > t) \tag{50}$$

(Section 2.4.1). For the pinning chain $\Omega^{(0)}, ..., \Omega^{(n)}$,

$$E_{\Omega^{(n)}}[W^I] = E_{\Omega^{(0)}}[W^I] + \sum_{k=0}^{n-1} (E_{\Omega^{(k+1)}}[W^I] - E_{\Omega^{(k)}}[W^I]) \tag{51}$$

. Each summand equals $\sum_{t=1}^{x} (P_{\Omega^{(k+1)}}(U^I > t) - P_{\Omega^{(k)}}(U^I > t))$. By Section 3.1.10, every term in the t-sum is ≥ 0, hence

$$E_{\Omega^{(n)}}[W^I] \geq E_{\Omega^{(0)}}[W^I] \tag{52}$$

.

Using this to handle non-monotone t-terms. Equation 89 need not hold termwise for every $t \leq x$ when x is large (Section 3.8.3). The telescoping identity separates

$$E_{\Omega^{(n)}}[W^I] - E[W^{II}] = \sum_{t=1}^{x} (P_{\Omega^{(n)}}(U^I > t) - P(U^{II} > t)) \tag{53}$$

into (i) the fully pinned comparison at $t \leq p_i - m_i$ (Equation 84 after Section 3.9.3 on grids, or directly on the line at $n = 1$), plus (ii) finitely many increments from $\Omega^{(0)}$ to $\Omega^{(n)}$, plus (iii) possibly negative terms $(P_{\Omega^{(n)}}(U^I > t) - P(U^{II} > t))$ for $t > p_i - m_i$. The pin steps (ii) add a nonnegative buffer $\sum_{k,t} (P_{\Omega^{(k+1)}}(U^I > t) - P_{\Omega^{(k)}}(U^I > t))$; for the budget cutoff $x \leq P - M$, Section 3.1.22 uses Section 3.1.20 (Section 3.1.17) rather than this cancellation.

3.1.12 Pinned consecutive tail beats the geometric rate (all n)

Fix $n \geq 1$, consecutive primes $p_1 = 2 < ... < p_n$, $m_i \geq 2$, and $q_i = \frac{m_i}{p_i}$. On Ω^{pin} with $0 \in S_i$ for every i, let U^I be the consecutive-scan first **ONE**. For each $j \in \{1, ..., n\}$ and each $1 \leq t \leq p_j - m_j$,

$$P(U^I > t) > (1 - q_j)^t \tag{54}$$

.

Proof by induction on n.

Base $n = 1$. Section 3.1.9.

Inductive step $n \Rightarrow n + 1$. Write P_{n+1} for the law with $n + 1$ pinned coordinates and P_n for the law with the first n pinned (coordinate $n + 1$ still unpinned in P_n). Section 3.1.10 gives $P_{n+1}(U^I > t) \geq P_{n(U^I > t)}$ for every $t \geq 1$.

Fix $j \leq n$. By the induction hypothesis, for $1 \leq t \leq p_j - m_j$,

$$P_{n(U^I > t)} > (1 - q_j)^t \tag{55}$$

, hence the same holds for P_{n+1}.

For $j = n + 1$ and $1 \leq t \leq p_{n+1} - m_{n+1}$, Section 3.1.8 gives

$$P_{n+1}(U^I > t) \geq P(\min\{s \geq 1 : s \bmod p_{n+1} \in S_{n+1}\} > t) \tag{56}$$

with $0 \in S_{n+1}$ pinned. The right-hand side is the pinned $n = 1$ consecutive scan on (p_{n+1}, m_{n+1}), so Section 3.1.9 yields

$$P_{n+1}(U^I > t) > (1 - q_{n+1})^t \tag{57}$$

.

Thus the tail inequality holds for every coordinate at depth $n + 1$. □

3.1.13 Consecutive survival is non-increasing

For the consecutive-scan first **ONE** $U^I = \min\{t \geq 1 : t \in D\}$ and any $t \geq 1$,

$$\{U^I > t + 1\} \subset \{U^I > t\} \tag{58}$$

, hence $P(U^I > t + 1) \leq P(U^I > t)$.

Proof. $\{U^I > t + 1\}$ means $\{1, ..., t + 1\} \cap D = \emptyset\}$, which implies $\{1, ..., t\} \cap D = \emptyset\}$, i.e. $U^I > t$. □

3.1.14 Early consecutive tails (comparison range)

Fix $n \geq 1$, consecutive primes $p_1 = 2 < ... < p_n$, $m_i \geq 2$, and $j \in \{1, ..., n\}$. On Ω^{pin}, for each $1 \leq t \leq p_j - m_j$,

$$P(U^I > t) > (1 - q_j)^t \tag{59}$$

. This is Section 3.1.12; recorded here for the split at $T = p_{j^*} - m_{j^*}$. □

3.1.15 Remark (late-t termwise comparison can fail)

Let j^* maximize $p_{j^*} - m_{j^*}$, $T := p_{j^*} - m_{j^*}$, and $q_{j^*} := \frac{m_{j^*}}{p_{j^*}}$. Section 3.1.12 gives $P(U^I > t) > (1 - q_{j^*})^t$ for $1 \leq t \leq T$. For $t > T$, Section 3.1.13 gives $P(U^I > t) \leq P(U^I > T)$, not $\geq$: longer prefixes are **harder** to avoid entirely. So one cannot extend the termwise inequality to every $t \leq P - M$ by monotonicity alone. Example on consecutive primes $p_1 = 2$, $p_2 = 3$, $p_3 = 5$ with $(m_1, m_2, m_3) = (1, 1, 3)$: $T = 2$, $P - M = 27$, and $P(U^I > 20) = 0 < (1 - q_{j^*})^{20}$ while $E[W^I] > E[W^{II}_{j^*}]$ still holds at $x = P - M$.

3.1.16 Pinned one-modulus survival sums to $\frac{p-m}{m}$

Fix (p, m) with $2 \leq m < p$, $N := p - 1$, $K := m - 1$, and $T := p - m = N - K$. On Ω^{pin} for this modulus, with $U^I = \min\{s \geq 1 : s \bmod p \in S\}$,

$$P(U^I > t) = \frac{\binom{N-t}{K}}{\binom{N}{K}} \tag{60}$$

for $1 \leq t \leq T$ (Equation 83), and $P(U^I > t) = 0$ for $t > T$. Then

$$\sum_{t=1}^{T} P(U^I > t) = \frac{p - m}{m}. \tag{61}$$

Proof. For $1 \leq t \leq T$ one has $N - t \geq K$, so every term is nonzero. With $j := N - t$,

$$\sum_{t=1}^{T} \binom{N - t}{K} = \sum_{j=K}^{N-1} \binom{j}{K} = \binom{N}{K + 1} \tag{62}$$

by the hockey-stick identity $\sum_{j=0}^{M} \binom{j}{K} = \binom{M+1}{K+1}$ (here $M = N - 1$ and $\binom{j}{K} = 0$ for $j < K$). Dividing by $\binom{N}{K}$ gives

$$\frac{\binom{N}{K+1}}{\binom{N}{K}} = \frac{N - K}{K + 1} = \frac{p - m}{m}. \tag{63}$$

□

3.1.17 Exact pinned head sum at $T = p - m$

Fix (p, m) with $2 \le m < p$, $q := \frac{m}{p}$, and $T := p - m$. On Ω^{pin} for this modulus,

$$\sum_{t=1}^{T} \left(P(U^I > t) - (1-q)^t\right) = \frac{(1-q)^{T+1}}{q}. \tag{64}$$

Proof. Equation 61 gives $\sum_{t=1}^{T} P(U^I > t) = \frac{p-m}{m}$. Section 2.4.2 gives $\sum_{t=1}^{T} (1-q)^t = (1-q)\frac{1-(1-q)^T}{q}$. Since $1 - q = \frac{p-m}{p}$,

$$\frac{p-m}{m} - (1-q)\frac{1-(1-q)^T}{q} = \frac{p-m}{m}\left(1-(1-q)^T\right) = (1-q)^T\frac{p-m}{m} = \frac{(1-q)^{T+1}}{q}. \tag{65}$$

□

3.1.18 Intersection head dominates the best coordinate head

Fix $n \ge 1$, consecutive primes $p_1 = 2 < ... < p_n$, $m_i \ge 2$, and $j^* \in \{1, ..., n\}$. Let U^I be the consecutive-scan first **ONE** on the full CRT intersection and $U^I_{j^*}$ the pinned one-modulus statistic on $\left(p_{j^*}, m_{j^*}\right)$. On Ω^{pin}, for every $t \ge 1$,

$$P(U^I > t) \ge P\left(U^I_{j^*} > t\right) \tag{66}$$

, hence

$$\sum_{t=1}^{T} \left(P(U^I > t) - \left(1 - q_{j^*}\right)^t\right) \ge \sum_{t=1}^{T} \left(P\left(U^I_{j^*} > t\right) - \left(1 - q_{j^*}\right)^t\right) \tag{67}$$

for $T := p_{j^*} - m_{j^*}$ and $q_{j^*} := \frac{m_{j^*}}{p_{j^*}}$.

Proof. Section 3.1.8 gives $P(U^I > t) \ge P\left(U^I_{j^*} > t\right)$ for each t. Subtract the same geometric terms and sum over $t = 1, ..., T$. □

3.1.19 Budget cutoff exceeds the comparison range for $n \ge 2$

Fix $n \ge 2$, consecutive primes $p_1 = 2 < ... < p_n$, $m_i \ge 2$, $P = \prod p_i$, $M = \prod m_i$, and $T := \max_j\left(p_j - m_j\right)$. Then $P - M > T$.

Proof. Each factor $p_j - m_j \ge 1$, and for $n \ge 2$ at least two moduli contribute, so

$$P - M = \sum_{j=1}^{n} \left(\prod_{i \ne j} p_i\right)\left(p_j - m_j\right) \ge p_{j^*} - m_{j^*} + (n-1) = T + (n-1) > T. \tag{68}$$

□

3.1.20 Head sum dominates the late geometric tail

Fix $n \ge 2$, consecutive primes $p_1 = 2 < ... < p_n$, $m_i \ge 2$, $P = \prod p_i$, $M = \prod m_i$, and j^* with

$$T := p_{j^*} - m_{j^*} = \max_j\left(p_j - m_j\right) \tag{69}$$

, $q_{j^*} := \frac{m_{j^*}}{p_{j^*}}$. For every x with $T < x \le P - M$ on Ω^{pin},

$$\sum_{t=1}^{T} \left(P(U^I > t) - \left(1 - q_{j^*}\right)^t\right) > \sum_{t=T+1}^{x} \left(1 - q_{j^*}\right)^t. \tag{70}$$

Proof. Write $H_{j^*} := \sum_{t=1}^{T} \left(P\left(U^I_{j^*} > t\right) - \left(1 - q_{j^*}\right)^t\right)$. Section 3.1.17 at $\left(p_{j^*}, m_{j^*}\right)$ gives $H_{j^*} = \frac{\left(1-q_{j^*}\right)^{T+1}}{q_{j^*}}$. Section 3.1.18 gives

$$\sum_{t=1}^{T} \left(P(U^I > t) - \left(1 - q_{j^*}\right)^t\right) \ge H_{j^*}. \tag{71}$$

For the geometric tail, Section 2.4.2 yields

$$\sum_{t=T+1}^{x}\left(1-q_{j^*}\right)^t=\frac{\left(1-q_{j^*}\right)^{T+1}}{q_{j^*}}\cdot\left(1-\left(1-q_{j^*}\right)^{x-T}\right)<\frac{\left(1-q_{j^*}\right)^{T+1}}{q_{j^*}}=H_{j^*} \tag{72}$$

because $0<1-q_{j^*}<1$ and $x>T$. Combining the two displays gives Equation 70. □

3.1.21 Extended consecutive capped moment

Fix $n\ge 2$, consecutive primes $p_1=2<...<p_n$, $m_i\ge 2$, $P=\prod p_i$, $M=\prod m_i$, and a cutoff $x\ge 1$. On Ω^{pin}, let U^I be the consecutive-scan first **ONE** and $W^I=\min(U^I-1,x)$. For each $j\in\{1,...,n\}$, let $U_j^{II}\sim\text{Geometric}(q_j)$ with $q_j=\frac{m_j}{p_j}$ and $W_j^{II}=\min(U_j^{II}-1,x)$.

1. **Comparison cutoff.** If $1\le x\le p_j-m_j$, then
$$E[W^I]>E[W_j^{II}] \tag{73}$$
by Section 2.4.1 and Section 3.1.12 (termwise $P(U^I>t)>P(U_j^{II}>t)$).

2. **Budget cutoff.** If $1\le x\le P-M$, then $L_0=U^I-1\le P-M$ (Section 2.7.4). Choose j^* with $p_{j^*}-m_{j^*}=\max_j(p_j-m_j)$. Section 3.1.22 gives $E[W^I]>E[W_{j^*}^{II}]$ (Section 3.1.20; Section 3.1.15).

3. **Infinite cutoff.** For every $x\ge P-M$, $E[W^I]=\frac{P-M}{M}$ (Section 2.7.5, Section 2.7.6) and $E[W^I]>E[W_{j^*}^{II}]$ (Section 3.1.22).

Coprime progression input. When coordinate j is not in the pinned block $\{1,...,k\}$, Section 3.1.7 explains how information on the line scan $1,2,...$ is transported along $p_j,2p_j,...$ while the other moduli (already pinned) cycle through all residue classes.

Contrast with the grid. On a CRT **grid**, unpinned empty runs are **longer** than pinned (Section 3.1.3), so the telescoping inequality reverses ($E[W^{\text{un}}]\ge E[W^{\text{pin}}]$ there). The consecutive line uses the opposite monotonicity (Section 3.1.10).

3.1.22 Corollary (budget cutoff with best comparison coordinate)

Fix $n\ge 2$, consecutive primes $p_1=2<...<p_n$, $m_i\ge 2$, $P=\prod p_i$, $M=\prod m_i$, and choose j^* with

$$p_{j^*}-m_{j^*}=\max_j(p_j-m_j) \tag{74}$$

. Let U^I be the consecutive-scan first **ONE** on Ω^{pin}, $W^I=\min(U^I-1,x)$, and $W_{j^*}^{II}=\min(U_{j^*}^{II}-1,x)$ for $U_{j^*}^{II}\sim\text{Geometric}(q_{j^*})$ with $q_{j^*}=\frac{m_{j^*}}{p_{j^*}}$.

For every $x\ge 1$,

$$E[W^I]>E[W_{j^*}^{II}] \tag{75}$$

.

Proof. Write $T:=p_{j^*}-m_{j^*}$ and $B:=P-M$. By Section 2.4.1 and Section 2.4.2,

$$E[W^I]-E[W_{j^*}^{II}]=\sum_{t=1}^{x}\left(P(U^I>t)-\left(1-q_{j^*}\right)^t\right)=\sum_{t=1}^{\min(x,B)}P(U^I>t)-\sum_{t=1}^{x}\left(1-q_{j^*}\right)^t \tag{76}$$

using Section 2.7.5 for $t>B$. If $x\le T$, the first block alone yields the strict inequality (Section 3.1.14). If $T<x\le B$, Section 3.1.20 gives

$$\sum_{t=1}^{T}\left(P(U^I>t)-\left(1-q_{j^*}\right)^t\right)>\sum_{t=T+1}^{x}\left(1-q_{j^*}\right)^t \tag{77}$$

, and adding $\sum_{t=T+1}^{x}P(U^I>t)\ge 0$ gives $E[W^I]-E[W_{j^*}^{II}]>0$. If $x>B$, Section 2.7.5 gives $E[W^I]=\frac{B}{M}$ (Section 2.7.6), while $E[W_{j^*}^{II}]$ is nondecreasing in x and strictly exceeds its value at $x = B$. The case $x = B$ is the budget cutoff already treated; for $x > B$, $E[W^I] = E[W^I]$ at $x = B > E[W_{j^*}^{II}]$ at $x = B \le E[W_{j^*}^{II}]$ at x. □

3.1.23 Corollary (infinite cutoff)
Under the hypotheses of Section 3.1.22, for every $x \ge 1$,

$$E[W^I] > E[W_{j^*}^{II}]. \tag{78}$$

Once $x \ge P - M$, $E[W^I] = \frac{P-M}{M}$ and is constant in x; the coordinate-rate geometric side satisfies

$$\lim_{x\to\infty} E[W_{j^*}^{II}] = \frac{p_{j^*} - m_{j^*}}{m_{j^*}} \tag{79}$$

.

Proof. Section 3.1.22. The plateau and Section 2.7.6 give the limit of $E[W^I]$; Section 2.4.2 gives the geometric limit. □

3.1.24 Remark (product rate at infinite cutoff)
Let $q_{\text{prod}} := \frac{M}{P} = \prod_{i=1}^{n} \frac{m_i}{p_i}$. The **uncapped** product-rate geometric mean is

$$\lim_{x\to\infty} E[\min(U^{II} - 1, x)] = \frac{1 - q_{\text{prod}}}{q_{\text{prod}}} = \frac{P - M}{M} \tag{80}$$

(Section 2.4.2). On the consecutive scan, Section 2.7.6 gives the **same** value for the intersection model:

$$\lim_{x\to\infty} E[W^I] = \frac{P - M}{M} \tag{81}$$

. Thus the infinite-cutoff first distribution matches the product-rate geometric benchmark **exactly**, while it strictly dominates the faster coordinate-rate model at j^* (Section 3.1.23). This is not a contradiction with Section 3.1.5: there the grid comparison at rate q_{prod} reverses because U_i^I has finite support $p_i - m_i$ per coordinate, whereas the consecutive scan exploits the full period budget $B = P - M$ before survival vanishes.

3.2 Case $n = 1$: capped distance from 0

We specialize to (p, m) in place of (p_1, m_1) with $1 \le m \le p$ and cutoff $x \ge 1$. Work in Ω^{pin}: $0 \in S$ and the first positive **ONE** is $U^I = \min\{t \ge 1 : t \in S\}$ along $1, 2, \ldots$, with distance $D^I = U^I - 1$ and $W^I = \min(D^I, x)$.

3.2.1 First model: run of zeros before the first positive ONE
Equivalently, $\{U^I > x\} = \{1, 2, \ldots, x\} \cap S = \emptyset\}$ among positive residues. With $T = S \setminus \{0\}$ uniform among $(m-1)$-subsets of $\{1, \ldots, p-1\}$,

$$P(U^I > x) = \frac{\binom{p-1-x}{m-1}}{\binom{p-1}{m-1}} \tag{82}$$

whenever $m - 1 \le p - 1 - x$ (otherwise $P(U^I > x) = 0$).

When $m - 1 \le p - 1 - x$,

$$P(U^I > x) = \prod_{j=0}^{m-2} \frac{p - 1 - x - j}{p - 1 - j} \tag{83}$$

3.2.2 Second model
Let $q := \frac{m}{p}$. Let $U^{II} \sim \text{Geometric}(q)$, $D^{II} = U^{II} - 1$, and $W^{II} := \min(D^{II}, x)$.

3.2.3 Target ($n = 1$, same rate $q = \frac{m}{p}$)

For $1 \le x \le p - m$,

$$E[W^{II}] < E[W^{I}] \tag{84}$$

i.e. $\sum_{t=1}^{x} P(U^{II} > t) < \sum_{t=1}^{x} P(U^{I} > t)$ by Section 2.4.1. Equivalently, $P(U^{II} > t) < P(U^{I} > t)$ for each $1 \le t \le p - m$ (Equation 89 with $k = 1$).

3.2.4 Proof of Equation 84

For $k = 1$, Section 3.5 gives Equation 89, and the corollary there yields $E[W^{II}] < E[W^{I}]$ for $1 \le x \le p - m$ via Section 2.4.1. □

This compares both models at the **same** rate $q = \frac{m}{p}$. For $n \ge 2$ on the consecutive-prime chain, Section 3.9.3 reduces each grid direction to the $n = 1$ problem on (p_i, m_i), and Equation 108 compares against $U_i^{II} \sim \text{Geometric}(q_i)$ at the coordinate rate q_i.

—

3.3 Case $n = 2$: CRT-aligned scan indices

Take consecutive primes $p_1 = 2$, $p_2 = 3$ and multiplicities (m_1, m_2) on that pair (Section 3.0.1). Work on the CRT line $\{0, 1, ..., p_1 p_2 - 1\}$ with indicator f from the first distribution (Equation 83 style, now joint over (S_1, S_2)).

The $n = 1$ scan used consecutive integers $1, 2, ..., x$. For $n = 2$, **which** indices matter depends on which modulus is in focus. In general (Section 3.8), modulus p_i uses the grid $P_i, 2P_i, ...$ with $P_i = \prod_{j \ne i} p_j$; for $n = 2$ this gives $p_1, 2p_1, ...$ and $p_2, 2p_2, ...$

Remark (CRT grid vs. consecutive scan). The full indicator f is defined on every integer: $1, 2, 3, 4, ...$ all carry **ONE** or **ZERO** labels, and Section 4 uses this consecutive scan for the early-**ONE** theorems (Section 4.2.42).

The $n = 2$ moment comparison (Section 3.6, Equation 102) uses only the CRT-aligned subsequence

$$p_1, 2p_1, 3p_1, ... \quad \text{or, symmetrically,} \quad p_2, 2p_2, 3p_2, ... \tag{85}$$

attached to one modulus. Integers strictly between grid points—such as $2, 3, 4$ when $p_1 = 3$—are not omitted from D; they are simply excluded from the waiting-time statistic of Section 3.

Grid reduction. With pinning ($0 \in S_1$), every multiple tp_1 is already compatible mod p_1; only the coordinate mod p_2 varies. Because $\gcd(p_1, p_2) = 1$, the map $t \longrightarrow tp_1 \bmod p_2$ is a bijection on $\frac{\mathbb{Z}}{p_2}\mathbb{Z}$ (Section 3.7.1), so scanning $t = 1, 2, ...$ on the p_1-grid is equivalent to scanning all residue classes mod p_2 in a permuted order—the same $n = 1$ waiting-time problem on (p_2, m_2).

Example ($p_1 = 2$, $p_2 = 3$, $P = 6$). The p_2-aligned grid is $2, 4, 6, 8, ...$ The consecutive scan also visits $1, 3, 5, 7, ...$; any of these can be **ONES** if they lie in D. Section 3 compares the first grid hit $\min\{t \ge 1 : t \cdot 2 \in D\}$ with the first consecutive hit $\min\{x \ge 1 : x \in D\}$ on the same set D.

—

3.4 Single modulus: capped k-th ONE vs. capped k-th success

Scope. One pair (p, m) only; no CRT intersection yet. Use Section 2.4 on the scan $t = 1, 2, \ldots$ (the **ONE** at 0 is automatic). Assume $1 \le k \le m - 1$ and cutoff $x \ge 1$.

3.4.1 First model (random subset, indicator ONES)

Draw $T = S \setminus \{0\}$ uniformly among $(m-1)$-subsets of $\{1, \ldots, p-1\}$. Set $f(t) = 1$ if $t \in T$ and 0 for $t \ge 1$ with $t \notin T$; always $f(0) = 1$. Let T_k^I be the 1-indexed position of the k-th positive **ONE** along $\{1, 2, \ldots\}$:

$$T_k^I = \min\{t \ge 1 : |\{1, \ldots, t\} \cap S| = k\} \tag{86}$$

. For $k = 1$ this is $T_1^I = \min\{t \ge 1 : t \in S\}$. The distance to the k-th positive **ONE** is $D_k^I := T_k^I - 1$, and $W_k^I := \min\left(D_k^I, x\right)$.

Let $H_t = |T \cap \{1, \ldots, t\}|$. Then H_t follows the hypergeometric law (Section 3.4.2):

$$P(H_t = j) = \binom{t}{j} \frac{\binom{p-1-t}{m-1-j}}{\binom{p-1}{m-1}} \tag{87}$$

for $0 \le j \le \min(t, m-1)$, and $P\left(T_k^I > t\right) = P(H_t \le k-1) = \sum_{j=0}^{k-1} P(H_t = j)$.

3.4.2 Hypergeometric count for pinned sampling without replacement

Draw T uniformly among $(m-1)$-subsets of $\{1, \ldots, p-1\}$ (the pinned model with $0 \in S$). For $0 \le t \le p-1$ and $0 \le j \le \min(t, m-1)$,

$$P(|T \cap \{1, \ldots, t\}| = j) = \binom{t}{j} \frac{\binom{p-1-t}{m-1-j}}{\binom{p-1}{m-1}} \tag{88}$$

.

Proof. Choose a subset of size $m-1$ from $p-1$ elements by first choosing j elements inside $\{1, \ldots, t\}$ and $m-1-j$ outside. There are $\binom{t}{j}\binom{p-1-t}{m-1-j}$ favorable choices out of $\binom{p-1}{m-1}$ total. □

3.4.3 Second model (independent trials)

Let $q := \frac{m}{p}$. Run independent Bernoulli(q) trials (Section 2.5.1) and let T_k^{II} be the 1-indexed trial of the k-th success. Define $D_k^{II} := T_k^{II} - 1$ and $W_k^{II} := \min\left(D_k^{II}, x\right)$. Let G_t be the number of successes in the first t trials; $G_t \sim \text{Binomial}(t, q)$. Then $P\left(T_k^{II} > t\right) = P(G_t \le k - 1) = \sum_{j=0}^{k-1} \binom{t}{j} q^j (1-q)^{t-j}$.

3.4.4 Target inequality

For $1 \le t \le p - m$ and $1 \le k \le m - 1$,

$$P\left(T_k^{II} > t\right) < P\left(T_k^I > t\right) \tag{89}$$

For any cutoff $x \ge 1$, Section 2.4.1 gives

$$E\left[W_k^{II}\right] = \sum_{t=1}^{x} P\left(T_k^{II} > t\right) \tag{90}$$

and $E\left[W_k^I\right] = \sum_{t=1}^{x} P\left(T_k^I > t\right)$. For $1 \le x \le p - m$ (**comparison cutoff**),

$$E\left[W_k^{II}\right] < E\left[W_k^I\right] \tag{91}$$

, using Equation 89 termwise.

At $k = 1$ (first positive **ONE**), Equation 89 gives Equation 84.

—

3.5 Proof of the capped k-th comparison (single modulus)

Write $N = p-1$, $K = m-1$, and $q = \frac{m}{p}$. For $0 \le \ell \le K$ and $t \ge 0$, define

$$F(t,\ell) := \sum_{j=0}^{\ell} \binom{t}{j} \frac{\binom{N-t}{K-j}}{\binom{N}{K}}, \quad G(t,\ell) := \sum_{j=0}^{\ell} \binom{t}{j} q^j (1-q)^{t-j} \tag{92}$$

(so $P\big(T_k^I > t\big) = F(t, k-1)$ and $P\big(T_k^{II} > t\big) = G(t, k-1)$). We prove $F(t,\ell) > G(t,\ell)$ whenever $1 \le t \le N-K$ ($t \le p-m$) and $0 \le \ell \le K-1$.

3.5.1 Lemma (empty run)

If $1 \le s \le N-K$, then

$$\frac{\binom{N-s}{K}}{\binom{N}{K}} > (1-q)^s = \left(\frac{p-m}{p}\right)^s \tag{93}$$

.

Proof. Induct on s. For $s = 1$, both sides equal $\frac{N-K}{N} = \frac{p-m}{p-1} > \frac{p-m}{p}$. If $1 < s \le N-K$, then

$$\frac{\binom{N-s}{K}}{\binom{N}{K}} = \frac{\binom{N-s+1}{K}}{\binom{N}{K}} \cdot \frac{N-s-K+1}{N-s+1} \tag{94}$$

. By induction the first factor exceeds $\left(\frac{N-K}{N}\right)^{s-1}$, and for $s \ge 2$ one has

$$\frac{N-s-K+1}{N-s+1} > \frac{N-K}{N} \tag{95}$$

because $N(N-s-K+1) - (N-K)(N-s+1) = K(s-1) > 0$. Multiplying gives the claim. □

3.5.2 Lemma (hypergeometric is stochastically smaller)

For $1 \le t \le N-K$ and $0 \le \ell \le K$,

$$F(t,\ell) = P(H_t \le \ell) > P(G_t \le \ell) = G(t,\ell) \tag{96}$$

.

Proof. Because $\frac{K}{N} = \frac{m-1}{p-1} < \frac{m}{p} = q$, each hypergeometric draw has a smaller per-step hit rate than Bernoulli sampling once hits are present. Formally, induct on t.

$t = 1$: $F(1,0) = \frac{N-K}{N} > 1-q = G(1,0)$ and $F(1,\ell) = 1 = G(1,\ell)$ for $\ell \ge 1$.

Assume $F(t-1,\ell) \ge G(t-1,\ell)$ for all ℓ. When passing from $t-1$ to t, the hypergeometric update adds a hit with conditional probability $\frac{K-j}{N-t+1}$ if $H_{t-1} = j$, whereas the binomial update uses q. For $0 \le j \le \min(t-1, K)$ and $t \le N-K$,

$$\frac{K-j}{N-t+1} \le \frac{K}{N-t+1} \le \frac{K}{N} < q \tag{97}$$

. So each transition pushes less mass upward than the Bernoulli transition.

Writing the standard recursions

$$P(H_t = j) = P(H_{t-1} = j)\frac{N-K-j}{N-t+1} + P(H_{t-1} = j-1)\frac{K-j+1}{N-t+1}, \tag{98}$$

$$P(G_t = j) = P(G_{t-1} = j)(1-q) + P(G_{t-1} = j-1)q, \tag{99}$$

and summing $j = 0, \ldots, \ell$, the induction hypothesis gives $F(t,\ell) > G(t,\ell)$ for $\ell = 0$ by Section 3.5.1, and for $\ell \ge 1$ the same recursions increase $F(t,\ell) - G(t,\ell)$ strictly once $F(t-1,\ell-1) > G(t-1,\ell-1)$. In particular $F(t,k-1) > G(t,k-1)$ for $1 \le k \le K$. □

3.5.3 Theorem
For $1 \le t \le p - m$ and $1 \le k \le m - 1$, Equation 89 holds.

Proof. $P\left(T_k^I > t\right) = F(t, k-1)$ and $P\left(T_k^{II} > t\right) = G(t, k-1)$. Apply Section 3.5.2. □

3.5.4 Corollary (capped moments)
If $1 \le x \le p - m$, then $E\left[W_k^{II}\right] < E\left[W_k^I\right]$.

Proof. By Section 2.4.1,

$$E\left[W_k^{II}\right] = \sum_{t=1}^{x} P\left(T_k^{II} > t\right) \tag{100}$$

and $E\left[W_k^I\right] = \sum_{t=1}^{x} P\left(T_k^I > t\right)$. For $1 \le t \le x \le p - m$, Equation 89 gives $P\left(T_k^{II} > t\right) < P\left(T_k^I > t\right)$. Summing over $t = 1, ..., x$ yields the claim. □

> **Range of t and x.** The hypothesis $t \le p - m$ matches $m - 1 \le p - 1 - t$, i.e. the hypergeometric model still has room for a length-t run of zeros. For $t > p - m$, one has $P\left(T_k^I > t\right) = 0$ while $P\left(T_k^{II} > t\right) > 0$ may hold, so the tail inequality is stated only in the range where both models are comparable.

—

3.6 Case $n = 2$: first intersection ONE vs. coordinate-rate geometric

Take consecutive primes $p_1 = 2$, $p_2 = 3$ with $1 \le m_i \le p_i$ and $0 \in S_1, 0 \in S_2$ (pinned on both coordinates, as in Ω^{pin}; Section 3.0.1). Let $q_i := \frac{m_i}{p_i}$. Draw (S_1, S_2) with $0 \in S_i$ and $|S_i| = m_i$; let f be the intersection indicator on $\{0, ..., p_1 p_2 - 1\}$.

3.6.1 Scan model (the p_2-aligned grid)
Use Section 2.4 on the grid tp_1 for $t = 1, 2, ...$ Because $0 \in S_1$, every grid point tp_1 is automatically in D modulo p_1; the first positive-grid **ONE** is

$$U_2^I := \min\{t \ge 1 : tp_1 \in D(S_1, S_2)\} \tag{101}$$

, with distance $D_2^I = U_2^I - 1$ and $W_2^I := \min\left(D_2^I, x\right)$.

3.6.2 Random model (coordinate rate)
Let $U_2^{II} \sim \text{Geometric}(q_2)$, $D_2^{II} = U_2^{II} - 1$, and $W_2^{II} := \min\left(D_2^{II}, x\right)$ as in Section 2.4.

3.6.3 Target (first positive-grid ONE, same rate q_2)
Fix a cutoff $x \ge 1$ with $1 \le x \le p_2 - m_2$ (in particular whenever $m_2 \ge 2$),

$$E\left[W_2^{II}\right] < E\left[W_2^I\right] \tag{102}$$

equivalently $E\left[W_2^I\right] > E\left[W_2^{II}\right]$: the capped first intersection **ONE** on the p_2-grid is **slower on average** than the capped first success at rate $q_2 = \frac{m_2}{p_2}$.

Edge case $\boldsymbol{m_2 = 1}$. Then only $0 \in S_2$, the grid scan hits only when $tp_1 \equiv 0 \bmod p_2$, i.e. at $t = p_2, 2p_2, ...,$ so the first-**ONE** problem degenerates and is excluded.

—

3.7 Proof of Equation 102

3.7.1 Step 1: CRT reduction to $n = 1$ on (p_2, m_2)

By Section 3.1.2 (with $i = 2$ and $0 \in S_1$) and $\gcd(p_1, p_2) = 1$, for every $t \geq 1$

$$tp_1 \in D(S_1, S_2) \quad \Leftrightarrow \quad tp_1 \equiv s \bmod p_2 \quad \text{for some} \quad s \in S_2 \tag{103}$$

. The map $t \longrightarrow tp_1 \bmod p_2$ is a bijection on $\frac{\mathbb{Z}}{p_2}\mathbb{Z}$, so U_2^I has the same law as the $n = 1$ positive-distance waiting time for (p_2, m_2) (with $0 \in S_2$ fixed).

Proof. If $tp_1 \in D$, then $(tp_1) \bmod p_1 = 0 \in S_1$ automatically and $(tp_1) \bmod p_2 \in S_2$. Conversely, if $(tp_1) \bmod p_2 \in S_2$, then $tp_1 \in D$. The bijection $t \longrightarrow tp_1 \bmod p_2$ preserves the order of scan indices, so the first hitting index t matches the $n = 1$ problem on modulus p_2. □

3.7.2 Step 2: apply Equation 84

On (p_2, m_2) with $0 \in S_2$ and rate $q_2 = \frac{m_2}{p_2}$, Equation 84 gives

$$E[W_2^{II}] < E[W_2^I] \tag{104}$$

for $1 \leq x \leq p_2 - m_2$, which is Equation 102. □

The symmetric p_1-grid (tp_2, benchmark $U_1^{II} \sim \text{Geometric}(q_1)$) follows by swapping indices.

—

3.8 General $n \geq 2$: induction on the number of moduli

Fix $n \geq 2$, consecutive primes $p_1 = 2 < p_2 = 3 < ... < p_n$, and multiplicities m_i with $1 \leq m_i \leq p_i$ (Section 3.0.1). Write

$$q_i := \frac{m_i}{p_i}, \quad q := \prod_{i=1}^{n} q_i \tag{105}$$

. For each index $i \in \{1, ..., n\}$, define the **CRT step size**

$$P_i := \prod_{j=1, j \neq i}^{n} p_j \tag{106}$$

.

3.8.1 Grid scan attached to modulus p_i

Use Section 2.4 on the grid tP_i for $t = 1, 2, ...$. The first positive-grid **ONE** in direction i is

$$U_i^I := \min\{t \geq 1 : tP_i \in D(S_1, ..., S_n)\} \tag{107}$$

, with distance $D_i^I = U_i^I - 1$ and $W_i^I := \min(D_i^I, x)$. For each direction i, let $U_i^{II} \sim \text{Geometric}(q_i)$, $D_i^{II} = U_i^{II} - 1$, and $W_i^{II} := \min(D_i^{II}, x)$.

3.8.2 Target ($n \geq 2$, coordinate rate q_i)

Fix a cutoff $x \geq 1$. For each i with $m_i \geq 2$ and $1 \leq x \leq p_i - m_i$ (**comparison cutoff**; see Section 2.7.7),

$$E[W_i^{II}] < E[W_i^I] \tag{108}$$

equivalently $E[W_i^I] > E[W_i^{II}]$, the same ordering as Equation 84 on the reduced coordinate (p_i, m_i). If $x > p_i - m_i$, the grid-side first distribution contributes no further terms to $E[W_i^I]$ while $E[W_i^{II}]$ can still increase.

The $n = 1$ case is Equation 84 (Section 3.5); Section 3.9.3 extends it to every CRT grid direction once all coordinates are pinned.

3.8.3 Remark (comparison cutoff vs. budget cutoff)

Yes: x is the cutoff in $W = \min(D, x)$ (Section 2.4). The theorems below use two different upper limits on that parameter:

1. **Comparison cutoff** ($1 \le x \le p_i - m_i$): where $E[W_i^{II}] < E[W_i^{I}]$ is proved (Equation 108).
2. **Budget cutoff** ($1 \le x \le P - M$): where the consecutive-scan first gap satisfies $L_0 = U - 1 \le P - M$ (Section 2.7.4), so the cap is not below the true zero run in the period budget sense.
3. **Infinite cutoff** ($x \to \infty$): $E[W^I]$ is constant for $x \ge P - M$ and equals $\frac{P-M}{M}$ (Section 2.7.6); the coordinate-rate geometric benchmark keeps growing to $\frac{p_{j^*} - m_{j^*}}{m_{j^*}}$, while the product-rate geometric benchmark matches $\frac{P-M}{M}$ (Section 3.1.24).

Grid / reduced law (sharp comparison range). After Section 3.9.3, U_i^I is the pinned $n = 1$ statistic on (p_i, m_i). For $t > p_i - m_i$, $P(U_i^I > t) = 0$ but $P(U_i^{II} > t) = (1 - q_i)^t > 0$, so Equation 89 fails and $E[W_i^I]$ stops growing in x while $E[W_i^{II}]$ keeps growing. Induction on n does **not** widen this comparison range for grid U_i^I.

Consecutive scan (budget cutoff extends). On $t = 1, 2, \ldots$, more moduli increase $P - M$ (Section 2.7.4), so one may take cutoff $x \le P - M$ without contradicting the gap identity. For $x > p_i - m_i$, $P(U^I > t)$ can stay positive on the line scan while the **grid-reduced** law at coordinate i has $P(U_i^I > t) = 0$ for $t > p_i - m_i$. Comparing against $U_i^{II} \sim \text{Geometric}(q_i)$ at a fixed coordinate i need not be termwise monotone when $x > p_i - m_i$; use j^* with $p_{j^*} - m_{j^*} = \max_j (p_j - m_j)$ instead (Section 3.1.22, Section 3.1.15).

Status (consecutive scan, pinned).
1. Comparison cutoff $1 \le x \le p_i - m_i$: Section 3.1.21, Section 3.1.12 (termwise).
2. Budget cutoff $1 \le x \le P - M$: Section 3.1.22 (Section 3.1.20, Section 3.1.17, Equation 61).
3. Infinite cutoff $x \ge P - M$: $E[W^I] = \frac{P-M}{M}$ (Section 2.7.6, Section 3.1.23); matches product-rate geometric, dominates coordinate-rate geometric (Section 3.1.24).
4. Coprime progression (Section 3.1.7) transports line-scan information to $p_j, 2p_j, \ldots$ when p_j is the active modulus.
5. Grid: unpinned $\ge$ pinned (Section 3.1.3); consecutive: pinned $\ge$ unpinned (Section 3.1.10).

—

3.9 Proof of Section 3.8 by induction on n

We prove Equation 108 by induction on $n \ge 2$. The base case $n = 2$ is Equation 102. For the inductive step, adjoining the next consecutive prime p_{a+1} repeats the same pinning + CRT reduction and empty-run comparison (Section 3.1.6); the induction hypothesis is not used in the analytic part.

3.9.1 Base case $n = 2$

This is Equation 102 (proved above).

3.9.2 Inductive step: $n = a \Rightarrow n = a + 1$ ($a \ge 2$)

Fix $a \ge 2$ and adjoin the next consecutive prime (p_{a+1}, m_{a+1}) in the chain $2, 3, 5, \ldots$ with $0 \in S_{a+1}$. We prove Equation 108 for the enlarged family; it suffices to check one direction i (the others are identical).

Case $i = a + 1$. Here $P_{a+1} = \prod_{j=1}^{a} p_j$. By Section 3.9.3, U_{a+1}^{I} has the same law as the $n = 1$ positive-distance waiting time on (p_{a+1}, m_{a+1}). With $U_{a+1}^{II} \sim \text{Geometric}(q_{a+1})$, Equation 84 gives $E[W_{a+1}^{II}] < E[W_{a+1}^{I}]$ for $1 \le x \le p_{a+1} - m_{a+1}$.

Case $i \le a$. Here $P_i = p_{a+1} \prod_{j \ne i, j \le a} p_j$. Again Section 3.9.3 reduces U_i^I to the $n = 1$ waiting time on (p_i, m_i); Equation 84 at rate q_i gives $E[W_i^{II}] < E[W_i^I]$.

Thus Equation 108 holds for $n = a + 1$. □

3.9.3 CRT grid reduction (all n)

Fix $i \in \{1, ..., n\}$ and $P_i = \prod_{j \ne i} p_j$. Because $0 \in S_j$ for every j and $\gcd(P_i, p_i) = 1$ (pairwise coprimality of the p_j),

$$tP_i \in D(S_1, ..., S_n) \quad \Leftrightarrow \quad tP_i \equiv s \bmod p_i \;\text{ for some }\; s \in S_i \tag{109}$$

. The map $t \longrightarrow tP_i \bmod p_i$ is a bijection on $\frac{\mathbb{Z}}{p_i}\mathbb{Z}$, so U_i^I has the same law as the $n = 1$ waiting time $\min\{t \ge 1 : t \in S_i\}$ on (p_i, m_i).

Proof. For $j \ne i$, $P_i \equiv 0 \bmod p_j$, so $(tP_i) \bmod p_j = 0 \in S_j$. Thus $tP_i \in D$ <=> $(tP_i) \bmod p_i \in S_i$. Since $\gcd(P_i, p_i) = 1$, $t \longrightarrow tP_i \bmod p_i$ is bijective and order-preserving on scan indices. □

3.9.4 Corollary

If every $m_i \ge 2$, then Equation 108 holds for every $n \ge 2$ and every modulus direction i in Ω^{pin}. For $n = 1$, use Equation 84 (Section 3.5) instead. If some $m_i = 1$, direction i degenerates (grid hits only at multiples of p_i). □

—

4 Early ONE witnesses

Section 4 has two layers: a **general** CRT bound $U \le P$ (Section 4.1), and a **prime-sieve / Hardy–Littlewood** bound $U(\boldsymbol{S}(\boldsymbol{h})) \le p_n$ once $p_n \ge W(\boldsymbol{h})$ (Section 4.2.42). The universal claim $U \le p_n$ for **every** $\boldsymbol{S} \in \Omega$ is false (Section 4.2.44).

4.1 General case: pairwise coprime moduli

Fix $n \ge 1$ and moduli $p_1, ..., p_n \ge 2$ with $\gcd(p_i, p_j) = 1$ for $i \ne j$. Fix multiplicities $1 \le m_i \le p_i$ and the configuration space Ω from Section 2.3. Let $P = \prod_{i=1}^{n} p_i$, $D(\boldsymbol{S})$ the CRT intersection, and $U(\boldsymbol{S}) = \min\{t \ge 1 : t \in D(\boldsymbol{S})\}$ on the consecutive scan.

4.1.1 Witness before the period

If $m_i \ge 1$ for every i, then for every $\boldsymbol{S} \in \Omega$ there exists $x \in \{1, ..., P\}$ with $x \in D(\boldsymbol{S})$.

Proof. Fix $\boldsymbol{S}$. If some coordinate has $0 \notin S_j$, pick $s_j \in S_j \setminus \{0\}$ for that j and arbitrary $s_i \in S_i$ elsewhere. Section 2.6 gives $x \in \{0, ..., P - 1\}$ with $x \equiv s_i \bmod p_i$ for all i. Then $x \not\equiv 0 \bmod p_j$, so $x \ne 0$; hence $1 \le x \le P - 1$.

If $0 \in S_i$ for every i, choose $s_i = 0$ for all i; then $0 \in D(\boldsymbol{S})$. Also $P \equiv 0 \bmod p_i$ and $0 \in S_i$ for each i, so $P \in D(\boldsymbol{S})$. For $n = 1$ one has $P = p_1 \ge 2$; for $n \ge 2$, $P \ge 2p_n \ge 2$. Thus $P \in \{1, ..., P\}$ in both cases. □

4.1.2 Corollary (first ONE before P)

Under the hypotheses of Section 4.1, for every $\boldsymbol{S} \in \Omega$,

$$U(\boldsymbol{S}) \le P \tag{110}$$

.

Proof. Section 4.1.1 gives $x \in \{1, ..., P\} \cap D(\boldsymbol{S})$, so the first positive **ONE** is at most P. □

4.1.3 Empty runs longer than the period
For $t \geq 1$, let $N_{\text{bad}}(t)$ count $\boldsymbol{S} \in \Omega$ with $\{1, ..., t\} \cap D(\boldsymbol{S}) = \emptyset$. If $t \geq P$, then $N_{\text{bad}}(t) = 0$.

Proof. Section 4.1.2 shows every configuration has $U \leq P$, so each has a **ONE** in $\{1, ..., P\} \subset \{1, ..., t\}$ when $t \geq P$. □

4.1.4 Probability form on the general model
For $t \geq P$,

$$P(E_t) = 0 \tag{111}$$

and Section 4.2.30 holds with t in place of p_n^2.

Proof. $P(E_t) = N_{\text{bad}} \frac{t}{|}\Omega| = 0$ by Section 4.1.3. □

—

4.2 Prime sieve: consecutive primes and uniform tail gap

This subsection specializes to consecutive primes $p_1 = 2 < p_2 = 3 < ...$ and the uniform tail gap $p_i - m_i = K$ for $i \geq \ell$. The intersection model is unchanged; the improvement over Section 4.1.2 is proved only for the Hardy–Littlewood configuration $\boldsymbol{S}(\boldsymbol{h})$ (Section 4.2.42), not for all of Ω.

4.2.1 Parameters
Let $p_1 < p_2 < ... < p_n$ be the first n **consecutive primes** ($p_1 = 2$, $p_2 = 3$, $p_3 = 5$, ...). Fix a natural number $\ell \geq 1$ and admissible multiplicities

$$0 \leq m_i \leq p_i \tag{112}$$

for $i = 1, ..., \ell - 1$. Fix an integer $K \geq 0$ and impose a **uniform tail gap** for $i \geq \ell$:

$$p_i - m_i = K \tag{113}$$

(equivalently $m_i = p_i - K$). Then $0 \leq m_i \leq p_i$ holds automatically; we need $m_i \geq 1$ on the tail, i.e. $K \leq p_i - 1$ (true for every tail prime once K is fixed).

Draw $\boldsymbol{S} = (S_1, ..., S_n)$ independently with $S_i \in \Omega_i$ as in Section 2.3 (each S_i is any m_i-subset of $\frac{\mathbb{Z}}{p_i}\mathbb{Z}$; **no pinning**). Let $P = \prod_{i=1}^{n} p_i$, $D(\boldsymbol{S}) \subset \{0, ..., P-1\}$ be the CRT intersection, and $f_{\boldsymbol{S}}$ the indicator from Equation 9. On the **consecutive** scan $1, 2, 3, ...$, define the first positive **ONE**

$$U := \min\{t \geq 1 : f_{\boldsymbol{S}}(t) = 1\} \tag{114}$$

. (Equivalently, U is the 1-indexed first success time; the distance from 0 is $U - 1$ as in Section 2.4.)

The joint configuration space has

$$|\Omega| = \prod_{i=1}^{n} \binom{p_i}{m_i} \tag{115}$$

, and a fixed deterministic tuple such as the Hardy–Littlewood pattern $\boldsymbol{S}(\boldsymbol{h})$ from Section 5.2 lies in Ω as soon as $|S_i| = m_i$ for every i.

4.2.2 All-ZEROES before p_n^2
For $t \geq 1$, let

$$E_t := \{U > t\} = \{f_{\boldsymbol{S}}(x) = 0 \ \text{ for every } \ x \in \{1, 2, ..., t\}\} \tag{116}$$

be the event that the first t positive positions are all **ZEROES** (no **ONE** yet). Equivalently, $\{1, ..., t\} \cap D(\boldsymbol{S}) = \emptyset$.

The probability is a **rational** with denominator $|\Omega|$:

$$P(E_t) = N_{\text{bad}} \frac{t}{|} \Omega| \tag{117}$$

where $N_{\text{bad}}(t) \in \{0, 1, 2, ...\}$ counts bad configurations (Section 4.1.3).

4.2.3 Configuration count (not $\binom{P}{M}$)

The configuration space is

$$\Omega = \prod_{i=1}^{n} \Omega_i, \quad |\Omega| = \prod_{i=1}^{n} \binom{p_i}{m_i} \tag{118}$$

(Section 2.3). Each **full pattern** $\boldsymbol{S} = (s_1, ..., s_n)$ has probability

$$P(\boldsymbol{S}) = \frac{1}{|} \Omega| = \frac{1}{\prod_{i=1}^{n}} \binom{p_i}{m_i} \tag{119}$$

.

This is **not** the same as

$$\binom{P}{M} = \binom{\prod p_i}{\prod m_i} \tag{120}$$

, which counts M-subsets of $\frac{\mathbb{Z}}{P}\mathbb{Z}$ **without** the coordinate product structure. The integer hole below compares $P(E_t)$ to $\frac{1}{|}\Omega| = \frac{1}{\prod} \binom{p_i}{m_i}$; replacing $|\Omega|$ by $\binom{P}{M}$ would be the wrong denominator.

4.2.4 Remark (consecutive primes; not a generic coprime grid)

Throughout Section 4.2 and the lemmas below, $p_1 < p_2 < ... < p_n$ are the **first $\boldsymbol{n}$ consecutive primes** ($p_1 = 2,\ p_2 = 3,\ p_3 = 5,\ ...$) with HL multiplicities (m_i) from Section 5.2 on that same prime list. Pairwise coprimality is automatic, but the **ordered** sizes $p_1 = 2,\ p_{i+1} \leq p_i^2$ (Section 4.2.46), and $\log P = \sum_{i=1}^{n} \log p_i = \Theta(p_n)$ are what enter the period and cutoff arguments. The domination and sieve targets below are **not** stated for an abstract coprime tuple (p_i, m_i) with unrelated primes.

4.2.5 Remark (two difficulties: hole versus domination)

The sieve target on the **first** configuration space Ω is

$$P_\Omega\left(E_{p_n^2}\right) < \frac{1}{|}\Omega| = \frac{1}{\prod_{i=1}^{n}} \binom{p_i}{m_i} \tag{121}$$

(Section 4.2.30, Section 4.2.3). Two logically separate steps appear in every second-distribution approach:

1. **Parameter hole (easy for HL in the product model).** Compare a miss probability built only from $(P, M) = (\prod p_i, \prod m_i)$ to $\frac{1}{|}\Omega|$. In the coordinate-product second law below (Section 4.2.7, Section 4.2.11), this is essentially trivial at $t_n = p_n^2$ once $p_n^2 > p_i$ for every $i \leq n$.

2. **Cutoff transfer (hard; not permanent).** Show that the **intersection** first law on Ω is dominated by that second miss rate: $P_\Omega(U^I > t) \leq P^{II}(U^{II} > t)$. This **fails in general** (Section 4.2.15): the first event $\{1, ..., t\} \cap D(\boldsymbol{S}) = \emptyset$ is **larger** than the coordinate-product miss event, so $P_\Omega(E_t) \geq P^{II,\text{prod}}(E_t)$, not $\leq$.

The aggregated hypergeometric model (Section 4.2.8) reverses the difficulty: Section 4.2.16 is checked on small HL data, but Equation 163 often fails for moderate n (twins: first near $n \approx 19$) while $N_{\text{bad}}(p_n^2) = 0$ still holds (Section 4.2.37). Neither the geometric bound $\left(1 - \frac{M}{P}\right)^t$ nor the aggregated hole alone closes HL for all depths; the unconditional input is the finite head check and periodic budget (Section 4.2.52).

4.2.6 Remark (aggregated hole versus configuration count)

The target denominator is always $\frac{1}{|}\Omega| = \frac{1}{\prod \binom{p_i}{m_i}}$, **not** $\frac{1}{\binom{P}{M}}$ on Ω^{agg} (Section 4.2.3). For moderate n, Equation 163 may fail even when $N_{\text{bad}}(p_n^2) = 0$; see Section 4.2.37 and Section 4.2.5.

4.2.7 Product-coordinate second law at (P, M) (not intersection)

Collapse the coordinates as in Section 2.1:

$$P = \prod_{i=1}^{n} p_i, \quad M = \prod_{i=1}^{n} m_i \tag{122}$$

. The **first distribution** remains $\boldsymbol{S} = (S_1, ..., S_n) \in \Omega$ with CRT intersection $D(\boldsymbol{S})$ and

$$E_t^I = \{U^I > t\} = \{\{1, ..., t\} \cap D(\boldsymbol{S}) = \emptyset\} \tag{123}$$

(Section 4.2.2).

Define the **product-coordinate second** event at the same (P, M) but **without** the intersection: for $R_i(t) := \{x \bmod p_i : 1 \le x \le t\}$ and $k_i(t) := |R_i(t)|$,

$$E_t^{II,\text{prod}} := \{S_i \cap R_i(t) = \emptyset \ \text{ for every } \ i = 1, ..., n\}. \tag{124}$$

On the product space Ω, with $\boldsymbol{S}$ uniform,

$$P_\Omega\left(E_t^{II,\text{prod}}\right) = \prod_{i=1}^{n} \frac{\binom{p_i - k_i(t)}{m_i}}{\binom{p_i}{m_i}}. \tag{125}$$

This depends only on (p_i, m_i) and t; it does **not** pass through $D(\boldsymbol{S})$ or $\binom{P}{k,M}$.

4.2.8 Aggregated hypergeometric second (intersection-style on one modulus)

With the same P, M, define Ω^{agg} to be the family of M-subsets $T \subset \frac{\mathbb{Z}}{P}\mathbb{Z}$ with $|T| = M$, uniform, and

$$U^{II,\text{agg}} := \min\{t \ge 1 : t \bmod P \in T\} \tag{126}$$

, $E_t^{II,\text{agg}} := \left\{U^{II,\text{agg}} > t\right\}$.

This is an **intersection-style** benchmark on one modulus: a single $T \subset \frac{\mathbb{Z}}{P}\mathbb{Z}$ plays the role of $D(\boldsymbol{S})$. It is **not** the same as Equation 125.

4.2.9 Pinned aggregated second distribution

In the same setting as Section 4.2.8, define the **pinned** aggregated space

$$\Omega^{\text{agg,pin}} := \left\{T \subset \frac{\mathbb{Z}}{P}\mathbb{Z} : |T| = M, 0 \in T\right\} \tag{127}$$

, uniform with $|\Omega^{\text{agg,pin}}| = \binom{P-1}{M-1}$. Let $U^{II,\text{pin}} := \min\{t \ge 1 : t \bmod P \in T\}$ for T uniform on $\Omega^{\text{agg,pin}}$.

This matches the pinned $n = 1$ law (Equation 83) at modulus (P, M): one fixes $0 \in T$ and draws the remaining $M - 1$ residues uniformly from $\{1, ..., P - 1\}$.

4.2.10 Pinned product first distribution (sieve comparison)

For the **first** distribution on the consecutive scan, use the pinned product space

$$\Omega^{\mathrm{pin}} = \prod_{i=1}^{n} \Omega_i^{\mathrm{pin}} \tag{128}$$

from Section 2.3 ($0 \in S_i$ for every i), with

$$|\Omega^{\mathrm{pin}}| = \prod_{i=1}^{n} \binom{p_i - 1}{m_i - 1} \tag{129}$$

. Let $U^{I,\mathrm{pin}}$ be the first positive **ONE** for $\boldsymbol{S}$ uniform on Ω^{pin}. The unpinned law on Ω is still used for Section 4.2.31 on the full configuration space (Section 4.2.36); the pinned pair below is the correct apples-to-apples comparison with Section 4.2.9 (as in Section 3).

4.2.11 Product-coordinate hole for HL tail data

Fix admissible $\boldsymbol{h}$ with $\ell = \ell(\boldsymbol{h})$, tail gap $K = K_{\boldsymbol{h}}$, and HL multiplicities (p_i, m_i) from Section 5.2 on consecutive primes. Let $t_n = p_n^2$ and $n \geq \ell$.

1. If $p_n^2 \geq p_i$ for every $i \leq n$ (equivalently $t_n \geq p_n$, which holds for $n \geq 2$ by Section 4.2.46), then $k_i(t_n) = p_i$ for each $i \leq n$, hence $\binom{p_i - k_i(t_n)}{m_i} = 0$ and
$$P_\Omega\Big(E_{t_n}^{II,\mathrm{prod}}\Big) = 0 < \frac{1}{|}\Omega| \tag{130}$$
.
2. On the tail $i \geq \ell$ with $p_i - m_i = K$ and $p_i > K$, the factor $\frac{\binom{p_i - k_i(t_n)}{m_i}}{\binom{p_i}{m_i}}$ is zero once $k_i(t_n) = p_i$; the head contributes only finitely many positive factors.

Proof. For $t_n = p_n^2 \geq p_n \geq p_i$, every residue mod p_i appears among $\{1, ..., t_n\}$, so $R_i(t_n) = \frac{\mathbb{Z}}{p_i}\mathbb{Z}$ and $k_i(t_n) = p_i$. Then $\binom{p_i - p_i}{m_i} = \binom{0}{m_i} = 0$ for $m_i \geq 1$. Equation 125 gives $P_\Omega\Big(E_{t_n}^{II,\mathrm{prod}}\Big) = 0$. Since $|\Omega| \geq 1$, $0 < \frac{1}{|}\Omega|$. □

This shows the **parameter-level** hole is easy in the product-coordinate model; it does **not** bound $P_\Omega\Big(E_{t_n}^{I}\Big)$ without a separate domination argument (Section 4.2.15).

4.2.12 Aggregated miss probability (unpinned)

In Section 4.2.8, let $R_t := \{x \bmod P : x \in \{1, ..., t\}\}$ and $k_t := |R_t| \leq \min(t, P)$. Then

$$P\Big(E_t^{II,\mathrm{agg}}\Big) = \frac{\binom{P - k_t}{M}}{\binom{P}{M}} \leq \left(1 - \frac{M}{P}\right)^{k_t} \leq \left(1 - \frac{M}{P}\right)^{t}. \tag{131}$$

Proof. Avoiding all **ONES** in $\{1, ..., t\}$ means choosing an M-subset of $\frac{\mathbb{Z}}{P}\mathbb{Z}$ disjoint from R_t, giving the exact count $\frac{\binom{P-k_t}{M}}{\binom{P}{M}}$. Equation 157 with $(p, m, k) = (P, M, k_t)$ yields the two inequalities. □

4.2.13 Pinned aggregated miss probability

In Section 4.2.9, let $R_t^+ := \{x \bmod P : x \in \{1, ..., t\}\}$ and $k_t^+ := |R_t^+|$. For $t < P$ (so $0 \notin R_t^+$),

$$P(U^{II,\mathrm{pin}} > t) = \frac{\binom{P-1-k_t^+}{M-1}}{\binom{P-1}{M-1}}. \tag{132}$$

For general t, with $\ell_t := |R_t^+ \cup \{0\}|$,

$$P(U^{II,\mathrm{pin}} > t) = \frac{\binom{P-\ell_t}{M-1}}{\binom{P-1}{M-1}} \tag{133}$$

when $M - 1 \leq P - \ell_t$ (otherwise the probability is 0).

Proof. Fix $t < P$. A pinned M-set T misses every positive $x \in \{1, ..., t\}$ iff $T \cap R_t^+ = \emptyset$ and $0 \in T$, so the free part is an $(M-1)$-subset of $\frac{\mathbb{Z}}{P}\mathbb{Z} \setminus \{0\} \cup R_t^+$, which has $P - 1 - k_t^+$ elements. Counting gives Equation 132. The general ℓ_t case is the same with $P - \ell_t$ available residues. □

4.2.14 Bad configurations inject into aggregated misses

Let $N_{\text{bad}}(t)$ be as in Section 4.2.2, $k_t = |R_t|$ as in Section 4.2.12, and $M = \prod_{i=1}^{n} m_i$. Then

$$N_{\text{bad}}(t) \le \binom{P - k_t}{M} \tag{134}$$

.

Proof. If $\boldsymbol{S} \in \Omega$ has no **ONE** in $\{1, ..., t\}$, then $D(\boldsymbol{S}) \cap \{1, ..., t\} = \emptyset$, so $D(\boldsymbol{S}) \cap R_t = \emptyset$. By Section 2.7, $|D(\boldsymbol{S})| = M$. By Section 2.7.1, $\boldsymbol{S} \mapsto D(\boldsymbol{S})$ is injective on Ω, hence

$$\in \Omega : D(\boldsymbol{S}) \cap R_t = \emptyset\}| = |\left\{T \subset \frac{\mathbb{Z}}{P}\mathbb{Z} : |T| = M, T \cap R_t = \emptyset, T = D(\boldsymbol{S}) \text{ for some } \boldsymbol{S}\right\}| \le |\left\{T \subset \frac{\mathbb{Z}}{P}\mathbb{Z} : |T| = M, T \cap R_t = \emptyset\right\}| \tag{135}$$

□

4.2.15 First versus second: inclusion and domination

With E_t^I, $E_t^{II,\text{prod}}$, and $E_t^{II,\text{agg}}$ as above:

1. **Coordinate product versus intersection.** $E_t^{II,\text{prod}} \subset E_t^I$: if every S_i misses all residues $x \bmod p_i$ with $x \in \{1, ..., t\}$, then no $x \in \{1, ..., t\}$ can lie in $D(\boldsymbol{S})$. Hence
$$P_\Omega\left(E_t^{II,\text{prod}}\right) \le P_\Omega\left(E_t^I\right) \tag{136}$$
. The **reverse** inequality $P_\Omega\left(E_t^I\right) \le P_\Omega\left(E_t^{II,\text{prod}}\right)$ **fails in general**: a configuration can have every S_i meeting some residue from $R_i(t)$ while still having no global CRT point in $\{1, ..., t\}$.
2. **Aggregated versus first.** The comparison $P_\Omega\left(E_t^I\right) \le P\left(E_t^{II,\text{agg}}\right)$ is **not** a consequence of subset relations on the same sample space (Section 4.2.21, Section 4.2.26). Do **not** assume $P_\Omega(U^I > t) \le P^{II}(U^{II} > t)$ for all t, n, and HL data.
3. **Contrast with Section 4.2.11.** There $P_\Omega\left(E_{p_n^2}^{II,\text{prod}}\right) = 0$ for $n \ge 2$, while $P_\Omega\left(E_{p_n^2}^I\right)$ is typically positive on the random model. So the first law need not dominate the product-coordinate second **forever**; the easy HL inequality is $0 < \frac{1}{|}\Omega|$, not $P_\Omega(E^I) \le P^{II,\text{prod}}$.

4.2.16 Count bound and aggregated comparison (partial)

In the setting of Section 4.2.8,

$$P_\Omega(U^I > t) = N_{\text{bad}} \frac{t}{\prod_{i=1}^{n}} \binom{p_i}{m_i} \le \frac{\binom{P-k_t}{M}}{\prod_{i=1}^{n}} \binom{p_i}{m_i}. \tag{137}$$

When additionally

$$P_\Omega\left(E_t^I\right) \le P\left(E_t^{II,\text{agg}}\right) = \frac{\binom{P-k_t}{M}}{\binom{P}{M}}, \tag{138}$$

Equation 138 together with Equation 163 yields Section 4.2.36 only where both inequalities hold; the unconditional HL input remains Section 4.2.37 (Section 4.2.5). Equation 138 is **not** proved in general (Section 4.2.26, Section 4.2.22); the only complete case in this file is $n = 1$ (Section 4.2.20).

Proof. Section 4.2.14 gives Equation 137. Section 4.2.12 identifies $P\left(E_t^{II,\text{agg}}\right)$. □

4.2.17 Domination cutoff $t \le Kp_n$ (same K as tail gap)

Fix admissible $\boldsymbol{h}$ with $\ell = \ell(\boldsymbol{h})$, uniform tail gap $K = K_{\boldsymbol{h}}$ (so $p_i - m_i = K$ for $i \ge \ell$), and HL multiplicities on the first n consecutive primes. Write $t_n^K \coloneqq Kp_n$ and $t_n^2 \coloneqq p_n^2$.

The **domination cutoff** is the integer window $\{1, ..., t_n^K\}$: one asks whether

$$P_\Omega(E_t^I) \le P\left(E_t^{II,\text{agg}}\right) \quad \text{for every} \quad t \in \{1, ..., t_n^K\}, \tag{139}$$

or at least at the endpoint $t = t_n^K$. This uses the **same** K as the tail condition $p_i - m_i = K$; it is **not** the sieve cutoff $t_n^2 = p_n^2$ from Section 4.2.30.

4.2.18 Period dominates Kp_n on consecutive primes

In the setting of Section 4.2 with consecutive primes $p_1 = 2 < p_2 = 3 < ... < p_n$ and $P = \prod_{i=1}^n p_i$:

1. For $n \ge 3$, $Kp_n < P$ for every integer $K \ge 1$.
2. For $n = 2$, $P = 6$ and $Kp_2 = 3K$; hence $Kp_2 < P$ iff $K = 1$, while $Kp_2 = P$ when $K = 2$.

Proof. For $n = 2$, $P = 2 \times 3 = 6$ and $Kp_2 = 3K$; the table is immediate.

For $n \ge 3$, $P = \prod_{i=1}^n p_i \ge p_1 p_2 p_3 = 30$. Also $p_n \ge p_3 = 5$, so $Kp_n \le Kp_n$ with fixed K is maximized at the largest n in a given range; it suffices to check $30 > Kp_3 = 5K$ for $n = 3$, i.e. $K < 6$, which holds for every fixed HL tail gap $K = K_{\boldsymbol{h}}$. For $n \ge 4$, $P \ge 30\frac{p_4}{p_3} = 42 > 6p_n$ since $p_n \ge 7$, hence $P > Kp_n$ for every $K \ge 1$. □

4.2.19 Distinct residues when $t = Kp_n$ is at most the period

In the setting of Section 4.2, let $P = \prod_{i=1}^n p_i$ and $t = Kp_n$. If $t \le P$ (in particular when $Kp_n < P$ by Section 4.2.18, or when $n = 2$ and $Kp_2 = P = 6$), then

$$k_t = |\{x \bmod P : 1 \le x \le t\}| = t = Kp_n \tag{140}$$

.

Proof. For $1 \le x < y \le t \le P$, the residues $x \bmod P$ and $y \bmod P$ are distinct: if $x \equiv y \bmod P$, then $P \,\texttt{mid}\, (y - x)$ with $0 < y - x < P$, impossible. Thus each $x \in \{1, ..., t\}$ contributes a distinct class and $k_t = t$. □

4.2.20 Domination at $n = 1$ (modulus $p_1 = 2$)

In the consecutive-prime chain, $n = 1$ means $p_1 = 2$ only. Then $P = p_1$, $M = m_1$, $|\Omega| = \binom{p_1}{m_1} = \binom{P}{M}$, and $D(S_1) = S_1$. For every $t \ge 1$ with $k_t = |\{x \bmod P : 1 \le x \le t\}|$,

$$P_\Omega(E_t^I) = \frac{\binom{P-k_t}{M}}{\binom{P}{M}} = P\left(E_t^{II,\text{agg}}\right). \tag{141}$$

Proof. Each configuration is an m_1-subset $S_1 \subset \frac{\mathbb{Z}}{p_1}\mathbb{Z}$; $\{1, ..., t\} \cap D(S_1) = \emptyset$ iff $S_1 \cap R_t = \emptyset$. Counting gives $N_{\text{bad}}(t) = \binom{p_1 - k_t}{m_1}$ and $|\Omega| = \binom{p_1}{m_1}$, hence the first probability equals $\frac{\binom{P-k_t}{M}}{\binom{P}{M}}$. The aggregated second law is the same count on Ω^{agg} (Section 4.2.12). □

4.2.21 Domination as a density on CRT images

In the setting of Section 4.2 (Section 4.2.4), let $\Phi : \Omega \to 2^{\{\{0,...,P-1\}\}}$ be $\Phi(\boldsymbol{S}) = D(\boldsymbol{S})$ (Section 2.7.1), R_t and k_t as in Section 4.2.12, and

$$\mathcal{A}_t \coloneqq \left\{T \subset \frac{\mathbb{Z}}{P}\mathbb{Z} : |T| = M, T \cap R_t = \emptyset, T \in \Phi(\Omega)\right\}. \tag{142}$$

Then $N_{\text{bad}}(t) = |\mathcal{A}_t|$ and

$$P_\Omega(E_t^I) = |\mathcal{A}_t|/|\Omega|, \quad P\left(E_t^{II,\mathrm{agg}}\right) = \frac{\binom{P-k_t}{M}}{\binom{P}{M}}. \tag{143}$$

Thus Equation 138 is equivalent to

$$|\mathcal{A}_t|/|\Omega| \le \frac{\binom{P-k_t}{M}}{\binom{P}{M}}, \tag{144}$$

i.e.
CRT images that miss R_t are not over-represented among **all** M-subsets that miss R_t.

Proof. Section 4.2.14 gives $\mathcal{A}_t = \{\Phi(\boldsymbol{S}) : \boldsymbol{S} \in \Omega, \Phi(\boldsymbol{S}) \cap R_t = \emptyset\}$ with $|\mathcal{A}_t| = N_{\mathrm{bad}}(t)$. Injectivity identifies $N_{\mathrm{bad}}(t)$ with the numerator in Equation 144. Section 4.2.12 is the denominator identity. □

4.2.22 Injectivity does not imply domination

In the setting of Section 4.2.21, Section 4.2.14 gives only

$$|\mathcal{A}_t| \le \binom{P-k_t}{M} \tag{145}$$

. Dividing by $|\Omega| = |\Phi(\Omega)|$ yields Equation 137, but **not** Equation 144, because in general

$$\binom{P}{M} > |\Omega| \tag{146}$$

(Section 4.2.3). Indeed, whenever $\mathcal{A}_t = \Phi(\Omega)$ (every CRT image misses R_t), the left side of Equation 144 equals 1 while the right side is $\frac{\binom{P-k_t}{M}}{\binom{P}{M}} < 1$ once $k_t > 0$ and $M \le P - k_t$. So any proof of Equation 138 must use structure of $\Phi(\Omega)$ beyond Section 2.7.1.

4.2.23 Remark (domination cutoff versus sieve cutoff)

Two cutoffs must not be conflated (Section 4.2.5):

1. **Domination cutoff** $t_n^K = Kp_n$. Here $k_t = Kp_n$ is asymptotically negligible relative to P for large n, so $P\left(E_t^{II,\mathrm{agg}}\right) = \frac{\binom{P-k_t}{M}}{\binom{P}{M}}$ is typically **large** (most M-subsets of $\frac{\mathbb{Z}}{P}\mathbb{Z}$ still miss the small set R_t). Equation 138 is designed for this **transfer** step: compare intersection miss rate on Ω to the aggregated hypergeometric rate (Section 4.2.21). It is **not** proved here for general n and $t \le Kp_n$ (Section 4.2.26).
2. **Sieve hole cutoff** $t_n^2 = p_n^2$. Here $k_{t_n^2} = p_n^2$ once $t_n^2 < P$ (Section 4.2.35), and the target is $P_\Omega\left(E_{t_n^2}^I\right) < 1/|\Omega|$, equivalently Equation 163 on the aggregated side **after** domination. For twins, Equation 163 first becomes true near $n \approx 19$ (Section 4.2.6), while $N_{\mathrm{bad}}(p_n^2) = 0$ is checked directly for $n + 1 \le 6$ (Section 4.2.37).

So domination at Kp_n is an **earlier, easier** comparison range than the aggregated parameter hole at p_n^2; closing HL for all depths still uses the finite head and periodic budget, not Equation 139 alone.

4.2.24 Coordinate all-miss forces intersection miss

In the setting of Section 4.2.34, $E_t^{\mathrm{coord}} \subset E_t^I$, hence $N_{\mathrm{bad}}(t) \ge |\{\boldsymbol{S} \in \Omega : S_i \cap R_i(t) = \emptyset, \quad \forall i\}| = \prod_{i=1}^n \binom{p_i - k_i(t)}{m_i}$.

Proof. Already noted in Section 4.2.34: if $S_i \cap R_i = \emptyset$ for every i and $x \in \{1, ..., t\}$, then $x \bmod p_i \in R_i(t)$ implies $x \bmod p_i \notin S_i$, so $x \notin D(\boldsymbol{S})$. □

4.2.25 Two consecutive primes: product-type residue lists

Take $n = 2$ in the consecutive-prime chain, so $p_1 = 2$, $p_2 = 3$, $P = 6$. Let $R_i := \{x \bmod p_i : 1 \leq x \leq t\}$, $k_i := |R_i|$, and $R_t := \{x \bmod P : 1 \leq x \leq t\}$. If $t \leq 6$ and every pair $(a, b) \in R_1 \times R_2$ is realized by some $x \in \{1, ..., t\}$ (equivalently $|R_t| = k_1 k_2$; this holds for $t = 6 = P$ and fails only for smaller t when some CRT pair is missing from $\{1, ..., t\}$), Then for every $\boldsymbol{S} = (S_1, S_2) \in \Omega$,

$$D(\boldsymbol{S}) \cap R_t = \emptyset \Longleftrightarrow S_i \cap R_i = \emptyset \quad \text{for} \quad i = 1, 2. \tag{147}$$

Hence $N_{\text{bad}}(t) = \binom{p_1 - k_1}{m_1}\binom{p_2 - k_2}{m_2}$ and

$$P_\Omega(E_t^I) = \prod_{i=1}^{2} \frac{\binom{p_i - k_i}{m_i}}{\binom{p_i}{m_i}} \tag{148}$$

.

Proof. Forward is Section 4.2.24. For the converse, if $S_1 \cap R_1 \neq \emptyset$ and $S_2 \cap R_2 \neq \emptyset$, pick $a \in S_1 \cap R_1$ and $b \in S_2 \cap R_2$. By hypothesis there exists $x \in \{1, ..., t\}$ with $x \equiv a \bmod p_1$ and $x \equiv b \bmod p_2$; then $x \in D(\boldsymbol{S}) \cap R_t$, contradiction. The count identity is the product of the two coordinate miss counts. □

4.2.26 Open domination target at Kp_n

Fix $\ell \geq 1$, $K \geq 1$, and head multiplicities $(m_i)_{i<\ell}$ on consecutive primes, with tail $p_i - m_i = K$ for $i \geq \ell$. The desired **domination cutoff** (Section 4.2.17) is

$$P_\Omega\left(E_{Kp_n}^I\right) \leq \frac{\binom{P - k_{Kp_n}}{M}}{\binom{P}{M}} \quad \text{once} \quad Kp_n < P \quad \text{and} \quad k_{Kp_n} = Kp_n \tag{149}$$

(Section 4.2.19). This is the endpoint case of Equation 139. It does **not** follow from Section 4.2.14 or Section 4.2.22. A proof would bound the CRT image density Equation 144 using the **consecutive-prime** structure of (p_i, m_i): ordered sizes $p_1 = 2 < ... < p_n$, Section 4.2.46, $P = \prod_{i=1}^n p_i$, marginal constraints $|\pi_i(D(\boldsymbol{S}))| = m_i$, and the tail gap $p_i - m_i = K$ for $i \geq \ell$. Generic coprime moduli without this ordering are outside the HL sieve model (Section 4.2.4). Until such a bound is in place, Equation 138 must be treated as an explicit hypothesis wherever it is invoked (Section 4.2.36, Section 4.2.54).

4.2.27 Pinned first: coordinate factorization

On Ω^{pin} from Section 4.2.10, for $E_t = \{U^{I,\text{pin}} > t\}$,

$$P_{\Omega^{\text{pin}}}(E_t) = \prod_{i=1}^{n} \frac{\binom{p_i - 1 - k_i^+}{m_i - 1}}{\binom{p_i - 1}{m_i - 1}}, \tag{150}$$

where $k_i^+ := |\{x \bmod p_i : x \in \{1, ..., t\}\}|$ (for $t < p_i$, this is $|R_t^+|$ at modulus p_i).

Proof. The same factorization as Equation 161, with each coordinate drawn uniformly among m_i-subsets that contain 0. For fixed $t < p_i$, a miss at i is equivalent to choosing the $(m_i - 1)$ positive residues in $S_i \setminus \{0\}$ disjoint from R_t^+, giving the stated binomial factor. □

4.2.28 Pinned first dominated by pinned aggregated

Fix consecutive primes $p_1 = 2 < ... < p_n$ as in Section 4.2, $m_i \geq 2$, and $t \geq 1$ with $t < \min_i p_i$. For $t = p_n^2$ with $n \geq 3$, $p_n^2 < p_{n+1}$ by Section 4.2.46 and $p_n^2 < p_1 p_2 p_3 \leq P$. The depth-$n =$ 2 case is the separate consecutive-prime model $p_1 = 2,\ p_2 = 3,\ P = 6$ (Section 4.2.25). On Section 4.2.10 and Section 4.2.9,

$$P_{\Omega^{\text{pin}}}(U^{I,\text{pin}} > t) \le P(U^{II,\text{pin}} > t) = \frac{\binom{P-1-k_t^+}{M-1}}{\binom{P-1}{M-1}}. \tag{151}$$

Proof. Equation 150 gives an exact product formula for the pinned first law. Equation 132 gives the pinned aggregated identity at the same k_t^+. The same density issue as Section 4.2.22 applies: injectivity on Ω^{pin} gives a count bound but not Equation 151 in general. Equation 151 is not proved here beyond the $n = 1$ specialization (Section 4.2.20). This lemma records the correct pinned comparison target aligned with Section 3. □

4.2.29 Remark (pinned vs. unpinned second model)

1. **Section 3 (moments).** Ω^{pin} is compared to **coordinate-rate** geometric models at $q_i = \frac{m_i}{p_i}$ (Equation 108), not to $\Omega^{\text{agg,pin}}$.
2. **Section 4 (sieve).** The **unpinned** transfer $P_\Omega(U^I > t) \le \frac{\binom{P-k_t}{M}}{\binom{P}{M}}$ (Section 4.2.16) feeds Section 4.2.31 on $|\Omega| = \prod \binom{p_i}{m_i}$.
3. **Pinned pair.** Section 4.2.28 is the correct hypergeometric comparison at (P, M) with 0 pinned on both sides; it matches the Section 3 philosophy (Equation 83) after collapsing coordinates.

4.2.30 Target inequality (sieve hole on Ω)

For $t_n := p_n^2$, the **sieve hole** target on the product configuration space is

$$P_\Omega\left(E_{t_n}\right) < \frac{1}{|}\Omega| = \frac{1}{\prod_{i=1}^n}\binom{p_i}{m_i} \tag{152}$$

.

On the tail $p_i - m_i = K$, one has $\binom{p_i}{m_i} = \binom{p_i}{K}$ and $|\Omega| = \prod_{i=1}^n \binom{p_i}{K}$ (up to head factors $i < \ell$). The asymptotic target Equation 163 compares the aggregated miss rate $\frac{\binom{P-k_t}{M}}{\binom{P}{M}}$ on Ω^{agg} directly to $\frac{1}{|}\Omega|$; Section 4.2.16 and Section 4.2.31 link this to P_Ω on the CRT configuration space.

4.2.31 From a tiny probability to a mandatory ONE

If $P(E_t) < \frac{1}{|}\Omega| = \frac{1}{\prod_{i=1}^n}\binom{p_i}{m_i}$, then $N_{\text{bad}}(t) = 0$ and **every** configuration has at least one **ONE** in $\{1, ..., t\}$ (equivalently $U \le t$).

Proof. $N_{\text{bad}}(t) = |\Omega| \cdot P(E_t) < 1$. Since $N_{\text{bad}}(t)$ is a non-negative integer, $N_{\text{bad}}(t) = 0$. Thus no configuration has all **ZEROES** on $\{1, ..., t\}$; each has $U \le t$. □

4.2.32 Product-rate upper bound (auxiliary; not used in the sieve hole)

In the setting of Section 4.2.2, let $M = \prod_{i=1}^n m_i$ and $P = \prod_{i=1}^n p_i$. For every $t \ge 1$,

$$P_\Omega(E_t) = P_\Omega(U^I > t) \le \left(1 - \frac{M}{P}\right)^t. \tag{153}$$

Proof. For $x \in \{1, ..., t\}$, let $A_x := \{x \in D(\boldsymbol{S})\}$. Each A_x is an **increasing** event in the coordinates $(S_1, ..., S_n)$: enlarging some S_i can only add elements to $D(\boldsymbol{S})$. On a product probability space, increasing events are positively correlated (the FKG / Harris inequality; see (Fortuin et al. 1972; Harris 1960)). Hence

$$P_\Omega(\cup_{x=1}^t A_x) \ge \prod_{x=1}^t P_\Omega(A_x). \tag{154}$$

For a fixed x, choosing $\boldsymbol{S}$ uniformly in Ω and using independence across coordinates,

$$P_\Omega(A_x) = P_\Omega(x \in D(\boldsymbol{S})) = \prod_{i=1}^{n} P(x \bmod p_i \in S_i) = \prod_{i=1}^{n} \frac{m_i}{p_i} = \frac{M}{P}. \tag{155}$$

Therefore

$$P_\Omega(E_t) = P_\Omega(\cap_{x=1}^{t} A_x^c) = 1 - P_\Omega(\cup_{x=1}^{t} A_x) \le 1 - \prod_{x=1}^{t}\left(\frac{M}{P}\right) = \left(1 - \frac{M}{P}\right)^t. \tag{156}$$

□

4.2.33 One-coordinate miss bound

Fix integers $p > m \ge 1$ and $k \ge 0$ with $m \le p - k$. Let S be uniform among m-subsets of $\frac{\mathbb{Z}}{p}\mathbb{Z}$. For any k-element set $R \subset \frac{\mathbb{Z}}{p}\mathbb{Z}$,

$$P(S \cap R = \emptyset) = \frac{\binom{p-k}{m}}{\binom{p}{m}} \le \left(1 - \frac{m}{p}\right)^k. \tag{157}$$

Proof. If $m > p - k$, then $\binom{p-k}{m} = 0$ and Equation 157 holds. Assume $m \le p - k$. Set $a_k := \frac{\binom{p-k}{m}}{\binom{p}{m}}$. Then $a_0 = 1$ and, for $k \ge 1$ with $m \le p - k$,

$$\frac{a_k}{a_{k-1}} = \frac{\binom{p-k}{m}}{\binom{p-k+1}{m}} = \frac{p-k-m+1}{p-k+1}. \tag{158}$$

The one-step ratio bound

$$\frac{p-k-m+1}{p-k+1} \le \frac{p-m}{p} = 1 - \frac{m}{p} \tag{159}$$

is equivalent to $p(p-k-m+1) \le (p-k+1)(p-m)$, i.e.
$-kp \le -kp$. By induction, $a_k \le \left(1 - \frac{m}{p}\right)^k$. The left-hand side of Equation 157 is exactly a_k when $|R| = k$ and R is disjoint from the chosen m-set. □

4.2.34 Coordinate all-miss factorization (auxiliary)

Fix $n \ge 1$, consecutive primes $p_1 = 2 < ... < p_n$, and $m_i \ge 1$ (Section 4.2). Let $R_i := \{x \bmod p_i : 1 \le x \le t\}$ and $k_i := |R_i| \le \min(t, p_i)$. Define the **coordinate all-miss** event

$$E_t^{\text{coord}} := \{S_i \cap R_i = \emptyset \ \text{ for every } \ i = 1, ..., n\} \tag{160}$$

. Then

$$P_\Omega\left(E_t^{\text{coord}}\right) = \prod_{i=1}^{n} \frac{\binom{p_i-k_i}{m_i}}{\binom{p_i}{m_i}} \le \left(1 - \frac{M}{P}\right)^t, \tag{161}$$

where $M = \prod m_i$ and $P = \prod p_i$. Moreover $E_t^{\text{coord}} \subset E_t$ (if every coordinate misses every residue hit by $\{1, ..., t\}$, then no $x \in \{1, ..., t\}$ lies in $D(\boldsymbol{S})$), so

$$P_\Omega(E_t) \ge P_\Omega\left(E_t^{\text{coord}}\right) \tag{162}$$

. The **consecutive-scan** upper bound $P_\Omega(E_t) \le \left(1 - \frac{M}{P}\right)^t$ is Equation 153 (Section 4.2.32), not the coordinate product formula.

Proof. Independence across coordinates gives the product identity for E_t^{coord}. Apply Equation 157 on each factor with $|R| = k_i$ to obtain the product bound $\le \left(1 - \frac{M}{P}\right)^t$. If $S_i \cap R_i = \emptyset$ for every i and $x \in \{1, ..., t\}$, then for each such x there exists i with $x \bmod p_i \in R_i$ and hence $x \bmod p_i \notin S_i$, so $x \notin D(\boldsymbol{S})$ and $E_t^{\text{coord}} \subset E_t$. □

4.2.35 Aggregated second miss rate below the configuration weight

Fix $\ell \geq 1$, $K \geq 0$, and tail gap $p_i - m_i = K$ for $i \geq \ell$ on consecutive primes. Let $P = \prod p_i$, $M = \prod m_i$, $k_t = |\{x \bmod P : 1 \leq x \leq t\}|$, $t_n = p_n^2$, and $|\Omega| = \prod_{i=1}^{n} \binom{p_i}{m_i}$. Then there exists $N_0 = N_0\left(\ell, K, (m_i)_{i<\ell}\right)$ such that for every $n \geq N_0$,

$$\frac{\binom{P-k_{t_n}}{M}}{\binom{P}{M}} < \frac{1}{|}\Omega| = \frac{1}{\prod_{i=1}^{n}} \binom{p_i}{m_i}. \tag{163}$$

This is the **aggregated** parameter hole; it is **not** the same as Section 4.2.11 (product-coordinate, typically 0 at $t_n = p_n^2$). Even when Equation 163 holds, it does **not** imply $P_\Omega\left(E_{t_n}^I\right) < \frac{1}{|}\Omega|$ without Equation 138 (Section 4.2.5). For HL, Equation 163 often fails for moderate n (twins: first near $n \approx 19$); the unconditional sieve input is Section 4.2.37.

Proof. We compare logarithms of the two sides of Equation 163. Set

$$L := \sum_{i=1}^{n} \log\binom{p_i}{m_i} = \log|\Omega| \tag{164}$$

and

$$A := \log\binom{P - k_{t_n}}{M} - \log\binom{P}{M} \tag{165}$$

, so Equation 163 is equivalent to $A < -L$.

Tail gap $\boldsymbol{p_i - m_i = K}$ for $\boldsymbol{i \geq \ell}$. Then $m_i = p_i - K$ and $\binom{p_i}{m_i} = \binom{p_i}{K}$. For $p_i > K$ (automatic once $p_i \geq p_\ell > K$ on the tail, and on the finite head by direct check),

$$\log\binom{p_i}{m_i} = \log\binom{p_i}{K} \geq K \log p_i - \log(K!) - O(1) \tag{166}$$

. Hence

$$L = \sum_{i<\ell} \log\binom{p_i}{m_i} + \sum_{i\geq\ell} \log\binom{p_i}{K} \geq K \sum_{i=\ell}^{n} \log p_i - O(n) \tag{167}$$

, with the $O(n)$ term coming only from the fixed head $i < \ell$.

Step 1 (the residue count $\boldsymbol{k_{t_n}}$). For $t_n = p_n^2$, each $x \in \{1, ..., t_n\}$ gives a distinct class $x \bmod P$ once $t_n < P$, because $1 \leq x < y \leq t_n < P$ implies $x \not\equiv y \bmod P$. Now $\log P = \sum_{i=1}^{n} \log p_i = \Theta(p_n)$ by the prime number theorem ((Hardy and Wright 2008)), while $t_n = p_n^2 = o(P)$. Thus $t_n < P$ and $k_{t_n} = t_n$ for all sufficiently large n.

Step 2 (hypothesis for Equation 157). Equation 157 requires $M \leq P - k_{t_n}$. On the tail, $\frac{M}{P} = \left(\prod_{i<\ell} \frac{m_i}{p_i}\right) \cdot \prod_{i=\ell}^{n}\left(1 - \frac{K}{p_i}\right)$. The tail product converges to a constant $c_{\text{tail}} \in (0, 1)$ because $\sum_{i\geq\ell} \frac{K}{p_i} < \infty$, and the head factor is fixed. So $\frac{M}{P} \to c_{\text{head}} c_{\text{tail}} \in (0, 1)$, while $\frac{k_{t_n}}{P} = \frac{p_n^2}{P} = o(1)$. Therefore $M \leq P - k_{t_n}$ for all large n.

Step 3 (upper bound on $\boldsymbol{A}$). Apply Equation 157 at $(p, m, k) = \left(P, M, k_{t_n}\right)$:

$$A \leq k_{t_n} \log\left(1 - \frac{M}{P}\right) = p_n^2 \log\left(1 - \frac{M}{P}\right). \tag{168}$$

Since $0 < \frac{M}{P} \leq c_* < 1$ for some $c_* < 1$ independent of large n, there exists $c_1 > 0$ with $\log\left(1 - \frac{M}{P}\right) \leq -c_1\left(\frac{M}{P}\right)$, hence

$$A \leq -c_1 p_n^2 \left(\frac{M}{P}\right) \tag{169}$$

. Because $\frac{M}{P} \geq c_{\min} > 0$ for all large n (again by the tail product and fixed head), there exists $c_2 > 0$ such that

$$A \leq -c_2 p_n^2 \tag{170}$$

.

Step 4 (lower bound on L). By the prime number theorem, $\sum_{i=1}^{n} \log p_i = \Theta(p_n)$; in particular $\sum_{i=\ell}^{n} \log p_i \geq c_3 p_n$ for some $c_3 > 0$ and all large n. Combining with the tail-gap estimate above gives $L \geq K c_3 p_n - O(n)$. Since $p_n \to \infty$, there exists $c_4 > 0$ such that $L \geq c_4 p_n$ for all large n.

Step 5 (closure). We have $A \leq -c_2 p_n^2$ and $L \geq c_4 p_n$. For n large enough, $c_2 p_n^2 > c_4 p_n$, hence $A < -L$ and Equation 163 holds.

Edge case $K = 0$ on the tail. Then $m_i = p_i$ for $i \geq \ell$, so $M = P$ and $\binom{P-k_{t_n}}{M} = 0$ whenever $k_{t_n} > 0$, which holds for n large. The left-hand side of Equation 163 is then $0 < \frac{1}{|}\Omega|$. □

4.2.36 Aggregated sieve hole (conditional on domination)

Suppose $n \geq N_0$ from Section 4.2.35 and Equation 138 holds at $t_n = p_n^2$ (hypothesis; not proved in general, Section 4.2.26, Section 4.2.15). Then

$$P_\Omega(U^I > t_n) \leq \frac{\binom{P-k_{t_n}}{M}}{\binom{P}{M}} < \frac{1}{\prod_{i=1}^{n} \binom{p_i}{m_i}} \tag{171}$$

by Equation 163, and Section 4.2.31 gives $N_{\mathrm{bad}}(t_n) = 0$.

Proof. Section 4.2.16, Section 4.2.35, Section 4.2.31. □

The unconditional HL cutoff uses Section 4.2.37 for $\ell \leq n < N_0$ and does not require Section 4.2.36 (Section 4.2.54).

4.2.37 Finite head check

Fix admissible $\boldsymbol{h}$ with tail data $\ell = \ell(\boldsymbol{h})$, $K = K_{\boldsymbol{h}}$, and HL multiplicities (p_i, m_i) from Section 5.2 on the first n consecutive primes. Let $t_n = p_n^2$ and $|\Omega| = \prod_{i=1}^{n} \binom{p_i}{m_i}$ (Section 4.2.3). For each n with $\ell \leq n < N_0\big(\ell, K, (m_i)_{i<\ell}\big)$, where N_0 is as in Section 4.2.35,

$$N_{\mathrm{bad}}(t_n) = 0. \tag{172}$$

Equivalently, every $\boldsymbol{S} \in \Omega$ has a **ONE** in $\{1, ..., t_n\}$, so $U(\boldsymbol{S}) \leq t_n$; in particular $U(\boldsymbol{S}(\boldsymbol{h})) \leq t_n$.

Verification procedure. $|\Omega|$ is finite. For each $\boldsymbol{S} = (S_1, ..., S_n) \in \Omega$, test whether $\{1, ..., t_n\} \cap D(\boldsymbol{S}) = \emptyset$; increment $N_{\mathrm{bad}}(t_n)$ when it is empty. This is a finite exhaustive search over $\prod_{i=1}^{n} \binom{p_i}{m_i}$ configurations.

Twin primes $\boldsymbol{h} = (0, 2)$. Here $\ell = 2$, $K = 2$, and $m_1 = 1$, $m_i = p_i - 2$ for $i \geq 2$. The script scripts/verify_sieve_hole.py reproduces the counts below (run: python3 scripts/verify_sieve_hole.py –pattern twins –max-n 6).

n	p_n	$t_n = p_n^2$	config count	$N_{\mathrm{bad}}(t_n)$
2	3	9	6	0
3	5	25	60	0
4	7	49	1260	0
5	11	121	69300	0
6	13	169	5405400	0

For these depths, Equation 163 often **fails** (the asymptotic aggregated hole is not yet active), yet $N_{\text{bad}}(t_n) = 0$ holds directly. For twin primes, the script reports the first depth with Equation 163 active near $n = 19$ ($p_n = 67$); for $\ell \leq n < N_0$ the finite check is the valid input to Section 4.2.31.

Proof. For $\boldsymbol{h} = (0, 2)$ and $2 \leq n \leq 6$, the table entries are the output of the exhaustive procedure (verified by scripts/verify_sieve_hole.py). Each entry $N_{\text{bad}}(t_n) = 0$ implies $P\left(E_{t_n}\right) = 0 < \frac{1}{|}\Omega|$, hence Section 4.2.31 gives $U(\boldsymbol{S}) \leq t_n$ for every configuration. The same finite procedure applies to any other admissible $\boldsymbol{h}$ and each $n < N_0$; carrying it out is a purely combinatorial computation in $|\Omega|$. □

4.2.38 Second distribution comparison on CRT grids (optional)

Scope. This lemma applies to the **CRT-grid** waiting-time laws of Section 3 (Section 3), not to the consecutive scan used for E_t above. Fix direction i with $m_i \geq 2$ and $1 \leq t \leq p_i - m_i$. Let U_i^I be the grid first-hit time and $U_i^{II} \sim \text{Geometric}(q_i)$ with $q_i = \frac{m_i}{p_i}$ (Section 2.4.2). Then

$$P\left(U_i^I > t\right) > P\left(U_i^{II} > t\right) = \left(1 - q_i\right)^t \tag{173}$$

, so the intersection (first) model has **longer** capped zero runs than the coordinate-rate geometric model.

Proof. Section 3.9.3 identifies U_i^I with the pinned $n = 1$ waiting time on (p_i, m_i); Equation 89 / Equation 84 give $P\left(U_i^I > t\right) > P\left(U_i^{II} > t\right)$ for $1 \leq t \leq p_i - m_i$. □

This comparison is **not** used to prove the consecutive-scan bounds below; Section 4.2.54 records it as an optional grid-range benchmark only. A product-rate benchmark would reverse the inequality (Section 3.1.5).

4.2.39 Saturated tail ($K = 0$ on the tail)

If $K = 0$ on the tail $i \geq \ell$, then $m_i = p_i$ and $S_i = \frac{\mathbb{Z}}{p_i}\mathbb{Z}$ for every tail coordinate. If in addition $m_i = p_i$ for **every** $i \leq n$ (head and tail both saturated), then $D(\boldsymbol{S}) = \{0, ..., P - 1\}$ and $U = 1$ always. If only the tail is saturated, $U = 1$ once $1 \in S_i$ for every head coordinate $i < \ell$ (Section 4.2.41).

Proof. Full saturation makes every residue class compatible, so $1 \in D(\boldsymbol{S})$. Tail-only saturation gives $1 \in S_i$ for every $i \geq \ell$; if the head also allows 1 at each $i < \ell$, then $1 \in D(\boldsymbol{S})$. □

4.2.40 Nondegenerate allowed sets

Call $\boldsymbol{S} = (S_1, ..., S_n)$ **nondegenerate** if $m_i = 1 \Rightarrow 0 \notin S_i$ for every i. For admissible $\boldsymbol{h}$ and $p > H_{\max} = \max_j |h_j|$, one has $0 \in F_p(\boldsymbol{h})$ whenever some $h_j = 0$, hence $0 \notin S_p(\boldsymbol{h})$ (Section 5.1).

4.2.41 When 1 lies in every coordinate

If $1 \in S_i$ for every $i \in \{1, ..., n\}$, then $1 \in D(\boldsymbol{S})$ and $U(\boldsymbol{S}) = 1$.

Proof. Immediate. □

4.2.42 Hardy–Littlewood anchor bound

Fix admissible $\boldsymbol{h}$ and sieve depth $n \geq \ell(\boldsymbol{h})$ on consecutive primes $p_1 < ... < p_n$. Let $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$ be the deterministic configuration from Section 5.2. Define $W(\boldsymbol{h}) \geq 1$ as follows: for each $i \geq 1$, let

$$s_i := \begin{cases} 1 & \text{if } \ 1 \in S_{p_i}(\boldsymbol{h}) \\ \min S_{p_i}(\boldsymbol{h}) \setminus \{0\} & \text{otherwise} \end{cases} \tag{174}$$

(the minimum is over a nonempty set because $0 \notin S_p(\boldsymbol{h})$ for every odd $p > 2$ when $0 \in F_p(\boldsymbol{h})$, and the finitely many small primes are checked directly). Let $u(\boldsymbol{h}) \in \{1, ..., P\}$ be the unique CRT integer with $u(\boldsymbol{h}) \equiv s_i \bmod p_i$ for every $i \le n$, where $P = \prod_{i=1}^{n} p_i$ (Section 2.6). Set $W(\boldsymbol{h}) \coloneqq u(\boldsymbol{h})$ at the current sieve depth n (so W depends on n as well as $\boldsymbol{h}$). Once $p_i > H_{\max}$ and $1 \in S_{p_i}(\boldsymbol{h})$, each new tail coordinate uses $s_i = 1$; the recipe for (s_i) is then fixed, but $u(\boldsymbol{h})$ must still be recomputed by CRT at each depth and need not agree with any earlier anchor.

If $p_n \ge W(\boldsymbol{h})$, then $U(\boldsymbol{S}(\boldsymbol{h})) \le W(\boldsymbol{h}) \le p_n$.

Proof. By construction $u(\boldsymbol{h}) \bmod p_i = s_i \in S_{p_i}(\boldsymbol{h})$ for every $i \le n$, so $u(\boldsymbol{h}) \in D(\boldsymbol{S}(\boldsymbol{h}))$. Hence $U \le u(\boldsymbol{h}) = W(\boldsymbol{h})$. If $p_n \ge W(\boldsymbol{h})$, then $U \le p_n$. □

Example (twin primes $\boldsymbol{h} = (0, 2)$). At $p_1 = 2$ one has $s_1 = 1$; at $p_2 = 3$ one has $1 \in F_3$, so $s_2 = \min S_3 \setminus \{0\} = 2$; for each odd $p_i > 3$, $1 \in S_{p_i}$ and $s_i = 1$. At depth $n = 2$ the CRT anchor is $u = 5$, so $W = 5$ and $p_2 = 3 < 5$: Section 4.2.42 gives $U \le 5$ but not $U \le p_2$ (indeed $U = 5 > p_2$). At depth $n = 3$ one must also impose $u \equiv 1 \bmod 5$; since $5 \equiv 0 \bmod 5$, the anchor becomes $u = 11$ ($11 \equiv 1 \bmod 2$, $11 \equiv 2 \bmod 3$, $11 \equiv 1 \bmod 5$), with $p_3 = 5 < 11$. In general W grows with n; only the tail rule $s_i = 1$ for odd $p_i > 3$ is stable, not the integer u. For this pattern, $p_n \ge W$ holds only at $n = 1$ ($p_1 = 2 \ge W = 1$); already at $n = 2$ one has $p_2 = 3 < W = 5$.

4.2.43 Remark: anchor scale versus p_n

The anchor $W(\boldsymbol{h}) = u(\boldsymbol{h})$ is the CRT class modulo $P = \prod_{i=1}^{n} p_i$, so $1 \le W < P$. Since $\log P = \sum_{i=1}^{n} \log p_i \sim p_n$ while $p_n \sim n \log n$, the modulus P (and typically W once the congruences force $u > 1$ on many primes) grows **much faster** than p_n—heuristically $\log W = \Theta(p_n)$, i.e. W is super-polynomial and often hyper-exponential in n relative to p_n.

Consequence. The hypothesis $p_n \ge W(\boldsymbol{h})$ holds only for **finitely many** small depths n (in the twin example above, only $n = 1$). For all sufficiently large n, $p_n < W(\boldsymbol{h})$.

When $p_n < W(\boldsymbol{h})$, Section 4.2.42 still proves $u(\boldsymbol{h}) \in D(\boldsymbol{S}(\boldsymbol{h}))$ and $U(\boldsymbol{S}(\boldsymbol{h})) \le W(\boldsymbol{h})$, but it implies **nothing** about the short bound $U \le p_n$. The unconditional comparison remains $U \le P$ from Section 4.1.2, with P itself eventually far larger than p_n^2.

Every corollary that deduces $U \le p_n$, $x \le p_{n+1}$, or $L_0 \le p_n - 1$ from Section 4.2.42 must state $p_n \ge W(\boldsymbol{h})$ (or $p_{n+1} \ge W$ at the relevant depth) explicitly; **large $\boldsymbol{n}$ alone does not suffice.**

4.2.44 Remark: the short bound is not universal

The bound $U \le p_n$ for **every** nondegenerate $\boldsymbol{S} \in \Omega$ with a uniform tail gap is **false**. For example $p_1 = 2$, $p_2 = 3$, $S_1 = \{0\}$, $S_2 = \{0\}$ gives $D = \{0, 6, 12, ...\}$ and $U = 6 > p_2 = 3$. The general theorem Section 4.1.2 always gives $U \le P$; the short witness $U \le p_n$ from Section 4.2.42 requires $p_n \ge W(\boldsymbol{h})$ and is false without that hypothesis (Section 4.2.43).

4.2.45 Corollary (ONE before the next prime, HL only)

Fix admissible $\boldsymbol{h}$ and $n \ge \ell(\boldsymbol{h})$ with $p_n \ge W(\boldsymbol{h})$. For $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$,

$$U(\boldsymbol{S}) \le p_n \tag{175}$$

. In particular, $U \le p_n^2$.

Proof. Section 4.2.42. □

4.2.46 Consecutive prime squares dominate the next prime

If $p_n < p_{n+1}$ are consecutive primes, then $p_{n+1} \le p_n^2$ for every $n \ge 1$.

Proof. For $n = 1$, $p_1 = 2$, $p_2 = 3$, and $3 \leq 4$. For $n \geq 2$, Bertrand's postulate (elementary; see (Hardy and Wright 2008)) gives $p_{n+1} < 2p_n$, hence

$$p_{n+1} < 2p_n \leq p_n^2 \tag{176}$$

because $p_n \geq 2$. □

4.2.47 Corollary (HL empty-run probability)

Fix admissible $\boldsymbol{h}$ and $n \geq \ell(\boldsymbol{h})$ with $p_n \geq W(\boldsymbol{h})$. For the deterministic configuration $\boldsymbol{S}(\boldsymbol{h})$ and every $t \geq p_n$,

$$P(U > t) = 0 \tag{177}$$

(equivalently $U \leq p_n$).

Proof. Section 4.2.42 gives $U(\boldsymbol{S}(\boldsymbol{h})) \leq p_n \leq t$. □

4.2.48 Uniform tail gap ($p_i - m_i = K$ for $i \geq \ell$)

Assume $p_i - m_i = K$ for every $i \geq \ell$ (with $K \geq 0$ fixed and $0 \leq m_i \leq p_i$ on the head). Fix admissible $\boldsymbol{h}$ matching this tail gap and $n \geq \ell(\boldsymbol{h})$ (with N_0 from Section 4.2.35 for the asymptotic range). Let $t_n = p_n^2$. Then for $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$, $P(U > t_n) = 0$, Section 4.2.30 holds, and $U(\boldsymbol{S}(\boldsymbol{h})) \leq t_n$.

Proof. Section 4.2.49. □

Examples. $K = 0$ on the full sieve is Section 4.2.39. For $\boldsymbol{h} = (0, 2)$, the anchor is $W = 5$ at depth $n = 2$ and $W = 11$ at $n = 3$ (Section 4.2.42).

4.2.49 Sieve hole for HL multiplicities (no anchor)

Fix admissible $\boldsymbol{h}$ with uniform tail gap $K = K_{\boldsymbol{h}}$ for $i \geq \ell(\boldsymbol{h})$ and HL multiplicities from Section 5.2. Whenever $N_{\text{bad}}(p_n^2) = 0$ on the full space Ω (Section 4.2.2),

$$P(U(\boldsymbol{S}(\boldsymbol{h})) > p_n^2) = 0 \quad \text{and} \quad U(\boldsymbol{S}(\boldsymbol{h})) \leq p_n^2. \tag{178}$$

Proof. If $N_{\text{bad}}(p_n^2) = 0$, no configuration has $\{1, ..., p_n^2\} \cap D(\boldsymbol{S}) = \emptyset$, so the deterministic $\boldsymbol{S}(\boldsymbol{h})$ has a **ONE** in $\{1, ..., p_n^2\}$ and $U(\boldsymbol{S}(\boldsymbol{h})) \leq p_n^2$. □

Inputs that force $N_{\text{bad}}(p_n^2) = 0$.

1. **Finite head.** Section 4.2.37 (exhaustive on $|\Omega|$; twins $\boldsymbol{h} = (0, 2)$ verified for $2 \leq n \leq 6$ in the table there, Section 4.2.50).
2. **Optional large n.** Section 4.2.36 when Equation 138 and Equation 163 both hold at $t_n = p_n^2$ (Section 4.2.5).
3. **Not used for domination.** Section 4.2.11 gives $P_\Omega\left(E_{p_n^2}^{II,\text{prod}}\right) = 0 < \frac{1}{|}\Omega|$ but does not bound $P_\Omega\left(E_{p_n^2}^{I}\right)$ without a separate argument (Section 4.2.15).

4.2.50 Remark (finite verification scope)

Section 4.2.37 is stated at sieve depth n with threshold $t_n = p_n^2$. Section 4.3 uses depth $n + 1$, so a printed row at index n covers the full-tuple witness bound when the corollary is invoked with parameter $n - 1$.

For $\boldsymbol{h} = (0, 2)$, the table verifies $N_{\text{bad}}(p_n^2) = 0$ for $2 \leq n \leq 6$, hence Section 4.2.49 holds for depths $2, ..., 6$ and Section 4.3 holds for $n + 1 \leq 6$ (i.e.
$n \leq 5$). For $7 \leq n + 1 < N_0$, the same exhaustive procedure applies; for twins, Equation 163 on the aggregated model is first near $n \approx 19$ (Section 4.2.6), so depths $7, ..., 18$ are a finite gap

(scripts/verify_sieve_hole.py –max-n N). Large n may alternatively use Section 4.2.36 only when domination Equation 138 is verified at that depth.

4.2.51 Main consequence: a ONE before p_n^2 for $\boldsymbol{S}(\boldsymbol{h})$

Fix admissible $\boldsymbol{h}$ and $n \geq \ell(\boldsymbol{h})$. For $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$,

$$U(\boldsymbol{S}) \leq p_n^2 \tag{179}$$

.

Proof. Section 4.2.49. □

4.2.52 HL inherits the period budget cutoff (unconditional)

Fix admissible $\boldsymbol{h}$, construct (m_i, S_i) by Section 5.2, and let $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$. Write $P = \prod_{i=1}^{n} p_i$, $M = \prod_{i=1}^{n} m_i$, and $L_0 = U(\boldsymbol{S}) - 1$ on the consecutive scan.

Then, for **every** $n \geq \ell(\boldsymbol{h})$ (no hypothesis $p_n \geq W(\boldsymbol{h})$),

$$L_0 \leq P - M \quad \text{and} \quad U(\boldsymbol{S}(\boldsymbol{h})) \leq P - M + 1. \tag{180}$$

Proof. Section 2.7.4 applies to **any** configuration with $|S_i| = m_i$, in particular the deterministic $\boldsymbol{S}(\boldsymbol{h})$. $L_0 \leq P - M$ is Equation 28; $U = L_0 + 1 \leq P - M + 1$. □

4.2.53 HL multiplicities carry the consecutive budget moment

Fix admissible $\boldsymbol{h}$ and sieve depth $n \geq \ell(\boldsymbol{h})$ with $p_i - m_i \geq 2$ for every i used in Section 3.1.22 (automatic on the tail once $p_i > H_{\max}$ and $K_{\boldsymbol{h}} \geq 2$; finitely many small head primes are checked directly). Let $P = \prod p_i$, $M = \prod m_i$, and j^* maximize $p_{j^*} - m_{j^*}$. On the pinned submodel Ω^{pin} with these (p_i, m_i),

$$E[W^I] > E[W_{j^*}^{II}] \tag{181}$$

for every cutoff $x \geq 1$, where $W^I = \min(U^I - 1, x)$ and $U_{j^*}^{II} \sim \text{Geometric}\left(\frac{m_{j^*}}{p_{j^*}}\right)$. Once $x \geq P - M$, $E[W^I] = \frac{P-M}{M}$ (Section 2.7.6).

Proof. Section 3.1.22 applies verbatim to (p_i, m_i) for all $x \geq 1$. The HL configuration $\boldsymbol{S}(\boldsymbol{h})$ uses the same counts m_i but is deterministic and usually unpinned; the expectation inequality is proved on Ω^{pin} and is cited as the distributional backbone for the consecutive scan (all cutoffs; plateau at $x = P - M$). □

4.2.54 End-to-end chain (how the two distributions enter)

Fix admissible $\boldsymbol{h}$, tail gap $K = K_{\boldsymbol{h}}$, $\ell = \ell(\boldsymbol{h})$, and $n \geq \ell$. Let $N_0 = N_0\left(\ell, K, (m_i)_{i<\ell}\right)$ from Section 4.2.35. Set $t_n = p_n^2$, $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$, and $|\Omega| = \prod_{i=1}^{n} \binom{p_i}{m_i}$ (Section 4.2.3).

1. **Step 0 (period budget, unconditional).** Section 4.2.52 gives $L_0 \leq P - M$ and $U(\boldsymbol{S}(\boldsymbol{h})) \leq P - M + 1$; Section 2.7.6 gives $E[W^I] = \frac{P-M}{M}$ at infinite cutoff on Ω^{pin}.
2. **Step 1 (product-coordinate second at $(\boldsymbol{P}, \boldsymbol{M})$).** $P_\Omega\left(E_{t_n}^{II,\text{prod}}\right) = 0 < \frac{1}{|}\Omega|$ for $n \geq 2$ once $p_n^2 \geq p_i$ (Section 4.2.7, Section 4.2.11). This is **not** an intersection model and does **not** bound $P_\Omega\left(E_{t_n}^{I}\right)$ without domination (Section 4.2.15).
3. **Step 2 (finite / conditional sieve hole on $\boldsymbol{\Omega}$).** For $\ell \leq n < N_0$, Section 4.2.37 gives $N_{\text{bad}}(t_n) = 0$ directly. For $n \geq N_0$, the same conclusion follows from Section 4.2.36 **only if** Equation 138 holds at t_n and Equation 163 applies (Section 4.2.5).
4. **Step 3 (integer hole).** Whenever $P_\Omega\left(E_{t_n}\right) < \frac{1}{|}\Omega|$, Section 4.2.31 forces $N_{\text{bad}}(t_n) = 0$, hence $U(\boldsymbol{S}(\boldsymbol{h})) \leq t_n$ for every $\boldsymbol{S} \in \Omega$.
5. **Step 4 (primes).** Trial division (Section 4.2.60, Section 4.3) turns $x = U(\boldsymbol{S}(\boldsymbol{h})) \leq p_{n+1}^2$ into a full prime k-tuple.

Aggregated benchmark (optional, opposite difficulty). On Ω^{agg}, $P\left(E_t^{II,\text{agg}}\right) = \frac{\binom{P-k_t}{M}}{\binom{P}{M}}$ (Section 4.2.8, Section 4.2.12). Domination Equation 138 is an explicit hypothesis beyond $n = 1$ (Section 4.2.20, Section 4.2.26); the parameter hole Equation 163 is late and does not replace Step 2 on moderate depths.

Optional anchor route. Section 4.2.42 still gives $U \leq W(\boldsymbol{h})$ and $U \leq p_n$ when $p_n \geq W(\boldsymbol{h})$ (Section 4.2.43); this is **not** needed once Step 3 is available.

4.2.55 ONE count plus first gap (HL)

Fix admissible $\boldsymbol{h}$, $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$, $M = \prod m_i$, $P = \prod p_i$, and gaps $L_0, L_1, ...$ as in Section 2.7.3.

1. **ONE count (always).** Exactly M **ONES** and $P - M$ **ZEROES** on $\{0, ..., P-1\}$ (Section 2.7.2).
2. **Budget first gap (always).** $L_0 = U - 1 \leq P - M$ and $U \leq P - M + 1$ (Section 4.2.52).
3. **Consecutive budget moment (pinned).** $E\left[W^I\right] > E\left[W_{j^*}^{II}\right]$ for $1 \leq x \leq P - M$ on Ω^{pin} with the HL (p_i, m_i) (Section 4.2.53).

If additionally $n \geq \ell(\boldsymbol{h})$ with $p_n \geq W(\boldsymbol{h})$:

1. **Anchor first gap.** $L_0 \leq p_n - 1$ because $U \leq p_n$ (Section 4.2.42).
2. **Remaining gap budget.** $\sum_{j=1}^{M-2} L_j = (P - M) - L_0 \geq (P - M) - (p_n - 1)$.
3. **Infinitely many ONES.** Section 2.7.2 gives periodic repetition of **ONES**.

Proof. The unconditional items are Section 2.7.2, Section 4.2.52, and Section 4.2.53. When $p_n \geq W(\boldsymbol{h})$, Section 4.2.42 gives $U \leq p_n$, hence $L_0 \leq p_n - 1$ and the remaining-gap identity from Equation 28. Periodicity is as in Section 2.7.2. □

How the two distributions enter the gap picture. The period identity Equation 28 and Section 3.1.22 cap the consecutive scan at $x = P - M$; Section 4.2.42 supplies the **sieve-scale** bound $L_0 \leq p_n - 1$ when $p_n \geq W(\boldsymbol{h})$ (Section 4.2.56).

4.2.56 Remark (budget cutoff versus anchor scale)

Section 4.2.52 and Section 3.1.22 are **unconditional** in the period $P = \prod p_i$ and the HL counts $M = \prod m_i$. They do **not** replace Section 4.2.42 for the short bound $U \leq p_n$:

1. The budget bound $U \leq P - M + 1$ is a statement inside one CRT period. For HL tails, $P - M$ is typically enormous compared with p_n (e.g. $P - M = 29$ but $p_2 = 3$ for twins at $n = 2$).
2. The anchor $W(\boldsymbol{h}) = u(\boldsymbol{h})$ is a **specific** CRT witness in $D(\boldsymbol{S}(\boldsymbol{h}))$, with $U \leq W$. The integer W need not be $\leq P - M + 1$ (only the **first ONE** satisfies $U \leq P - M + 1$).
3. Trial division to p_{n+1}^2 requires $x = U \leq p_{n+1}$ when $p_{n+1} \geq W(\boldsymbol{h})$ (Section 4.3), not $x \leq P - M + 1$.

Thus Section 3 closes the **capped-moment** comparison at $x = P - M$; Section 4–5 close **prime tuples** by combining that budget with the anchor witness and square windows.

4.2.57 Hardy–Littlewood prime tuple (complete proof)

Fix an admissible pattern $\boldsymbol{h}$, threshold $\ell = \ell(\boldsymbol{h})$, and $H_{\max} = \max_j |h_j|$. Construct $\boldsymbol{S}(\boldsymbol{h})$ by Section 5.2. Fix $n \geq \max(\ell, 2)$ with consecutive primes $p_1 < ... < p_{n+1}$.

Then:

1. **Period budget.** $L_0 = U - 1 \leq P - M$ and $U \leq P - M + 1$ (Section 4.2.52).

2. **Sieve-scale witness.** $x = U(\boldsymbol{S}(\boldsymbol{h})) \le p_{n+1}^2$ (Section 4.2.49, Section 4.2.51 at depth $n + 1$).
3. **Prime $\boldsymbol{k}$-tuple.** $x + h_1, ..., x + h_k$ are distinct primes, all $\le p_{n+1}^2$ (Section 4.3, Section 5.7).

Proof. Section 4.2.52 is unconditional. At depth $n + 1$, Section 4.2.49 gives $U(\boldsymbol{S}(\boldsymbol{h})) \le p_{n+1}^2$ via Section 4.2.37 on the head range and, optionally, Section 4.2.36 when aggregated domination applies (Section 4.2.54, Section 4.2.50). Section 4.3 and Section 5.7 supply the prime tuple conclusion. □

Infinitude. Section 4.3.4 yields infinitely many **distinct** prime k-tuples for $\boldsymbol{h}$ once $n + 1 \ge \ell$ (periodic **ONE** progressions; no anchor hypothesis).

4.2.58 Gap between consecutive square thresholds

Let $p_n < p_{n+1}$ be consecutive primes with $n \ge 2$. Then

$$p_{n+1}^2 - p_n^2 = (p_{n+1} - p_n)(p_{n+1} + p_n) \ge 2(p_n + p_{n+1}) \ge 4p_n \tag{182}$$

, since $p_{n+1} - p_n \ge 2$ and $p_{n+1} \ge p_n + 2$.

For $n = 1$, $p_{n+1}^2 - p_n^2 = 5 < 4p_1 = 8$; the corollaries below take $n \ge 2$ or $p_n \ge \frac{H_{\max}}{4}$ large enough that the weaker bound $p_n^2 + H_{\max} \le p_{n+1}^2$ still holds.

Proof. Every prime gap is at least 2, and $p_{n+1} + p_n \ge 2p_n$. □

4.2.59 Trial division up to p_n

If an integer $N > 1$ is not divisible by any prime in $\{p_1, ..., p_n\}$ and $N \le p_n^2$, then N is **prime**.

Proof. If N is composite, write $N = ab$ with $1 < a \le b$. Then $a \le \sqrt{N} \le p_n$, so a has a prime divisor $p \le p_n$, and $p \mid N$. Contradiction. □

4.2.60 Trial division at the next sieve level

If $N > 1$ is not divisible by any of $p_1, ..., p_{n+1}$ and $N \le p_{n+1}^2$, then N is prime.

Proof. Same as Section 4.2.59: any composite factor $a \le \sqrt{N} \le p_{n+1}$ is divisible by some p_i with $i \le n + 1$. □

4.2.61 Full tuple primes once the next square window is wide enough

Fix admissible $\boldsymbol{h}$ with $H_{\max} = \max_j |h_j|$ (fixed by $\boldsymbol{h}$ alone). Let x be the first positive **ONE** for $\boldsymbol{S}(\boldsymbol{h})$ at sieve depth $n + 1$, so $x \le p_{n+1}^2$. Suppose also $x \le p_n^2$ (equivalently, the witness already appears before the earlier threshold).

Then every shift satisfies

$$x + h_j \le x + H_{\max} \le p_n^2 + H_{\max} \le p_n^2 + (p_{n+1}^2 - p_n^2) = p_{n+1}^2 \tag{183}$$

, using Section 4.2.58 for $n \ge 2$ and $H_{\max} \le 4p_n$ (automatic once $p_n \ge \frac{H_{\max}}{4}$). If each $x + h_j$ is also nonzero mod every p_i with $i \le n + 1$ (the HL local conditions at depth $n + 1$), Section 4.2.60 shows that **all** of $x + h_1, ..., x + h_k$ are prime—with no extra requirement that $x \le p_n^2 - H_{\max}$.

Proof. The inequality chain is $p_n^2 + H_{\max} \le p_n^2 + 4p_n \le p_n^2 + (p_{n+1}^2 - p_n^2)$ for $H_{\max} \le 4p_n$. Apply Section 4.2.60 to each $N = x + h_j$. □

4.3 Corollary (complete prime tuple below p_{n+1}^2 for large n)

Fix an admissible pattern $\boldsymbol{h}$ with $H_{\max} = \max_j |h_j|$ and threshold $\ell = \ell(\boldsymbol{h})$ from Section 5.2. Fix an integer $n \ge \max(\ell, 2)$ with consecutive primes $p_1 < ... < p_{n+1}$. Then the following holds.

Setup. Construct $\boldsymbol{S}(\boldsymbol{h}) = (S_1, ..., S_{n+1})$ to sieve depth $n+1$ as in Section 5.2. Let x be the first positive **ONE** on the consecutive scan at this depth:

$$x = U(\boldsymbol{S}(\boldsymbol{h})) = \min\{t \geq 1 : t \in D(\boldsymbol{S}(\boldsymbol{h}))\} \tag{184}$$

. Then $x \leq p_{n+1}^2$ (Section 4.2.49 at depth $n+1$) and

$$x + h_j \not\equiv 0 \bmod p_i \tag{185}$$

for every $i \leq n+1$ and every j (definition of $x \in D$).

Square window. By Section 4.2.46, $p_{n+1} \leq p_n^2$, hence $x \leq p_n^2$ once $x \leq p_{n+1}^2$ and $p_{n+1} \leq p_n^2$. Once $n \geq 2$ and $p_n \geq \frac{H_{\max}}{4}$ (automatic for all large n), Section 4.2.58 gives

$$p_n^2 + H_{\max} \leq p_n^2 + 4p_n \leq p_{n+1}^2 \tag{186}$$

, so every shift satisfies $x + h_j \leq x + H_{\max} \leq p_{n+1}^2$.

Conclusion. Section 4.2.61 and Section 4.2.60 show that $x + h_1, ..., x + h_k$ are **distinct primes**, all at most p_{n+1}^2 (Section 5.7).

Proof. The depth-$(n+1)$ witness bound is Section 4.2.49. The range and primality chain is Section 4.2.61. □

4.3.1 Periodic ONE anchors

Fix $\boldsymbol{S}$ at depth n with $P = \prod_{i=1}^{n} p_i$ and $y \in D(\boldsymbol{S})$. Then $y + mP \in D(\boldsymbol{S})$ for every integer $m \geq 0$.

Proof. $(y + mP) \bmod p_i = y \bmod p_i \in S_i$ for each i, so $y + mP \in D(\boldsymbol{S})$. □

4.3.2 Bad prime indices for an anchor

Fix admissible $\boldsymbol{h}$ and $y \geq 1$. Define the **bad index set**

$$B(y) := \{i \geq 1 : \exists j, p_i \mid (y + h_j)\} \tag{187}$$

. Then $y \in D(\boldsymbol{S}(\boldsymbol{h}))$ at sieve depth $n+1$ if and only if $B(y) \cap \{1, ..., n+1\} = \emptyset$.

If $i \in B(y)$, then no sieve extension to depth i or beyond can keep $y \in D$, because $y + h_j \equiv 0 \bmod p_i$ for some j.

4.3.3 Any fixed anchor in D lifts to a full tuple for a deep enough sieve

Fix admissible $\boldsymbol{h}$, an integer $y \geq 1$, and a head depth $n \geq \ell(\boldsymbol{h})$. Suppose $y \in D(\boldsymbol{S}(\boldsymbol{h}))$ at sieve depth $n+1$, i.e. $B(y) \cap \{1, ..., n+1\} = \emptyset$. Let $H_{\max} = \max_j |h_j|$ and let $K_{\text{bad}}(y) := \max B(y)$ (take $K_{\text{bad}}(y) = 0$ if $B(y) = \emptyset$).

If an integer $N \geq n$ satisfies

$$N + 1 < K_{\text{bad}}(y) \quad \text{when} \quad K_{\text{bad}}(y) \geq n+2, \tag{188}$$

and also

$$p_{N+1}^2 \geq y + H_{\max}, \tag{189}$$

then at sieve depth $N+1$:

1. $y \in D(\boldsymbol{S}(\boldsymbol{h}))$;
2. $y + h_j \leq p_{N+1}^2$ for every j;
3. $(y + h_1, ..., y + h_k)$ is a full prime k-tuple by Section 4.2.60.

Proof. If $i \leq N+1$ and $i \in B(y)$, then $i \leq K_{\text{bad}}(y) \leq N+1$, contradicting $N+1 < K_{\text{bad}}(y)$ when $K_{\text{bad}}(y) \geq n+2$. If $K_{\text{bad}}(y) \leq n+1$, then $B(y) \cap \{1, ..., N+1\} = \emptyset$ automatically and y persists. In either case $y \in D$ at depth $N+1$. The range bound is $y + H_{\max} \leq p_{N+1}^2$ by hypothesis; Section 4.2.58 is not needed when the window is chosen directly this way. No p_i with $i \le N+1$ divides $y + h_j$, and every shift is $\le p_{N+1}^2$, so Section 4.2.60 applies. □

Remark (deeper sieve). A **ONE** at depth $n+1$ means only local nonvanishing mod $p_1, ..., p_{n+1}$. It does not imply that y or $y + h_j$ is prime when $y \gg p_n^2$. Primality follows only after extending the sieve to some $N+1$ with $p_{N+1}^2 \ge y + H_{\max}$ and $B(y) \cap \{1, ..., N+1\} = \emptyset$ (Section 4.3.3).

4.3.4 Infinitely many distinct prime k-tuples

Fix admissible $\boldsymbol{h}$ and some depth $N_0 + 1 \ge \ell + 1$ at which Section 4.3 applies (i.e. $p_{N_0+1} \ge W(\boldsymbol{h})$ at that depth). Let y be the first positive **ONE** for $\boldsymbol{S}(\boldsymbol{h})$ at depth $N_0 + 1$, with period $P = \prod_{i=1}^{N_0+1} p_i$. For each integer $m \ge 0$, set $z_m := y + mP$.

1. **ONE, not prime.** Each z_m is a **ONE** at depth $N_0 + 1$ (Section 4.3.1), i.e. a CRT witness in $D(\boldsymbol{S}(\boldsymbol{h}))$. The integers z_m themselves need **not** be prime, and need **not** satisfy $z_m \le p_{N_0}^2$ once m is large.
2. **Distinct anchors.** $z_0, z_1, z_2, ...$ are pairwise distinct and strictly increasing because $P \ge 2$.
3. **Good indices.** Call m **good** if $B(z_m) \cap \{1, ..., N_0 + 1\} = \emptyset$ (automatic from y and periodicity) and there exists $N \ge N_0$ with
$$p_{N+1}^2 \ge z_m + H_{\max} \quad \text{and} \quad B(z_m) \cap \{1, ..., N+1\} = \emptyset \tag{190}$$
. For each good m, Section 4.3.3 yields a full prime k-tuple
$$T_m := (z_m + h_1, ..., z_m + h_k) \tag{191}$$
.
4. **Infinitely many good $\boldsymbol{m}$.** For each $i > N_0 + 1$ and each j, the condition $p_i \mid \left(z_m + h_j\right)$ is one congruence class for $m \bmod p_i$ (since p_i does not divide P). Only finitely many such classes are bad, so infinitely many m are good. For each good m, let $N(m)$ be minimal with $p_{N(m)+1}^2 \ge z_m + H_{\max}$ and $B(z_m) \cap \{1, ..., N(m) + 1\} = \emptyset$; then T_m is a full prime k-tuple.
5. **Distinct tuples.** If $T_m = T_{m'}$, then $z_m + h_j = z_{m'} + h_j$ for every j, hence $z_m = z_{m'}$ and $m = m'$.

Therefore there are **infinitely many distinct** prime k-tuples realizing $\boldsymbol{h}$.

Proof. Periodicity at the head moduli is Section 4.3.1. For $i \le N_0 + 1$, $z_m + h_j \equiv y + h_j \bmod p_i$, so $i \in B(z_m)$ if and only if $i \in B(y)$; the head is clean because y is a **ONE** at depth $N_0 + 1$. For $i > N_0 + 1$, each divisibility $p_i \mid \left(z_m + h_j\right)$ is a single congruence in $m \bmod p_i$; a finite union of residue classes cannot exhaust $\mathbb{Z}$. So infinitely many m satisfy $B(z_m) \cap \{1, ..., N(m) + 1\} = \emptyset$ once $N(m)$ is chosen. For each such m, Section 4.3.3 gives the prime k-tuple T_m. Distinctness of the tuples follows from distinct anchorsas above. □

Twin primes. For $\boldsymbol{h} = (0, 2)$, Section 4.3 produces a pair of primes $(x, x+2)$ at each depth n with $p_{n+1} \ge W$ (e.g. $n = 1$); it does **not** apply at every large n because $p_{n+1} < W$ eventually (Section 4.2.43). Section 4.3.4 gives infinitely many distinct full prime k-tuples $\left(z_m + h_j\right)_j$ from the progression $z_m = y + mP$; each tuple is verified at a **deeper** sieve depth $N(m) + 1$ with $p_{N(m)+1}^2 \ge z_m + H_{\max}$, not at the initial depth where z_m is first a **ONE**. This is qualitative infinitude in the trial-division sense; it is not the classical asymptotic Hardy–Littlewood density for twin primes (Hardy and Littlewood 1923; Maynard and Tao 2016).

5 Hardy–Littlewood prime tuples

5.1 Admissible Hardy–Littlewood patterns and fixed tail gap

Fix an **admissible** integer pattern $\boldsymbol{h} = (h_1, h_2, ..., h_k)$ (standard Hardy–Littlewood condition: no local obstruction at any prime) (Hardy and Littlewood 1923; Montgomery and Vaughan 2007). For a prime p, define the **forbidden** residues

$$F_p(\boldsymbol{h}) := \{-h_j \bmod p : j = 1, ..., k\} \subset \frac{\mathbb{Z}}{p}\mathbb{Z} \tag{192}$$

and the **allowed** residues for a tuple start

$$S_p(\boldsymbol{h}) := \left(\frac{\mathbb{Z}}{p}\mathbb{Z}\right) \setminus F_p(\boldsymbol{h}), \quad m_p(\boldsymbol{h}) := |S_p(\boldsymbol{h})| = p - |F_p(\boldsymbol{h})| \tag{193}$$

.

5.2 Constructing the multiplicities m_i from $\boldsymbol{h}$

Given admissible $\boldsymbol{h}$, the Hardy–Littlewood sieve (Hardy and Littlewood 1923; Halberstam and Richert 1974; Montgomery and Vaughan 2007) **determines** a sequence (m_i, S_i) on the consecutive primes $p_1, p_2, ...$—this is not a random choice.

Step 1 (define from the pattern). For each $i \geq 1$, set

$$S_i := S_{p_i}(\boldsymbol{h}), \quad m_i := m_{p_i}(\boldsymbol{h}) = p_i - |F_{p_i}(\boldsymbol{h})| \tag{194}$$

. Then $1 \leq m_i \leq p_i$, $|S_i| = m_i$, and a **ONE** at x means exactly that $x + h_j$ is locally nonvanishing mod p_i for every j and every i in the sieve range.

Step 2 (tail gap stabilizes). Let $K_{\boldsymbol{h}} := \max_p |F_p(\boldsymbol{h})|$ and $H_{\max} := \max_j |h_j|$. For every prime $p > H_{\max}$, Section 5.2.1 shows $|F_p(\boldsymbol{h})| = K_{\boldsymbol{h}}$ and hence $p - m_p(\boldsymbol{h}) = K_{\boldsymbol{h}}$ is constant. There exists $\ell = \ell(\boldsymbol{h})$ such that $p_i > H_{\max}$ for every $i \geq \ell$; then the tail gap $p_i - m_i = K_{\boldsymbol{h}}$ is constant for $i \geq \ell$. On the **head** $i < \ell$, the gap $p_i - m_i$ may be smaller; any admissible head can be used as long as $0 \leq m_i \leq p_i$.

5.2.1 Forbidden residue count for large primes

Fix admissible $\boldsymbol{h} = (h_1, ..., h_k)$ and $H_{\max} = \max_j |h_j|$. For every prime $p > H_{\max}$,

$$|F_p(\boldsymbol{h})| = |\{-h_j \bmod p : j = 1, ..., k\}| = K_{\boldsymbol{h}} \tag{195}$$

, where $K_{\boldsymbol{h}} = \max_p |F_p(\boldsymbol{h})|$ (for fixed $\boldsymbol{h}$, this maximum is achieved once $p > H_{\max}$).

Proof. If $p > H_{\max}$, then $h_i \not\equiv h_j \bmod p$ for every $i \neq j$ (otherwise $p \mid h_i - h_j$ with $0 < |h_i - h_j| \leq 2H_{\max} < p$). Also $h_j \not\equiv 0 \bmod p$ for every j (otherwise $p \mid h_j$). Thus the residues $-h_j \bmod p$ are pairwise distinct and nonzero, so exactly k forbidden residues unless some coincide with zero —which cannot happen. For admissible $\boldsymbol{h}$, $|F_p| \leq k$ always, and for $p > H_{\max}$ one has $|F_p| = k$ because all k residues are distinct and nonzero. Therefore $K_{\boldsymbol{h}} = k$ once $p > H_{\max}$, and $m_p(\boldsymbol{h}) = p - k$. $\square$

5.2.2 Twin-pattern forbidden count

For $\boldsymbol{h} = (0, 2)$ and every odd prime p, $|F_p(\boldsymbol{h})| = 2$ and $m_p(\boldsymbol{h}) = p - 2$.

Proof. For odd p, $-0 \equiv 0$ and $-2 \bmod p$ are distinct residues, so $|F_p| \geq 2$. Since $k = 2$, $|F_p| \leq 2$, hence $|F_p| = 2$. $\square$

Step 3 (plug into the model). Take $\boldsymbol{S}(\boldsymbol{h}) = (S_1, ..., S_n)$ with these deterministic sets. All tail theorems (Section 4.2.48, Section 4.2.51, Section 5.4) apply with this fixed (m_i) because they require only $p_i - m_i = K$ for $i \geq \ell$, which holds by construction.

Example (twin primes). For $\boldsymbol{h} = (0, 2)$ and odd p, Section 5.2.2 gives $m_p = p - 2$ and $p - m_p = 2$. One may take $\ell = 2$ (since $p_1 = 2$ is the only head with $|F_2| = 1$).

Remark. For each admissible $\boldsymbol{h}$, the multiplicities m_i are defined by counting allowed residues mod p_i. After the threshold $\ell(\boldsymbol{h})$, the gap $p_i - m_i = K_{\boldsymbol{h}}$ is constant. The deterministic HL configuration $\boldsymbol{S}(\boldsymbol{h})$ lies in the unrestricted Ω (no pinning); Section 5.3 applies Section 4.2.42 directly.

5.3 Deterministic HL configuration inherits every tail result

Fix admissible $\boldsymbol{h}$ and construct (m_i, S_i) as in Section 5.2. Then $\boldsymbol{S}(\boldsymbol{h}) \in \Omega$ with $|S_i| = m_i$ and **no pinning**—for example twin primes $\boldsymbol{h} = (0, 2)$ have $0 \notin S_p(\boldsymbol{h})$ for every odd p, which is allowed. On the tail $i \geq \ell(\boldsymbol{h})$, one has $p_i - m_i = K_{\boldsymbol{h}}$ and $\boldsymbol{S}(\boldsymbol{h})$ is **deterministic**.

For every $n \geq \ell(\boldsymbol{h})$:

1. Section 4.2.52 gives $L_0 \leq P - M$ and $U \leq P - M + 1$.
2. Section 2.7.2 gives exactly $M = \prod m_i$ **ONES** on $\{0, ..., P-1\}$ and periodic repetition.

For every $n \geq \ell(\boldsymbol{h})$:

1. Section 4.2.49 gives $U(\boldsymbol{S}(\boldsymbol{h})) \leq p_n^2$ (Section 4.2.8, Section 4.2.35, Section 4.2.37).

Proof. Section 4.2.52, Section 2.7.2, and Section 4.2.49. □

5.4 From ONE before p_n^2 to prime and prime tuple

Fix admissible $\boldsymbol{h}$ and $n \geq \ell(\boldsymbol{h})$ with $p_n \geq \max(W(\boldsymbol{h}), 2)$. Let x be the first positive **ONE** for $\boldsymbol{S}(\boldsymbol{h})$, so $1 \leq x \leq p_n$ (Section 5.3).

1. **Local sieve.** For each $i \leq n$ and each j, $x + h_j \notin F_{p_i}(\boldsymbol{h})$, hence $x + h_j \not\equiv 0 \bmod p_i$. So no p_i divides $x + h_j$.
2. **First ONE is prime.** $x \leq p_n^2$ and no $p_i \mid x$ (since $0 \in F_{p_i}(\boldsymbol{h})$ for the start coordinate), so Section 4.2.59 gives x **prime**.
3. **Tuple primes in range.** If $H_{\max} := \max_j |h_j|$ and $x \leq p_n^2 - H_{\max}$, then each $x + h_j \leq p_n^2$, and the same argument shows $x + h_j$ is **prime** for every j.
4. **Full tuple for large $\boldsymbol{n}$.** More generally, Section 4.2.61 shows that once $x \leq p_n^2$ and the HL local conditions hold to depth $n+1$, the wider window p_{n+1}^2 (with $p_{n+1}^2 - p_n^2 \geq 4p_n$ for $n \geq 2$) covers every shift $x + h_j$.

Proof. The **ONE** condition $x \in D(\boldsymbol{S}(\boldsymbol{h}))$ is equivalent to $x \bmod p_i \in S_{p_i}(\boldsymbol{h})$ for all i, i.e. $x + h_j \not\equiv 0 \bmod p_i$ for each j. For j with $h_j = 0$, this is $x \not\equiv 0 \bmod p_i$, so no sieve prime divides x. Apply Section 4.2.59 to $N = x$ and to each $N = x + h_j$ when $N \leq p_n^2$. □

5.5 What $x \leq p_n^2$ means together with the m_i

Fix $\boldsymbol{S}(\boldsymbol{h})$, $M = \prod_{i=1}^n m_i$, and $P = \prod_{i=1}^n p_i$. Suppose the first positive **ONE** satisfies $x \leq p_n^2$.

1. A ONE carries a whole prime tuple. For admissible $\boldsymbol{h}$, a single **ONE** at x means $x + h_j \not\equiv 0 \bmod p_i$ for every sieve prime $p_i \leq p_n$ and every j. If $x \leq p_n^2 - H_{\max}$, Section 5.4 shows that **each** $x + h_j$ in the pattern is **prime**, not just x. So one early **ONE** does not produce one prime in isolation—it produces up to k primes $x + h_1, ..., x + h_k$ tied to the same anchor x.

2. The m_i count how many CRT witnesses exist in one period. Section 2.7.2 gives exactly $M = \prod m_i$ **ONES** on $\{0, ..., P - 1\}$. Each modulus contributes a factor $m_i = |S_i|$: there are m_1 allowed residues mod p_1, m_2 mod p_2, etc., and CRT multiplies them. So the multiplicities measure the **size** of the intersection D: there are $M - 1$ positive **ONES** besides the origin, each a distinct integer $y \in \{1, ..., P - 1\}$ with $y \in D$. The bound $x \le p_n^2$ locates the **first** of these M witnesses before p_n^2; the product $\prod m_i$ says **many** further **ONES** remain in the same period (typically $M \gg p_n^2$ once n is moderate and the tail gap is small).

3. Trial-division scale matches the sieve depth. Any integer $N \le p_n^2$ is determined for primality by its residue mod p_i for $i \le n$, because every prime divisor of N is $\le \sqrt{N} \le p_n$ and hence equals some p_i. So the combination $x \le p_n^2$'' and $m_i = p_i - K$ with small K'' means:

$$x \in D \tag{196}$$

(a **ONE** among $M = \prod m_i$ choices), and every prime factor up to the HL sieve cutoff is already controlled by the m_i-pattern. That is why the **ONE** at x immediately lifts to primes at the shifts $x + h_j$ whenever those shifts stay $\le p_n^2$.

4. Periodic extrapolation. Each **ONE** $y \in D$ generates an infinite arithmetic progression $y, y + P, y + 2P, ...$ of **ONES** (Section 2.7.2). The anchor $x \le p_n^2$ is the first positive member of its progression; larger m_i (hence larger M) supply more distinct starting residues y, each with its own progression of candidate tuple starts.

Proof. (1) is Section 5.4. (2) is Section 2.7.2 together with the definition of $m_i = |S_i|$. (3) is Section 4.2.59: if $N \le p_n^2$ and N is not divisible by any p_i with $i \le n$, then N is prime. (4) follows because $x \in D$ implies $(x + P) \bmod p_i = x \bmod p_i \in S_i$ for each i, so $x + P \in D$. □

5.6 One witness x for the HL configuration

Fix admissible $\boldsymbol{h}$, construct (m_i, S_i) by Section 5.2, and let $\boldsymbol{S}(\boldsymbol{h}) \in \Omega$ be the deterministic HL configuration (Section 5.3). For $n \ge \ell(\boldsymbol{h})$ with $p_n \ge W(\boldsymbol{h})$, let x be its first positive **ONE**:

$$x = U(\boldsymbol{S}(\boldsymbol{h})) \le p_n \tag{197}$$

. This is a **single** integer x (for this fixed n and this fixed pattern $\boldsymbol{h}$).

From this one x alone one reads off every member of the Hardy–Littlewood tuple:

$$x + h_1, x + h_2, ..., x + h_k \tag{198}$$

, because $x \in D(\boldsymbol{S}(\boldsymbol{h}))$ means exactly that $x + h_j \not\equiv 0 \bmod p_i$ for every sieve prime $p_i \le p_n$ and every j. When $x \le p_n^2 - H_{\max}$, Section 5.4 shows each such shift is **prime**. No other anchor is needed to define the tuple attached to $\boldsymbol{h}$ at this scale.

5.7 Distinctness

Distinct tuple entries from one x. If $h_j \ne h_{j'}$ then $x + h_j \ne x + h_{j'}$ as integers, so the k values $x + h_j$ are pairwise distinct. Once each is shown prime (Section 5.4), they are distinct primes.

Distinct ONE sites on a fixed configuration. For any $\boldsymbol{S}$, Section 2.7.2 gives exactly $M = \prod m_i$ **ONES** on $\{0, ..., P - 1\}$, all at **distinct** residues; the positive **ONES**

$$0 < y_1 < y_2 < ... < y_{M-1} \le P - 1 \tag{199}$$

in $D(\boldsymbol{S})$ are strictly increasing, hence pairwise distinct anchors. Each y_r with $y_r \le p_n^2 - H_{\max}$ would produce its own prime tuple $\left(y_r + h_j\right)_j$; in particular the first **ONE** $x = y_1$ is one such anchor.

Distinct configurations. If $\boldsymbol{S} \ne \boldsymbol{S}'$ and $U(\boldsymbol{S}) \ne U(\boldsymbol{S}')$, the anchors differ. When $D(\boldsymbol{S}) \ne D(\boldsymbol{S}')$, some $t \ge 1$ has $f_{\boldsymbol{S}}(t) \ne f_{\boldsymbol{S}'}(t)$; if exactly one of the two has a **ONE** at that t, their first-positive-**ONE** indices differ.

Proof. The shift claim is immediate from $h_j \ne h_{j'}$. **ONE** distinctness follows because **ONE** positions in $D \cap \{0, ..., P-1\}$ are strictly increasing when listed in order. For configurations, if $U(\boldsymbol{S}) \ne U(\boldsymbol{S}')$ the anchors differ; otherwise if $D(\boldsymbol{S}) \ne D(\boldsymbol{S}')$, let $t \ge 1$ be minimal with $f_{\boldsymbol{S}}(t) \ne f_{\boldsymbol{S}'}(t)$ and compare first **ONE** locations. □

5.8 Qualitative Hardy–Littlewood from one x and many ONES

Fix admissible $\boldsymbol{h}$ and $n \ge \ell(\boldsymbol{h})$ large enough that $p_n \ge \max\left(W(\boldsymbol{h}), \frac{H_{\max}}{4}\right)$. Let $\boldsymbol{S} = \boldsymbol{S}(\boldsymbol{h})$ and $x = U(\boldsymbol{S}) \le p_n$ as in Section 5.6.

1. **Existence of a prime tuple at qualifying scales.** Section 4.3 gives a full prime k-tuple $x + h_j$ with all entries $\le p_{n+1}^2$ at each depth n with $p_{n+1} \ge W(\boldsymbol{h})$ (Section 4.2.43).
2. **Many tuple anchors from the same $\boldsymbol{h}$.** Section 2.7.2 gives $M = \prod m_i$ distinct **ONES** $y_r \in D(\boldsymbol{S})$ per period; any y_r that is a **ONE** at depth $n+1$ with $y_r \le p_n^2$ yields a prime tuple by the same trial-division chain as Section 4.3.
3. **Infinitely many distinct tuples.** Section 4.3.4 gives pairwise distinct full prime k-tuples $\left(z_m + h_j\right)_j$ from the periodic progression $z_m = y + mP$ (Section 4.3.1).
4. **Role of $|\Omega|$.** Section 4.1.2 gives $U(\boldsymbol{S}) \le P$ for every configuration; $\boldsymbol{S}(\boldsymbol{h})$ is the distinguished member carrying $\boldsymbol{h}$, with the sharper bound $U \le p_n$ from Section 4.2.42.

Infinitely many distinct prime $\boldsymbol{k}$-tuples. Let y and P be as in Section 4.3.4. The progression $z_m = y + mP$ supplies infinitely many distinct **ONE** anchors. Each **good** index m yields a full prime k-tuple $T_m = \left(z_m + h_j\right)_j$ only after extending the sieve to depth $N(m)+1$ large enough that $p_{N(m)+1}^2 \ge z_m + H_{\max}$ and no p_i with $i \le N(m)+1$ divides any shift $z_m + h_j$ (Section 4.3.3, Section 4.3.2). Large z_m are handled by a larger trial window, not by the initial bound $p_{N_0}^2$.

What is not concluded here. This does **not** identify the tuple frequencies with the singular series, nor prove the classical counting formula for $\pi_{\boldsymbol{h}}(x)$.

Proof. Section 4.3.4. □

> **Note on notation.** At a fixed sieve depth n, the first **ONE** $x = U(\boldsymbol{S}(\boldsymbol{h}))$ is the canonical anchor. The periodic progression $z_m = y + mP$ from Section 4.3.4 supplies infinitely many **distinct** anchors and tuples as m varies.

6 Conclusion

6.1 Summary

1. Tail data are encoded by a fixed gap $K = p_i - m_i$: only K residues are forbidden mod each large p_i.
2. Section 4.1 / Section 4.1.2: for every configuration in Ω with $m_i \ge 1$, $U \le P$.

3. Section 2.7.4 / Section 3.1.22 / Section 2.7.6 / Section 3.1.23 / Section 4.2.52 / Section 4.2.53: period budget $L_0 \le P - M$, $E[W^I] > E[W^{II}]$ for all $x \ge 1$, and $E[W^I] = \frac{P-M}{M}$ for $x \ge P - M$ (pinned; HL multiplicities).
4. Section 4.2.49 / Section 4.2.57 / Section 4.3: for $\boldsymbol{S}(\boldsymbol{h})$ and $n + 1 \ge \ell(\boldsymbol{h})$, a prime k-tuple below p_{n+1}^2 (sieve hole; Section 4.2.3).
5. Section 4.2.31: any probability bound below $\frac{1}{|\Omega|}$ forces $N_{\text{bad}} = 0$ (used on the general model when $t \ge P$).
6. Section 4.2.38: on CRT grids at rate q_i, optional grid benchmark ($E[W_i^I] > E[W_i^{II}]$ via Equation 108).
7. Section 2.7.2 / Section 4.2.55: exactly $\prod m_i$ **ONES** per period; $L_0 \le P - M$ always; $L_0 \le p_n - 1$ when $p_n \ge W(\boldsymbol{h})$.
8. Section 5.1 / Section 5.2 / Section 5.3: each admissible $\boldsymbol{h}$ determines m_i with constant tail gap $p_i - m_i = K_{\boldsymbol{h}}$ for $i \ge \ell(\boldsymbol{h})$.
9. Section 5.6: for $\boldsymbol{S}(\boldsymbol{h})$ with $p_n \ge W(\boldsymbol{h})$, a single anchor $x \le p_n$ determines the tuple $x + h_j$.
10. Section 5.7 / Section 5.8 / Section 4.2.61 / Section 4.3 / Section 4.3.4: complete prime tuples below p_{n+1}^2 at depths with $p_{n+1} \ge W(\boldsymbol{h})$, and infinitely many **distinct** such tuples via periodic **ONE** progressions once some qualifying depth exists.

6.2 Explicit scope and limitations

The following are proved unconditionally in this document (modulo the cited definitions and Section 4.2.46):

1. For every admissible $\boldsymbol{h}$, a **full prime $\boldsymbol{k}$-tuple** below p_{n+1}^2 at verified / finite sieve depths (Section 4.3, Section 4.2.37, Section 4.2.50); large-n asymptotics via Equation 163 are optional and conditional on domination (Section 4.2.36).
2. **Infinitely many distinct** such prime k-tuples for each admissible $\boldsymbol{h}$ (Section 4.3.4).
3. For every configuration $\boldsymbol{S} \in \Omega$ with $m_i \ge 1$, a first **ONE** satisfies $U \le P$ (Section 4.1.2); for $\boldsymbol{S}(\boldsymbol{h})$ and every $n \ge \ell(\boldsymbol{h})$, Section 4.2.49 gives $U \le p_n^2$ when $N_{\text{bad}}(p_n^2) = 0$ (Section 4.2.37).
4. Section 4.2.7 / Section 4.2.11: product-coordinate second at (P, M) (not intersection); easy $0 < \frac{1}{|\Omega|}$ at $t_n = p_n^2$.
5. Section 4.2.15 / Section 4.2.5: first need not dominate second forever; hole and domination split across models.
6. Section 4.2.8 / Section 4.2.12 / Section 4.2.16: aggregated intersection-style second; domination partial, hole late (Section 4.2.6).
7. Capped waiting-time comparisons: $E[W^I] > E[W^{II}]$ at the matching coordinate rate q_i for every $n \ge 1$ on pinned CRT grids (Equation 84, Equation 108), and on the consecutive scan for every cutoff $x \ge 1$ against the best coordinate rate at j^* (Section 3.1.22, Section 3.1.23); once $x \ge P - M$, $E[W^I] = \frac{P-M}{M}$ (Section 2.7.6), matching the product-rate geometric limit (Section 3.1.24).

The following are **not** proved:

1. Exhaustive finite verification for **every** admissible $\boldsymbol{h}$ and every $\ell \le n < N_0$ beyond the twin table in Section 4.2.37 (the procedure is finite; carrying it out for other patterns is computational).
2. Hardy–Littlewood asymptotic frequency or singular-series accuracy.
3. The classical counting formula $\pi_{\boldsymbol{h}}(x) \sim \mathfrak{S}(\boldsymbol{h}) \frac{x}{(\log x)^k}$.
4. Unconditional results beyond trial division to p_{n+1}^2 (no analytic sieve input).

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